

# Term counts in the derivatives of power towers

One framework and three sequences:

OEIS A293239 ( $x^x$ ), A290268 ( $x^{x^2}$ ), A281434 ( $x^{x^x}$ )

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## Abstract

For a function  $f$ , let  $a(n)$  be the number of distinct monomials in the fully collected  $n$ th derivative  $f^{(n)}$ . This article treats three instances of that question at once:  $f = x^x$  (OEIS A293239),  $f = x^{x^2}$  (A290268) and  $f = x^{x^x}$  (A281434).

Part 1 proves, once, the apparatus the three share: a canonical differential-polynomial form  $f^{(n)} = f \cdot P_n$  or  $f \cdot x^{-n} P_n$  with a first-order recurrence for  $P_n$  over  $\mathbb{Z}$ , integrality of its coefficients, and the envelope bound that turns an upper estimate into a cancellation count. It also proves the linear independence of logarithmic monomials that makes “the number of terms” an invariant of  $f$  rather than of a differentiation algorithm, and the resulting identity  $a(n) = |\text{supp } P_n|$  — for two of the three frames;  $x^{x^x}$  falls outside that statement and supplies its own, as Part 1 is careful to say. Two normalizations are stated, because neither one covers all three parts in the frames those parts use; Remark 1.3 is careful about what that does and does not mean.

The three remaining parts are the three original investigations, unabridged. They reach different answers by different routes. **None of the three headline questions is fully settled here, and this article does not obscure that.** Two of the three entries print a conjectured closed form and neither is proved; the third prints no closed form, and there the growth *order* is settled while the constant is not. For  $x^x$  the term count is reduced exactly to the zeros of the first-kind Lehmer–Comtet triangle A008296, giving  $a(n) = 1 + \binom{n+1}{2} - Z(n)$  and  $a(n) = \Theta(n^2)$ , but the principal closed formula needs a nonvanishing statement that is isolated and left open; separately, the recurrence printed in the entry with the range  $n > 6$  is *disproved* at  $n = 8$ . For  $x^{x^2}$  the conjectured expression is proved to be an upper bound, every predicted cancellation is explained, and  $a(n) = \Theta(n^2)$  is established, but the conjectured leading constant is not; a later addition to that part (Section 3.14) proves the missing nonvanishing statement at logarithmic deficits three and four, leaving deficits five and above open. For  $x^{x^x}$  two explicit polynomial bounds give  $a(n) = \Theta(n^3)$  unconditionally, while the sharper equivalent  $a(n) \sim \frac{2}{3}n^3$  is not proved — and there, unlike the other two, the obstruction is not a nonvanishing statement at all but a density one, because additional zeros provably exist. Part 5 sets the three side by side, says what the shared method does and does not buy, and records the one piece of structure they share beyond it: every cancellation any of them can prove is a parity zero of the same falling factorial.

All computations reported here are exact. Every modular certificate is confirmed by exact integer arithmetic before a coefficient is declared to vanish, so no claim rests on a probabilistic zero test; but every computation certifies a finite range and none of them is an all- $n$  proof.

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## How this article is arranged

Three separate reports were written on three OEIS sequences that ask the same question about three different functions. They are collected here as one article because they share a construction, and kept as four distinct parts because the questions they answer are different: three functions, three recurrences, three answers. They do share one thing beyond the construction, and Section 5.4 sets it out — all three reduce to integer zeros of derivatives and differences of the same falling factorial — but no argument in any part carries over to another. Nothing that was proved in any of the three has been dropped, weakened or strengthened in the merge.

- **Part 1** is new to this article. It states and proves the machinery that the three reports had each established separately, and is explicit about where that generality stops: the canonical form, the recurrence and the envelope bound cover all three, the independence statement covers two of the three. It settles none of the three counting problems and is not intended to: the content of the subject is in the nonvanishing questions, and those do not transfer between the three functions.
- **Parts 2, 3 and 4** are the three original articles. Each opens with a status box saying what it does and does not prove, and a note on how it instantiates Part 1. Each part retains the setup section it was written with, so it can still be read on its own; those sections are now instances of Part 1 rather than independent developments, and are flagged as such where they begin.
- **Part 5** compares the three.
- **Sections 3.14–3.26**, at the end of Part 3, were added on 29 September 2026 from a later manuscript on A290268, which closes that part’s missing step at logarithmic deficits three and four. They follow the part they extend, so that no number elsewhere in the article moved, and they use their own coordinates (Table 3).

The parts keep the notation of their sources, which is not uniform across them. Section 1.5 sets out what a reader crossing between parts needs. Two things are worth carrying in advance:  **$j$  is the exponent of  $\log x$  in Parts 2 and 3 but the exponent of  $x^{-1}$  in Part 4**, where  $\ell$  is the logarithm exponent; and  **$z$  is  $x^2$  in Part 3,  $\log x$  in Part 4, and a formal generating-function variable in Part 2**. Also  $a(n)$  means a different sequence in each of the three parts.

Each part ends with a description of the code and data that accompanied it. Those descriptions have been updated to the merged package, in which every script lives under `code/` and every generated table under `data/`, and the article is `article.tex` and `article.pdf`. The three groups of scripts do not collide and each was run from the merged layout before this article was built.

These are AI-assisted research notes. None of the three has been refereed or machine-checked.

## 1 The common framework

Each of the three parts that follow studies a different function, obtains a different recurrence and arrives at a different answer. What they share is the apparatus in this part: a canonical polynomial form for the  $n$ th derivative, a uniqueness statement that makes “the number of terms” well defined, a first-order recurrence on that form, and the reduction of the counting

problem to a nonvanishing problem. Each of the three source reports established this apparatus for its own function. Here it is proved once.

Nothing in this part decides any of the three counting problems. It is deliberately cheap: every statement below is an exercise, and the work in this subject is entirely in the nonvanishing questions that Parts 2, 3 and 4 take up.

## 1.1 Setting

Throughout, functions are real-valued and smooth on  $(0, \infty)$ , and  $f$  is positive there, so  $\log f$  is defined and  $f'/f$  makes sense.

**Definition 1.1** (Differential frame). A *differential frame* is a tuple  $w = (w_1, \dots, w_m)$  of smooth functions on  $(0, \infty)$  together with the polynomial ring  $R = \mathbb{Z}[w_1, \dots, w_m]$ , where the  $w_i$  are treated as independent indeterminates. We write  $\text{supp } P$  for the set of exponent vectors of the monomials occurring in  $P \in R$  with a nonzero coefficient, and  $|\text{supp } P|$  for its cardinality.

The distinction between  $R$  as an abstract polynomial ring and the functions the  $w_i$  denote is the point of Section 1.3: a priori a polynomial identity in  $R$  is a stronger statement than the corresponding identity of functions, and it is the converse that has to be proved.

## 1.2 The canonical form and its recurrence

**Theorem 1.2** (Canonical differential-polynomial form). *Let  $w$  be a differential frame with ring  $R = \mathbb{Z}[w_1, \dots, w_m]$ , and let  $D: R \rightarrow R$  be the  $\mathbb{Z}$ -derivation determined by  $Dw_i = \delta_i$  for some  $\delta_1, \dots, \delta_m \in R$ .*

(a) (Unnormalized form.) *Suppose*

$$w'_i = \delta_i(w) \quad (1 \leq i \leq m), \quad \frac{f'}{f} = M(w) \quad \text{for some } M \in R.$$

*Then for every  $n \geq 0$ ,*

$$f^{(n)} = f \cdot P_n(w), \quad P_0 = 1, \quad P_{n+1} = M P_n + D P_n, \quad (1.1)$$

*and  $P_n \in R$ ; in particular every coefficient of  $P_n$  is an integer.*

(b) (Form normalized by  $x^{-n}$ .) *Suppose instead*

$$w'_i = \frac{1}{x} \delta_i(w) \quad (1 \leq i \leq m), \quad x \frac{f'}{f} = M(w) \quad \text{for some } M \in R.$$

*Then for every  $n \geq 0$ ,*

$$f^{(n)} = f \cdot x^{-n} P_n(w), \quad P_0 = 1, \quad P_{n+1} = (M - n) P_n + D P_n, \quad (1.2)$$

*and again  $P_n \in R$ .*

*Proof.* For (a), induct on  $n$ . The case  $n = 0$  is  $P_0 = 1$ . Assuming  $f^{(n)} = f P_n(w)$  and differentiating,

$$f^{(n+1)} = f' P_n(w) + f \sum_{i=1}^m w'_i \frac{\partial P_n}{\partial w_i}(w) = f \left( M(w) P_n(w) + \sum_{i=1}^m \delta_i(w) \frac{\partial P_n}{\partial w_i}(w) \right),$$

which is  $f(MP_n + DP_n)(w)$ . Since  $M, \delta_i \in R$  and  $P_0 = 1 \in R$ , induction also gives  $P_n \in R$ , and  $R$  consists of polynomials with integer coefficients.

For (b), the case  $n = 0$  reads  $f = f x^0 \cdot 1$ . Assume  $f^{(n)} = f x^{-n} P_n(w)$  and differentiate:

$$f^{(n+1)} = f' x^{-n} P_n - n f x^{-n-1} P_n + f x^{-n} \sum_i w'_i \frac{\partial P_n}{\partial w_i}.$$

Multiplying by  $x^{n+1}/f$  and using  $x f'/f = M$  and  $x w'_i = \delta_i$  gives

$$\frac{x^{n+1} f^{(n+1)}}{f} = (M - n) P_n + \sum_i \delta_i \frac{\partial P_n}{\partial w_i} = ((M - n) P_n + DP_n)(w),$$

which is the assertion for  $n + 1$ . Integrality follows as before.  $\square$

*Remark 1.3* (Why both clauses are kept). Neither clause covers all three parts *in the frames those parts use*, and since the parts are reproduced here unchanged, both clauses are needed.

Part 3 uses the frame  $z = x^2$ ,  $L = \log x$ , for which  $z' = 2x \notin \mathbb{Z}[z, L]$ : clause (a) does not apply, and clause (b) applies verbatim. Conversely Part 4 uses  $u = x^x$ ,  $v = x^{-1}$ ,  $z = \log x$ , for which  $xu' = u(z + 1)/v \notin \mathbb{Z}[u, v, z]$ : clause (b) does not apply, while clause (a) does.

It would overstate the situation to say that no single clause *could* have covered all three. Clause (b) in fact does, once the frames are changed so that  $x$  appears in place of  $x^{-1}$ :

$$x^x : (x, \log x), \quad x^{x^2} : (x^2, \log x), \quad x^{x^x} : (x^x, x, \log x),$$

each of which satisfies  $xw'_i \in R$  and  $xf'/f \in R$ , and each of which gives the same term counts. For  $x^{x^x}$  this is legitimate only because the exponent of  $x^{-1}$  in that part's envelope never exceeds  $n$ , so multiplying by  $x^n$  really does clear it. What one cannot do is change the frames and keep the parts: each part's transition table, envelope, coefficient indices and data files are written in its own variables, and rewriting some two and a half thousand lines of dense manipulation to gain a cosmetic uniformity in this section would be a poor trade. So the parts keep their frames and this theorem keeps two clauses.

One genuine obstruction is worth recording separately, since it is about independence rather than convenience. In Part 3 one might try to make clause (a) apply by adjoining  $y = x^{-1}$  to the frame  $(x^2, \log x)$ . The derivatives then are polynomial, but  $zy^2 = 1$  identically, so the monomials of the enlarged frame are no longer independent as functions and  $|\text{supp } P_n|$  stops counting anything at all. That failure is not a matter of taste.

Each of the three parts instantiates Theorem 1.2 in one line; the resulting recurrences are collected in Table 1.

| Part | $f$       | frame and clause                              | recurrence for $P_{n+1}$                                                             |
|------|-----------|-----------------------------------------------|--------------------------------------------------------------------------------------|
| 2    | $x^x$     | $t = x^{-1}$ , $L = \log x$ ; (a)             | $(1 + L)P_n + t \partial_L P_n - t^2 \partial_t P_n$                                 |
| 3    | $x^{x^2}$ | $z = x^2$ , $L = \log x$ ; (b)                | $(z(2L + 1) - n)P_n + 2z \partial_z P_n + \partial_L P_n$                            |
| 4    | $x^{x^x}$ | $u = x^x$ , $v = x^{-1}$ , $z = \log x$ ; (a) | $u(z^2 + z + v)P_n + u(z + 1)\partial_u P_n - v^2 \partial_v P_n + v \partial_z P_n$ |

Table 1: The three instances. In each case the displayed recurrence is exactly the one derived independently in the corresponding part (Propositions 2.1, 3.3 and 4.2). Note that  $z$  denotes  $x^2$  in Part 3 and  $\log x$  in Part 4; see Section 1.5.

### 1.3 Independence, and why the count is well posed

Counting “the terms of the  $n$ th derivative” is meaningless until one fixes what a term is, because a symbolic differentiator can be made to produce almost any number of summands by factoring differently. The following classical statement is what makes the count an invariant of the function rather than of an algorithm. Parts 2 and 3 each prove the special case they need by this argument. Part 4’s frame is not of this form, for the reason given after Corollary 1.5, and it proves its own independence statement by a related but genuinely different asymptotic argument.

**Theorem 1.4** (Independence of logarithmic monomials). *Let  $(\lambda_1, j_1), \dots, (\lambda_N, j_N)$  be distinct pairs with  $\lambda_k \in \mathbb{R}$  and  $j_k \in \mathbb{Z}_{\geq 0}$ . Then the functions  $x^{\lambda_k}(\log x)^{j_k}$  are linearly independent over  $\mathbb{R}$  on every interval  $(a, \infty)$  with  $a \geq 0$ .*

*Proof.* Suppose  $\sum_k c_k x^{\lambda_k}(\log x)^{j_k} = 0$  on  $(a, \infty)$  with not all  $c_k$  zero. Substitute  $x = e^s$  and group by exponent: the identity becomes

$$\sum_{\lambda} e^{\lambda s} p_{\lambda}(s) = 0 \quad (s > \log \max(a, 1)), \quad (1.3)$$

where  $\lambda$  runs over the distinct values among the  $\lambda_k$  and  $p_{\lambda}(s) = \sum_{k: \lambda_k = \lambda} c_k s^{j_k}$  is a polynomial. Because the pairs are distinct,  $p_{\lambda}$  is not the zero polynomial for at least one  $\lambda$ . Let  $\Lambda$  be the largest such  $\lambda$ . Dividing (1.3) by  $e^{\Lambda s}$  gives

$$p_{\Lambda}(s) = - \sum_{\lambda < \Lambda} e^{(\lambda - \Lambda)s} p_{\lambda}(s) \xrightarrow{s \rightarrow +\infty} 0,$$

since each exponent  $\lambda - \Lambda$  is strictly negative and an exponential decay beats a polynomial. No nonzero polynomial tends to 0 at  $+\infty$ , so  $p_{\Lambda} = 0$ , a contradiction.  $\square$

**Corollary 1.5** (The count is well defined). *Let  $f$  and  $w$  be as in either clause of Theorem 1.2, and suppose that the monomials  $w^{\alpha}$  of the frame correspond to pairwise distinct pairs  $(\lambda(\alpha), j(\alpha))$  in the sense that  $w^{\alpha} = x^{\lambda(\alpha)}(\log x)^{j(\alpha)}$ . Then the representation of  $f^{(n)}$  in (1.1), respectively (1.2), is unique, and the number of distinct monomials in the fully collected  $n$ th derivative is*

$$a(n) = |\text{supp } P_n|. \quad (1.4)$$

*Proof.* Two representations would differ by a finite vanishing linear combination of the functions  $w^{\alpha}$ , hence of pairwise distinct  $x^{\lambda}(\log x)^j$ , which Theorem 1.4 forbids. Uniqueness turns (1.4) into a definition that does not refer to any particular way of performing the differentiation.  $\square$

The hypothesis of Corollary 1.5 is verified separately in each part, since it depends on the frame: for  $x^x$  the monomials  $t^k L^j$  are  $x^{-k}(\log x)^j$ ; for  $x^{x^2}$  the monomials  $z^k L^j$  carry the extra factor  $x^{-n}$  and give  $x^{2k-n}(\log x)^j$ ; for  $x^{x^x}$  the variable  $u = x^x$  is itself not of the form  $x^{\lambda}$  with constant  $\lambda$ , and Part 4 accordingly proves its independence statement by a related but genuinely different asymptotic argument, which is retained there in full.

### 1.4 Envelopes, upper bounds and exact computation

**Proposition 1.6** (Envelope bound). *Write  $P_{n+1} = \Phi_n(P_n)$  for either recurrence of Theorem 1.2, so that  $\Phi_n(P) = MP + DP$  in clause (a) and  $\Phi_n(P) = (M - n)P + DP$  in clause (b); note that in the second case the operator depends on  $n$ , although  $\text{supp}((M - n)P) \subseteq \text{supp}(MP) \cup \text{supp } P$  for every  $n$ , so the support behaviour does not. Suppose  $\mathcal{E}_0, \mathcal{E}_1, \dots$  are*

finite sets of exponent vectors with  $\text{supp } P_0 \subseteq \mathcal{E}_0$  and with the property that  $\Phi_n$  maps any polynomial supported in  $\mathcal{E}_n$  to a polynomial supported in  $\mathcal{E}_{n+1}$ . Then  $\text{supp } P_n \subseteq \mathcal{E}_n$  for every  $n$ ; and if moreover the count is well defined in the sense of Corollary 1.5, or by the part's own substitute for it, then

$$a(n) \leq |\mathcal{E}_n|.$$

*Proof.* Immediate induction for the support statement. Only the direction matters:  $\Phi_n$  sends each monomial to a bounded set of monomials, so a cell outside  $\mathcal{E}_{n+1}$  receives no contribution at all. Cancellation between contributions can remove a cell of  $\mathcal{E}_{n+1}$  from the support, but it can never create one outside. The bound on  $a(n)$  is then (1.4).  $\square$

The deficit

$$\Delta(n) = |\mathcal{E}_n| - a(n) \geq 0 \tag{1.5}$$

is therefore exactly the number of cells inside the envelope whose coefficient vanishes. This is the shape every one of the three problems takes: an envelope that is easy to count, and a cancellation count that is not. The three parts differ precisely in how much they can say about  $\Delta$ .

*Remark 1.7* (The parts count deficits differently). A reader moving between the parts must not compare their deficit numbers directly. Part 4 writes  $\Delta(n)$  for *all* vanishing cells of its envelope, including the ones it proves must vanish; that quantity is itself a published sequence, OEIS A352697. Part 2 writes  $E(n)$  for only the zeros *outside* the families it has already proved to vanish, so  $E$  is the residual defect and is conjectured to be identically zero, which  $\Delta$  certainly is not. Part 3 likewise subtracts its proved reflection zeros before counting. Both conventions are useful and each part keeps its own; the generic  $\Delta$  of (1.5) is the first kind.

**Proposition 1.8** (Exact and modular computation). *With notation as above,  $P_0, \dots, P_N$  can be computed in  $O(\sum_{n \leq N} |\mathcal{E}_n|)$  integer ring operations, using only integer arithmetic; the implied constant is the frame's transition count  $|\text{supp } M| + \sum_i |\text{supp } \delta_i|$ . Moreover, for any modulus  $q \geq 2$ , a coefficient of  $P_n$  whose residue modulo  $q$  is nonzero is nonzero.*

*Proof.* Each recurrence applies at most  $|\text{supp } M| + \sum_i |\text{supp } \delta_i|$  monomial-to-monomial transitions to each cell of the support — a constant depending on the frame and not on  $n$  — so one step costs  $O(|\mathcal{E}_n|)$  operations; the coefficients are integers by Theorem 1.2. Reduction modulo  $q$  is a ring homomorphism, so it commutes with the recurrence; a nonzero image forces a nonzero preimage.  $\square$

Proposition 1.8 is the reason the computations reported in the three parts are certifications and not samples: a nonzero residue settles nonvanishing outright. The converse direction is the weak one — a zero residue on its own certifies nothing — and the three parts close it in three different ways, which is worth stating precisely rather than in one sentence:

- Part 2 computes the triangle in exact integer arithmetic outright, so no residue has to be resolved; its modular lemma is used for a different purpose, to rule out integer roots of a fixed polynomial over a whole diagonal.
- Part 3 runs two moduli and checks that the two lists of vanishing residues are disjoint, so at least one residue of every coefficient is nonzero. Its exact integer run stops at  $n = 200$ ; the certification to 3000 is modular.
- Part 4 re-evaluates every vanishing residue with an independent exact coefficient formula, and never accepts a zero residue as evidence.

All three are deterministic. No statement anywhere in this article rests on a probabilistic zero test. None of them, however, is an all- $n$  proof: each certifies a finite range, and each part says so where it reports the range.

## 1.5 Notation across the parts

The three parts below retain the notation of the reports they come from. That notation is not consistent between them, and rewriting it would have meant altering some two and a half thousand lines of dense manipulation with no way to check the result; declaring the clash is safer than silently repairing it.

The table that matters is not a dictionary of letters — nearly every letter is reused somewhere as a summation index, a residue or a generating-function variable, and no short table can be both complete and correct about that. What a reader does need, before crossing from one part to another, is the canonical monomial each part counts, and in particular which of its indices is the exponent of  $\log x$ .

|                   | Part 2: $x^x$            | Part 3: $x^{x^2}$         | Part 4: $x^{x^x}$                 |
|-------------------|--------------------------|---------------------------|-----------------------------------|
| frame             | $t = x^{-1}, L = \log x$ | $z = x^2, L = \log x$     | $u = x^x, v = x^{-1}, z = \log x$ |
| monomial          | $x^{x-k}(\log x)^j$      | $x^{x^2+2k-n}(\log x)^j$  | $x^{x^x+kx-j}(\log x)^\ell$       |
| $\log x$ exponent | $j$                      | $j$                       | $\ell$                            |
| power of $x^{-1}$ | $k$                      | — (absorbed in $x^{-n}$ ) | $j$                               |
| normalization     | $f^{(n)} = x^x P_n$      | $f^{(n)} = f x^{-n} P_n$  | $f^{(n)} = f P_n$                 |

Two warnings follow from that table, and they are the two places where a reader who is skimming will go wrong.

1.  **$j$  changes role.** In Parts 2 and 3 the index  $j$  is the exponent of  $\log x$ . In Part 4 it is the exponent of  $x^{-1}$ , and the exponent of  $\log x$  is  $\ell$ . Every envelope, every bound and every cancellation family in Part 4 is indexed in the second convention.
2.  **$z$  has three meanings.** It is  $x^2$  in Part 3 and  $\log x$  in Part 4 — a direct contradiction — and in Part 2 it is neither: there it is the formal variable in which the Lehmer–Comtet numbers are defined, through  $\Phi(z) = (1+z)\log(1+z)$  and  $b(m, r) = \frac{m!}{r!} [z^m] \Phi(z)^r$ .

A third warning concerns only the later Sections 3.14–3.26 at the end of Part 3: they use the depth coordinates of their source and of the Lean development, in which  **$k$  is the exponent of  $\log x$**  and  **$j$  the exponent of  $x$**  — the reverse of the rest of that part — and in which  $d$  is the logarithmic deficit. Table 3 gives the full dictionary with the false readings.

Beyond these, the letters  $u, v, t, q, U, Z, E$  and  $\Delta$  all occur in more than one part with unrelated meanings, as frame variables in one place and as summation indices, residues, primes, proposed closed forms or generating-function variables in another. Each is introduced where it is used, and within any one paragraph the meaning is fixed; but nothing in this article guarantees that a letter carries the same meaning two parts later, and the reader should not assume it. Finally,  $a(n)$  denotes a different sequence in each of the three parts — its own — and the deficit conventions differ as described in Remark 1.7.

## 2 $x^x$ : OEIS A293239

**Framework.** This part instantiates Theorem 1.2(a) with the frame  $t = x^{-1}$ ,  $L = \log x$ ; Proposition 2.1 below is that instance, derived here independently. Lemma 2.2 is the special case of Theorem 1.4 this part needs. Both are retained in full so that the part can be read on its own. Throughout this part  $a(n)$  denotes the term count for  $x^x$ .

**Abstract of the original report.** *The sequence A293239 counts the nonzero terms in the fully expanded  $n$ th derivative of  $x^x$ . Its proposed generating function is rational, and its proposed eventual formula is a quadratic quasipolynomial of period four. We reduce the counting problem exactly to the zero set of the first-kind Lehmer–Comtet numbers. All cancellations predicted by the conjecture are proved: a symmetry family and one additional zero. The converse assertion, excluding every further zero, is not established here; consequently this report does not claim a complete proof of the principal closed formula. The unconditional results include an exact defect formula, upper and lower bounds proving quadratic-order growth, nonvanishing on every prime-offset diagonal with an exact prime-adic valuation, and complete zero classifications for the first three columns and offsets one through sixteen. Independently implemented exact computations confirm the formula through  $n = 3000$ . A separate conjectured recurrence in the OEIS entry is false with its printed range: it already fails at  $n = 8$ . For the proposed sequence its correct eventual starting point is  $n = 14$ . The accompanying code and certificates distinguish finite verification from statements proved for infinitely many indices.*

### 2.1 The question and the status of the answer

For  $x > 0$ , put  $f(x) = x^x = \exp(x \log x)$ . Let  $a(n)$  be the number of nonzero summands after  $f^{(n)}(x)$  is expanded into the monomials

$$x^{x-k}(\log x)^j, \quad k, j \in \mathbb{Z}_{\geq 0}, \quad (2.1)$$

and coefficients of identical monomials are collected. In particular, an unexpanded expression such as  $x^x(1 + \log x)^n$  is not one term for this purpose. This is the convention in OEIS A293239, introduced by Vladimir Reshetnikov in 2017 [1].

The entry attributes the rational generating function and closed formula to Colin Barker and the proposed asymptotic equivalent to Václav Kotešovec. To separate a proved object from the proposed answer, define

$$q(0) = 1, \quad q(n) = 1 + \frac{n(n+1)}{2} - \left\lfloor \frac{n-1}{4} \right\rfloor - \mathbf{1}_{\{n \geq 8\}} \quad (n \geq 1). \quad (2.2)$$

The principal conjecture is  $a(n) = q(n)$ . For  $n \geq 8$ , this says simply

$$a(n) \stackrel{?}{=} \frac{n(n+1)}{2} - \left\lfloor \frac{n-1}{4} \right\rfloor. \quad (2.3)$$

The proposed generating function is

$$Q(z) := \sum_{n \geq 0} q(n)z^n = \frac{1 + z^2 + z^3 + z^6 - z^8 + z^9 + z^{12} - z^{13}}{(1-z)^2(1-z^4)}. \quad (2.4)$$

The rational function in (2.4) is indeed the generating function of the explicitly defined sequence  $q$ . What remains to be proved is its identification with the derivative count  $a$ .

### 2.1.1 What is proved

The central unconditional identity will be

$$a(n) = q(n) - E(n), \quad (2.5)$$

where  $E(n)$  counts a precisely defined set of additional zero coefficients. In particular,  $E(n)$  is a nonnegative, nondecreasing integer. Thus the proposed answer is an upper bound, not a formula that can be exceeded by a previously overlooked cancellation.

We prove, for all  $n \geq 0$ ,

$$n + 1 + \left\lfloor \frac{n^2}{4} \right\rfloor \leq a(n) \leq q(n). \quad (2.6)$$

Consequently  $a(n) = \Theta(n^2)$ . This does not establish the sharper claim  $a(n) \sim n^2/2$ ; that claim is equivalent to  $E(n) = o(n^2)$ . We also prove infinite nonvanishing results that substantially restrict where additional zeros could lie. The remaining gap is stated explicitly in Conjecture 2.9.

### 2.1.2 A correction that is unconditional

The OEIS entry also proposes

$$a(n) = 2a(n-1) - a(n-2) + a(n-4) - 2a(n-5) + a(n-6) \quad (n > 6). \quad (2.7)$$

This range is incorrect. Exact low-order differentiation gives

$$a(2) = 4, \quad a(3) = 7, \quad a(4) = 11, \quad a(6) = 21, \quad a(7) = 28, \quad a(8) = 35,$$

whereas the right side of (2.7) at  $n = 8$  is

$$2 \cdot 28 - 21 + 11 - 2 \cdot 7 + 4 = 36 \neq 35.$$

For  $q$ , the recurrence holds for every  $n \geq 14$ , and it fails at  $n = 13$ . This correction does not disprove the proposed generating function or closed formula. Section 2.8 explains the distinction.

## 2.2 A canonical polynomial for each derivative

Set  $t = x^{-1}$  and  $L = \log x$ . There is a polynomial  $P_n \in \mathbb{Z}[t, L]$  such that

$$f^{(n)}(x) = x^x P_n(x^{-1}, \log x). \quad (2.8)$$

Differentiating this expression and using  $t' = -t^2$ ,  $L' = t$ , and  $(x^x)' = x^x(1 + L)$  gives the following exact recurrence.

**Proposition 2.1.** *The derivative polynomials satisfy*

$$P_0 = 1, \quad P_{n+1} = (1 + L)P_n + t \frac{\partial P_n}{\partial L} - t^2 \frac{\partial P_n}{\partial t}. \quad (2.9)$$

*In particular, they have integer coefficients and can be computed using integer arithmetic only.*

For example,

$$\begin{aligned} P_1 &= 1 + L, \\ P_2 &= 1 + 2L + L^2 + t, \\ P_3 &= 1 + 3L + 3L^2 + L^3 + 3t + 3tL - t^2. \end{aligned}$$

Thus  $a(3) = 7$ , consistently with the example in the entry.

**Lemma 2.2.** *The monomials in (2.1) are linearly independent over  $\mathbb{R}$  when only finitely many are used. Consequently the collected expansion is unique, and  $a(n) = |\text{supp } P_n|$ .*

*Proof.* Divide a putative relation by the positive function  $x^x$  and set  $x = e^s$ . The result is  $\sum_{k=0}^K e^{-ks} p_k(s) = 0$ , with polynomial  $p_k$ . If some  $p_k$  is nonzero, let  $k_0$  be the smallest such index. Multiplication by  $e^{k_0 s}$  shows that  $p_{k_0}(s)$  tends to zero as  $s \rightarrow +\infty$ , because all remaining summands decay exponentially times a polynomial. No nonzero polynomial has that property. This is a contradiction.  $\square$

The independence statement matters: the variables  $t$  and  $L$  are related by  $t = e^{-L}$  after substitution, but this relation causes no additional finite polynomial identities.

## 2.3 The Lehmer–Comtet coefficient triangle

### 2.3.1 Definition and generating function

Put

$$\Phi(z) = (1+z) \log(1+z) = z + \sum_{d \geq 1} \frac{(-1)^{d-1}}{d(d+1)} z^{d+1}. \quad (2.10)$$

Define

$$b(m, r) = \frac{m!}{r!} [z^m] \Phi(z)^r \quad (m, r \geq 0). \quad (2.11)$$

Here  $b(0, 0) = 1$ ,  $b(m, 0) = 0$  for  $m > 0$ , and  $b(m, r) = 0$  when  $r > m$ . These are the first-kind Lehmer–Comtet numbers, tabulated in OEIS A008296 [4]. The general derivative problem and these numbers belong to the classical literature of Comtet, Lehmer, and Gould [7, 8, 9]. A recent treatment of their identities is Jeong’s preprint [10]. We derive the identities needed here rather than assuming a nonvanishing conclusion from those formulas.

**Theorem 2.3** (Exact coefficient formula). *For every  $n \geq 0$ ,*

$$f^{(n)}(x) = x^x \sum_{m=0}^n \binom{n}{m} (\log x)^{n-m} \sum_{r=0}^m b(m, r) x^{r-m}. \quad (2.12)$$

*Equivalently,*

$$[t^k][L^j]P_n(t, L) = \binom{n}{j} b(n-j, n-j-k), \quad (2.13)$$

*with out-of-range entries interpreted as zero.*

*Proof.* Taylor expansion at a fixed  $x > 0$  gives

$$\sum_{n \geq 0} \frac{f^{(n)}(x)}{x^x} \frac{u^n}{n!} = \frac{(x+u)^{x+u}}{x^x} = \exp(u \log x) \exp(x \Phi(u/x)).$$

Using (2.11), expand the second exponential:

$$\begin{aligned} \exp(x \Phi(u/x)) &= \sum_{r \geq 0} \frac{x^r}{r!} \Phi(u/x)^r \\ &= \sum_{m \geq 0} \frac{u^m}{m!} \sum_{r=0}^m b(m, r) x^{r-m}. \end{aligned}$$

Multiplication by  $\exp(u \log x)$  and extraction of the coefficient of  $u^n$  proves (2.12). All manipulations can also be performed formally, since only finitely many terms affect any prescribed coefficient.  $\square$

### 2.3.2 A shifted falling-factorial formula

Let

$$F_m(y) = y^{\overline{m}} = \prod_{h=0}^{m-1} (y - h), \quad F_0(y) = 1.$$

**Proposition 2.4.** For  $0 \leq r \leq m$ ,

$$b(m, r) = \frac{F_m^{(r)}(r)}{r!} = [u^r] \prod_{h=0}^{m-1} (u + r - h). \quad (2.14)$$

In particular, all  $b(m, r)$  are integers.

*Proof.* The binomial-series identity is  $(1+z)^y = \sum_{m \geq 0} F_m(y) z^m / m!$ . Differentiate  $r$  times in  $y$  and set  $y = r$ . The left side becomes  $(1+z)^r \log(1+z)^r = \Phi(z)^r$ . Comparing coefficients gives the first formula. The second is the polynomial Taylor formula at  $y = r$ .  $\square$

Writing  $F_m(y) = \sum_{\ell=0}^m s(m, \ell) y^\ell$  in terms of the signed Stirling numbers of the first kind also gives

$$b(m, r) = \sum_{\ell=r}^m \binom{\ell}{r} r^{\ell-r} s(m, \ell). \quad (2.15)$$

This is another exact formula, but its alternating summands do not by themselves resolve whether the result can vanish.

### 2.3.3 An integer recurrence

**Proposition 2.5.** For  $m \geq 2$  and  $1 \leq r \leq m$ ,

$$b(m, r) = (r - m + 1)b(m - 1, r) + b(m - 1, r - 1) + (m - 1)b(m - 2, r - 1). \quad (2.16)$$

Together with rows  $b(0, 0) = 1$  and  $(b(1, 0), b(1, 1)) = (0, 1)$ , this determines the entire triangle.

*Proof.* Let  $B_r(z) = \Phi(z)^r / r!$ . Since  $(1+z)\Phi'(z) = 1 + z + \Phi(z)$ , we have

$$(1+z)B'_r(z) = (1+z)B_{r-1}(z) + rB_r(z).$$

Compare coefficients of  $z^{m-1} / (m-1)!$ . The result is

$$b(m, r) + (m - 1)b(m - 1, r) = b(m - 1, r - 1) + (m - 1)b(m - 2, r - 1) + rb(m - 1, r),$$

which rearranges to (2.16).  $\square$

For reference, the first rows, omitting column zero, are

| $m \backslash r$ | 1     | 2     | 3    | 4   | 5    | 6   | 7   | 8  | 9 |
|------------------|-------|-------|------|-----|------|-----|-----|----|---|
| 1                | 1     |       |      |     |      |     |     |    |   |
| 2                | 1     | 1     |      |     |      |     |     |    |   |
| 3                | -1    | 3     | 1    |     |      |     |     |    |   |
| 4                | 2     | -1    | 6    | 1   |      |     |     |    |   |
| 5                | -6    | 0     | 5    | 10  | 1    |     |     |    |   |
| 6                | 24    | 4     | -15  | 25  | 15   | 1   |     |    |   |
| 7                | -120  | -28   | 49   | -35 | 70   | 21  | 1   |    |   |
| 8                | 720   | 188   | -196 | 49  | 0    | 154 | 28  | 1  |   |
| 9                | -5040 | -1368 | 944  | 0   | -231 | 252 | 294 | 36 | 1 |

## 2.4 The exact zero-count reduction

**Definition 2.6.** For  $n \geq 0$ , let

$$Z(n) = \#\{(m, r) : 1 \leq r \leq m \leq n, b(m, r) = 0\}.$$

Column zero is not included: its zeros for  $m > 0$  are forced by the definition and never represent potential nonzero derivative terms.

**Theorem 2.7.** For every  $n \geq 0$ ,

$$a(n) = 1 + \frac{n(n+1)}{2} - Z(n). \quad (2.17)$$

*Proof.* In (2.12), the pair  $(m, r)$  gives the monomial

$$x^{x-(m-r)}(\log x)^{n-m}$$

with coefficient  $\binom{n}{m}b(m, r)$ . The map  $(m, r) \mapsto (m-r, n-m)$  is injective for fixed  $n$ : the second coordinate recovers  $m$ , and the first then recovers  $r$ . Thus different pairs do not cancel each other. The factor  $\binom{n}{m}$  is positive. There is one nonzero entry at  $(0, 0)$ , and  $m$  potentially nonzero entries  $1 \leq r \leq m$  in every row  $m \geq 1$ . Removing exactly the zero entries proves the formula.  $\square$

This argument isolates the main difficulty. Once the triangle is known, term counting is easy; proving that its list of zeros is complete is not a consequence of merely writing down the triangle's recurrence.

### 2.4.1 Cancellations that are proved

**Proposition 2.8** (Symmetry zeros). For every integer  $h \geq 1$ ,

$$b(4h+1, 2h) = 0. \quad (2.18)$$

*Proof.* Center the product in (2.14) at  $r = 2h$ :

$$F_{4h+1}(2h+u) = u \prod_{s=1}^{2h} (u^2 - s^2).$$

This polynomial is odd. Its coefficient of the even power  $u^{2h}$  is zero.  $\square$

This family is not claimed as new: it appears as Proposition 3.10(iii) in Jeong [10], with attribution there to Theorem 7 of Lehmer [8]. Its elementary proof is included to make the reduction self-contained.

There is one further zero required by the proposed formula:

$$b(8, 5) = 0. \quad (2.19)$$

It follows immediately from the factorization proved in Section 2.5:

$$b(m, m-3) = \frac{m(m-1)(m-2)(m-3)(m-5)(m-8)}{48}.$$

The root  $m = 5$  is the first symmetry zero;  $m = 8$  is genuinely additional.

### 2.4.2 The precise missing assertion

**Conjecture 2.9** (Complete zero classification). For integers  $1 \leq r \leq m$ ,

$$b(m, r) = 0 \iff (m, r) = (8, 5) \text{ or } (m, r) = (4h + 1, 2h) \text{ for some } h \geq 1. \quad (2.20)$$

The right-to-left implication is proved above. The left-to-right implication is the unproved step in this report.

Let

$$\mathcal{Z}_0 = \{(8, 5)\} \cup \{(4h + 1, 2h) : h \geq 1\}$$

be the proved set of zeros, and define

$$e_m = \#\{r : 1 \leq r \leq m, b(m, r) = 0, (m, r) \notin \mathcal{Z}_0\},$$

$$E(n) = \sum_{m=1}^n e_m.$$

Thus  $E$  counts only zeros not in the proved list.

**Theorem 2.10** (Exact defect formula). For every  $n \geq 0$ ,

$$a(n) = q(n) - E(n).$$

*The following are equivalent: Conjecture 2.9;  $E(n) = 0$  for all  $n$ ;  $a(n) = q(n)$  for all  $n$ ; and the proposed eventual formula (2.3) for every  $n \geq 8$ .*

*Proof.* For  $n \geq 1$ , the number of symmetry zeros with first coordinate at most  $n$  is  $\lfloor (n-1)/4 \rfloor$ . The zero  $(8, 5)$  contributes  $\mathbf{1}_{\{n \geq 8\}}$ . Substituting

$$Z(n) = \left\lfloor \frac{n-1}{4} \right\rfloor + \mathbf{1}_{\{n \geq 8\}} + E(n)$$

into (2.17) proves the identity for  $n \geq 1$ ;  $n = 0$  is immediate. The first three equivalences follow from the definitions. The eventual formula also forces  $E(n) = 0$  for every  $n \geq 8$ , and nonnegativity and monotonicity then force  $E(n) = 0$  for smaller  $n$  as well.  $\square$

A hypothetical new zero at row  $M$  would decrease every count  $a(n)$  for  $n \geq M$  relative to  $q(n)$ . It would not merely alter a single isolated term of the sequence. This persistent-deficit property is a useful safeguard when testing proposed proofs or numerical data.

## 2.5 Diagonal polynomials and exact nonvanishing

### 2.5.1 A polynomial for each fixed offset

Define

$$H(z) = \frac{\Phi(z)}{z} = 1 + \frac{z}{2} - \frac{z^2}{6} + \frac{z^3}{12} - \frac{z^4}{20} + \cdots, \quad T_d(Y) = [z^d]H(z)^Y. \quad (2.21)$$

For a fixed integer  $d \geq 0$ ,  $T_d$  is a polynomial over  $\mathbb{Q}$  in  $Y$ .

**Lemma 2.11.** For  $m \geq d$ ,

$$b(m, m-d) = m^d T_d(m-d). \quad (2.22)$$

For  $d \geq 1$ ,  $T_d(Y)$  has degree  $d$ , leading coefficient  $1/(2^d d!)$ , and a factor  $Y$ . Consequently  $T_d(Y)/Y$  is a nonzero polynomial of degree  $d-1$ .

*Proof.* Since  $\Phi(z)^{m-d} = z^{m-d}H(z)^{m-d}$ , definition (2.11) gives (2.22). Now assume  $d \geq 1$  and write  $h(z) = H(z) - 1$ . Then

$$T_d(Y) = \sum_{\ell=1}^d \binom{Y}{\ell} [z^d]h(z)^\ell. \quad (2.23)$$

Every summand is divisible by  $Y$ . The only term of degree  $d$  in  $Y$  comes from  $\ell = d$ , and  $[z^d]h(z)^d = 2^{-d}$ . This proves the remaining assertions.  $\square$

The first three cases are

$$b(m, m-1) = \frac{m(m-1)}{2}, \quad (2.24)$$

$$b(m, m-2) = \frac{m(m-1)(m-2)(3m-13)}{24}, \quad (2.25)$$

$$b(m, m-3) = \frac{m(m-1)(m-2)(m-3)(m-5)(m-8)}{48}. \quad (2.26)$$

For instance,

$$T_3(Y) = \frac{Y}{12} - \frac{Y(Y-1)}{12} + \frac{Y(Y-1)(Y-2)}{48} = \frac{Y(Y-2)(Y-5)}{48}.$$

This proves both zeros on the third offset and excludes all others on that entire diagonal, not just up to a numerical cutoff.

### 2.5.2 Prime offsets: an infinite family of nonvanishing results

For a nonzero rational number  $v$ , write  $v_p(v)$  for its exponent of the prime  $p$  in its prime factorization. A rational number is called  $p$ -integral when its denominator in lowest terms is not divisible by  $p$ .

**Theorem 2.12** (Prime-offset nonvanishing). *Let  $p$  be any prime and let  $m \geq p$ . Then*

$$b(m, m-p+1) \neq 0, \quad v_p(b(m, m-p+1)) = v_p\left(\left\lfloor \frac{m}{p} \right\rfloor\right). \quad (2.27)$$

*In particular, an additional zero cannot occur on any offset  $d = m - r$  for which  $d+1$  is prime.*

*Proof.* Put  $d = p-1$  and  $h_j = [z^j](H(z) - 1)$ . For  $1 \leq j \leq p-2$ , the number  $h_j = (-1)^{j-1}/(j(j+1))$  is  $p$ -integral. Moreover,

$$ph_{p-1} \equiv 1 \pmod{p}.$$

For odd  $p$ , this follows from  $ph_{p-1} = -1/(p-1)$ ; for  $p=2$ , it is  $2h_1 = 1$ .

Divide (2.23) by  $Y$ . Its  $\ell=1$  summand is  $h_{p-1}$ . For every  $2 \leq \ell \leq p-1$ , a composition of  $p-1$  into  $\ell$  positive parts has all parts at most  $p-2$ . Therefore every coefficient contributing to  $[z^{p-1}]h(z)^\ell$  is  $p$ -integral. Also,

$$\frac{1}{Y} \binom{Y}{\ell} = \frac{(Y-1) \cdots (Y-\ell+1)}{\ell!}$$

has  $p$ -integral coefficients, because  $\ell!$  is not divisible by  $p$ . It follows, as a polynomial congruence over the  $p$ -integral rationals, that

$$p \frac{T_{p-1}(Y)}{Y} \equiv 1 \pmod{p}. \quad (2.28)$$

For every positive integer  $Y$ , the left side is therefore nonzero and has  $p$ -adic valuation zero. Hence  $v_p(T_{p-1}(Y)) = v_p(Y) - 1$ .

Set  $Y = m - p + 1 > 0$  and use (2.22). This yields

$$v_p(b(m, m - p + 1)) = v_p(m^{\underline{p}}) - 1.$$

Among the  $p$  consecutive positive integers  $m - p + 1, \dots, m$ , exactly one is divisible by  $p$ : it is  $p \lfloor m/p \rfloor$ . Its valuation is  $1 + v_p(\lfloor m/p \rfloor)$ , proving (2.27).  $\square$

This proof explains, for example, why the residual polynomial  $3m - 13$  in (2.25) is a nonzero constant modulo 3, and why analogous residual polynomials for offsets 4, 6, 10, 12, 16 are nonzero constants modulo 5, 7, 11, 13, 17, respectively.

### 2.5.3 Finite certificates that cover infinitely many row indices

For a fixed  $d \geq 1$ , put

$$S_d(X) = T_d(X - d).$$

The factor  $X - d$  is forced by Lemma 2.11. When  $d \geq 3$  is odd, the symmetry zero also forces  $X - (2d - 1)$ ; for  $d = 3$  there is in addition the factor  $X - 8$ . Remove these factors from  $S_d$ , clear rational denominators, divide out the greatest common divisor of the integer coefficients, and normalize the leading coefficient to be positive. Call the resulting polynomial  $R_d(X)$ . Thus, with a positive rational constant  $c_d$ ,

$$S_d(X) = c_d(X - d) \begin{cases} (X - 5)(X - 8)R_d(X), & d = 3, \\ (X - (2d - 1))R_d(X), & d \geq 5 \text{ odd}, \\ R_d(X), & \text{otherwise.} \end{cases} \quad (2.29)$$

**Lemma 2.13** (Modular certificate). *If  $R \in \mathbb{Z}[X]$  satisfies  $R(t) \not\equiv 0 \pmod{p}$  for every  $t \in \{0, 1, \dots, p - 1\}$ , then  $R$  has no integer root.*

*Proof.* An integer root, reduced modulo  $p$ , would be one of those residues and would give a zero value modulo  $p$ .  $\square$

**Theorem 2.14** (Computer-assisted diagonal classification). *Conjecture 2.9 holds for every pair  $1 \leq r \leq m$  with  $m - r \leq 16$ .*

*Proof.* For offset zero,  $b(m, m) = 1$ . For  $1 \leq d \leq 16$ , compute  $S_d$  exactly from (2.23) and factor it as in (2.29). The supplied `data/diagonal_certificates.json` records the full rational coefficients of  $S_d$ , the scalar  $c_d$ , all removed linear factors, the full integer coefficients of  $R_d$ , and every value of  $R_d(t)$  modulo the certifying prime. All of those values are nonzero. The following summary identifies the certificates:

| $d$ | $\deg R_d$ | $p$ | roots mod $p$ | $d$ | $\deg R_d$ | $p$ | roots mod $p$ |
|-----|------------|-----|---------------|-----|------------|-----|---------------|
| 1   | 0          | 2   | 0             | 9   | 7          | 7   | 0             |
| 2   | 1          | 3   | 0             | 10  | 9          | 11  | 0             |
| 3   | 0          | 2   | 0             | 11  | 9          | 13  | 0             |
| 4   | 3          | 5   | 0             | 12  | 11         | 13  | 0             |
| 5   | 3          | 3   | 0             | 13  | 11         | 11  | 0             |
| 6   | 5          | 7   | 0             | 14  | 13         | 23  | 0             |
| 7   | 5          | 5   | 0             | 15  | 13         | 19  | 0             |
| 8   | 7          | 11  | 0             | 16  | 15         | 17  | 0             |

The standard-library program `code/verify_diagonals.py` reconstructs every polynomial using rational arithmetic and checks the factorization and each residue. The modular lemma excludes every integer root of every  $R_d$ . For  $m > d$ , the removed factor  $m - d$  cannot vanish; the remaining removed factors give exactly the stated symmetry and exceptional zeros.  $\square$

These certificates are not extrapolations from finitely many values of  $m$ : each modular argument excludes integer roots for *all*  $m$  on its fixed diagonal. What is finite is the set of offsets treated. For a small example not requiring the machine-readable file,

$$\begin{aligned} R_4(X) &= 15X^3 - 390X^2 + 3245X - 8638 \equiv 2 \pmod{5}, \\ R_5(X) &= 3X^3 - 103X^2 + 1118X - 3776 \equiv 2X^2 + 2X + 1 \pmod{3}. \end{aligned}$$

The second reduced polynomial takes the values 1, 2, 1 at the three residues modulo 3. Hence there are no additional zeros on offset 4, and the only zero on offset 5 is the symmetry zero  $m = 9$ .

## 2.6 The first three columns for all row indices

The diagonal results keep  $m - r$  fixed. A complementary calculation keeps  $r$  fixed and proves nonvanishing in a different infinite region.

**Proposition 2.15.** *Column  $r = 1$  has no zero. Column  $r = 2$  has exactly the zero  $m = 5$ . Column  $r = 3$  has no zero.*

*Proof.* For  $r = 1$ , (2.14) gives

$$b(1, 1) = 1, \quad b(m, 1) = (-1)^m(m-2)! \quad (m \geq 2).$$

For  $r = 2$ , put  $t = m - 3 \geq 0$ . The shifted product is

$$F_m(2+u) = u(u+1)(u+2) \prod_{j=1}^t (u-j).$$

The coefficient of  $u^2$  is

$$b(m, 2) = (-1)^m(m-3)!(2H_{m-3} - 3), \quad m \geq 3, \quad (2.30)$$

where  $H_t = \sum_{j=1}^t 1/j$  and  $H_0 = 0$ . The harmonic numbers strictly increase, and  $H_2 = 3/2$ , so the only zero is  $m - 3 = 2$ . The remaining boundary entry is  $b(2, 2) = 1$ .

For  $r = 3$ , the boundary entry is  $b(3, 3) = 1$ . Put  $t = m - 4 \geq 0$  and define

$$H_t^{(2)} = \sum_{j=1}^t \frac{1}{j^2}, \quad g(t) = 6 - 11H_t + 3(H_t^2 - H_t^{(2)}).$$

Extracting the coefficient of  $u^3$  from  $u(u+1)(u+2)(u+3) \prod_{j=1}^t (u-j)$  gives

$$b(m, 3) = (-1)^t t! g(t). \quad (2.31)$$

The elementary identity

$$g(t+1) - g(t) = \frac{6H_t - 11}{t+1} \quad (2.32)$$

controls its entire sign pattern. We have  $g(0) = 6$ ,  $g(1) = -5$ , and  $H_3 = 11/6$ . Thus  $g$  decreases through  $t = 3$ , has  $g(3) = g(4) = -49/6$ , and is strictly increasing thereafter. Finally, exact rational evaluation gives

$$g(19) = -\frac{4571462267}{97772875200} < 0, \quad g(20) = \frac{537826687}{1150269120} > 0.$$

It follows that  $g(t) < 0$  for  $1 \leq t \leq 19$ , whereas  $g(t) > 0$  for  $t \geq 20$ . There is no integer zero.  $\square$

The last argument illustrates a distinction from a numerical search: exact values at two adjacent integers, together with a proved monotonicity statement outside a finite initial segment, settle the entire column.

## 2.7 Unconditional quadratic growth

**Theorem 2.16.** *For every integer  $n \geq 0$ ,*

$$n + 1 + \left\lfloor \frac{n^2}{4} \right\rfloor \leq a(n) \leq q(n). \quad (2.33)$$

*In particular,*

$$\frac{1}{4} \leq \liminf_{n \rightarrow \infty} \frac{a(n)}{n^2} \leq \limsup_{n \rightarrow \infty} \frac{a(n)}{n^2} \leq \frac{1}{2}, \quad a(n) = \Theta(n^2).$$

*Proof.* The upper bound is Theorem 2.10. For the lower bound, count nonzero entries along diagonals of the triangle in Theorem 2.7. The main diagonal, including  $(0, 0)$ , contributes  $n + 1$  entries.

Fix  $1 \leq d \leq n - 1$ . The diagonal  $m - r = d$  has  $n - d$  relevant positions,  $m = d + 1, \dots, n$ . By Lemma 2.11, its zero positions are roots of the nonzero polynomial  $T_d(m - d)/(m - d)$  of degree  $d - 1$ . There are therefore at most  $d - 1$  such positions, and at least

$$\max(0, n - d - (d - 1)) = \max(0, n - 2d + 1)$$

nonzero entries on the diagonal. Summing over all  $d \geq 1$  gives

$$a(n) \geq n + 1 + \sum_{d \geq 1} \max(0, n - 2d + 1) = n + 1 + \left\lfloor \frac{n^2}{4} \right\rfloor.$$

The last equality follows by summing the odd numbers  $1, 3, \dots, 2s - 1$  when  $n = 2s$ , and the even numbers  $2, 4, \dots, 2s$  when  $n = 2s + 1$ . The asymptotic bounds follow immediately.  $\square$

**Corollary 2.17** (A stronger explicit lower bound). *For every  $n \geq 0$ ,*

$$a(n) \geq n + 1 + \left\lfloor \frac{n^2}{4} \right\rfloor + \sum_{\substack{p \leq n \\ p \text{ prime}}} \min(p - 2, n - p + 1). \quad (2.34)$$

*Proof.* On offset  $d = p - 1$ , Theorem 2.12 guarantees all  $n - d = n - p + 1$  entries, rather than just  $\max(0, n - 2p + 3)$ . The improvement is  $\min(p - 2, n - p + 1)$ . These diagonals are disjoint, so their improvements add.  $\square$

The prime-enhanced bound is useful without invoking any theorem about the density of primes. It still does not close the factor-of-two gap in the leading quadratic coefficient. In terms of the exact defect,

$$a(n) \sim \frac{n^2}{2} \iff E(n) = o(n^2). \quad (2.35)$$

Thus the proposed asymptotic equivalent is potentially a weaker target than the complete zero classification: it would allow a sufficiently sparse set of additional zeros.

## 2.8 The rational function, quasipolynomial, and recurrence

### 2.8.1 Deriving the proposed generating function

There is one potential term at  $(0, 0)$  and  $m$  potential terms in row  $m$ . The cumulative potential count therefore has generating function

$$\sum_{n \geq 0} \left(1 + \frac{n(n+1)}{2}\right) z^n = \frac{1}{1-z} + \frac{z}{(1-z)^3}.$$

The symmetry zeros are born at rows 5, 9, 13, ..., and each remains absent from every later derivative. Their cumulative generating function is  $z^5/((1-z)(1-z^4))$ . The exceptional zero is born at row 8 and contributes  $z^8/(1-z)$ . Subtracting these two *proved* families gives

$$\begin{aligned} Q(z) &= \frac{1}{1-z} + \frac{z}{(1-z)^3} - \frac{z^5}{(1-z)(1-z^4)} - \frac{z^8}{1-z} \\ &= \frac{1 + z^2 + z^3 + z^6 - z^8 + z^9 + z^{12} - z^{13}}{(1-z)^2(1-z^4)}. \end{aligned} \quad (2.36)$$

This explains every correction term in the proposed rational function.

**Theorem 2.18** (Exact generating-function correction). *With  $A(z) = \sum_{n \geq 0} a(n)z^n$  and  $\mathcal{E}(z) = \sum_{m \geq 1} e_m z^m$ ,*

$$A(z) = Q(z) - \frac{\mathcal{E}(z)}{1-z}. \quad (2.37)$$

*Consequently the proposed rational function equals  $A(z)$  if and only if Conjecture 2.9 is true.*

*Proof.* Sum  $a(n) = q(n) - \sum_{m \leq n} e_m$  as a formal power series. Each  $e_m$  contributes  $e_m z^m / (1-z)$  to the cumulative sum.  $\square$

### 2.8.2 The real form of the complex expression

For  $n \geq 8$ , division into the four residue classes gives

$$q(n) = \frac{n^2}{2} + \frac{n}{4} + \begin{cases} 1, & n \equiv 0 \pmod{4}, \\ \frac{1}{4}, & n \equiv 1 \pmod{4}, \\ \frac{1}{2}, & n \equiv 2 \pmod{4}, \\ \frac{3}{4}, & n \equiv 3 \pmod{4}. \end{cases} \quad (2.38)$$

The same periodic correction is

$$\frac{5 + (-1)^n + (1-i)(-i)^n + (1+i)i^n}{8}, \quad i^2 = -1.$$

Substitution yields exactly the complex-valued-looking expression printed in OEIS, whose total value is real and integral. This is an algebraic identity for  $q$ , independent of the unresolved equality  $a = q$ .

For all  $n \geq 0$ , another convenient form is

$$q(n) = \frac{n(n+1)}{2} - \left\lfloor \frac{n-1}{4} \right\rfloor + \mathbf{1}_{\{1 \leq n \leq 7\}}.$$

Here  $\lfloor -1/4 \rfloor = -1$  correctly gives  $q(0) = 1$ . The first terms are

$$1, 2, 4, 7, 11, 15, 21, 28, 35, 43, 53, 64, 76, 88, 102, 117, 133, \dots$$

### 2.8.3 Why the recurrence starts at fourteen

Set

$$\begin{aligned} D(z) &= (1-z)^2(1-z^4) = 1 - 2z + z^2 - z^4 + 2z^5 - z^6, \\ N(z) &= 1 + z^2 + z^3 + z^6 - z^8 + z^9 + z^{12} - z^{13}. \end{aligned}$$

The identity  $D(z)Q(z) = N(z)$  implies, for  $n \geq 6$ ,

$$q(n) - 2q(n-1) + q(n-2) - q(n-4) + 2q(n-5) - q(n-6) = [z^n]N(z). \quad (2.39)$$

The right side vanishes for every  $n \geq 14$ , but at  $n = 13$  it equals  $-1$ . Thus the least starting index from which the homogeneous recurrence holds for *every* subsequent index of  $q$  is 14. For  $7 \leq n \leq 13$ , its only nonzero residuals are

| $n$      | 8  | 9 | 12 | 13 |
|----------|----|---|----|----|
| residual | -1 | 1 | 1  | -1 |

This proves the range correction asserted at the start of this part. If the original derivative sequence satisfies the corrected recurrence for all  $n \geq 14$ , its directly verified initial values  $a(0), \dots, a(13)$  force  $a = q$  by induction. Consequently the corrected recurrence for  $a$  is itself equivalent to the main conjecture, once those initial values are supplied; it is not an independent proved consequence for arbitrary orders.

The second forward differences  $q(n+2) - 2q(n+1) + q(n)$  begin

$$1, 1, 1, 0, 2, 1, 0, 1, \underbrace{2, 1, 1, 0, 2, 1, 1, 0, \dots}_{n \geq 8}$$

Again, this is proved for  $q$ ; transferring the entire pattern to  $a$  requires the missing nonvanishing statement.

## 2.9 Reproducible verification

### 2.9.1 What was actually checked

All reported checks use exact arithmetic. No floating-point threshold was used to decide that a coefficient was zero.

The C++ implementation with GMP computes the recurrence (2.16) through row 3000. It tests all 4,501,500 entries with  $1 \leq r \leq m \leq 3000$ . Exactly 750 are zero: 749 symmetry zeros and (8, 5). There are no other zeros in that range, and therefore

$$a(n) = q(n) \quad (0 \leq n \leq 3000), \quad a(3000) = 4,500,751. \quad (2.40)$$

These are finite computational results, not an induction proving all  $n$ .

A separate Python standard-library implementation computes the triangle through row 1500 and agrees with the C++ term counts at all 1501 indices  $0, \dots, 1500$ . Further independent checks include the complete polynomials from the differential recurrence (2.9) through order 60, all 496 shifted-product coefficients (2.14) with  $0 \leq r \leq m \leq 30$ , and 11,865 instances of the prime-adic valuation formula. All 54 values displayed directly in the OEIS entry agree with the exact computation. The separately linked OEIS table through 500 was not used as an independent data source for this report.

The sixteen diagonal certificates were reconstructed and checked with a third, rational-polynomial computation. In addition to showing no unexpected zeros in a finite triangular region, these certificates prove there are no unexpected zeros on those fixed diagonals for any row index.

### 2.9.2 Algorithm and complexity

Only two preceding rows are required for (2.16). A schematic implementation is

```
previousprevious = [1]
previous = [0, 1]
count = 2
for m = 2, ..., N:
    current[0] = 0
    for r = 1, ..., m:
        current[r] = (r-m+1)*previous[r]
            + previous[r-1] + (m-1)*previousprevious[r-1]
    count += number of nonzero current[r] for 1 <= r <= m
    rotate the three rows
```

Indices beyond the end of a row mean zero. The supplied implementations handle these boundaries explicitly.

The computation uses  $O(N^2)$  arithmetic operations and stores  $O(N)$  integers. These are not bit-complexity claims: the integers grow with  $N$ . From the product formula,

$$|b(m, r)| \leq \prod_{h=0}^{m-1} (1 + |r - h|) \leq (m+1)^m,$$

so a coefficient has  $O(m \log m)$  bits. The rolling-row implementation thus has an  $O(N^2 \log N)$ -bit storage bound. A crude schoolbook arithmetic bound for the running time is  $O(N^3 \log^2 N)$  bit operations.

### 2.9.3 Files and commands

The archive contains `article.tex`, the compiled `article.pdf`, a build script, both triangle verifiers, the rational certificate verifier, the complete count and zero-position tables through 3000, and the exact verification reports. From the extracted directory, run

```
python3 code/verify.py -max-n 1500 -data-dir data
python3 code/verify_diagonals.py -output-dir data -check-only
```

The C++ implementation requires a C++17 compiler and GMP development files:

```
c++ -O3 -std=c++17 code/verify_gmp.cpp -lgmpxx -lgmp -o verify_gmp
./verify_gmp 3000 data
```

The Python programs use only the standard library. The diagonal checker constructs the coefficients afresh rather than trusting the saved residue tables. Its proof obligations are elementary rational polynomial identities and finite modular evaluations; it does not require a general-purpose computer algebra system.

## 2.10 What remains to finish the requested proof

The derivation of the candidate rational function is complete, but the identification of that function with the derivative count is conditional. The exact outstanding statement is

$$[u^r] \prod_{h=0}^{m-1} (u + r - h) \neq 0$$

for all  $1 \leq r \leq m$  outside the explicitly proved zero set  $\mathcal{Z}_0$ . The preceding results settle this for  $r \leq 3$ , for  $m - r \leq 16$ , and whenever  $m - r + 1$  is prime, in addition to all rows through 3000 by exact computation. They do not cover every remaining pair.

The diagonal-polynomial formulation gives a particularly precise route to a complete proof. For every offset  $d$ , it would suffice to show that the residual polynomial  $R_d$  in (2.29) has no integer root  $m > d$ . Irreducibility would be stronger than necessary. A uniform family of modular nonvanishing certificates would also suffice, but the fact that such certificates work for  $d \leq 16$  does not establish that they work for all  $d$ .

For the asymptotic conjecture alone, an exclusion of every additional zero is unnecessary: by (2.35), it suffices to bound their cumulative number by  $o(n^2)$ . The present root-counting argument proves quadratic order but not this sharper estimate.

**Conclusion.** The printed recurrence with range  $n > 6$  is disproved, and its correct range for the proposed sequence is identified. The principal closed formula is supported by an exact structural reduction, several infinite nonvanishing theorems, and extensive reproducible verification. A complete all-order proof of that formula is *not* supplied: the missing converse in Conjecture 2.9 is left explicit rather than being replaced by an inference from numerical agreement.

### 3 $x^{x^2}$ : OEIS A290268

**Framework.** This part instantiates Theorem 1.2(b) — the clause normalized by  $x^{-n}$  — with the frame  $z = x^2$ ,  $L = \log x$ ; Proposition 3.3 below is that instance, derived here independently, and the  $-n$  in its recurrence is the normalizer’s contribution. Lemma 3.2 is the special case of Theorem 1.4 this part needs.

**Notation.** In this part  $z$  **denotes**  $x^2$ , not  $\log x$ ; compare Part 4, where  $z$  is  $\log x$ . See Section 1.5. Throughout this part  $a(n)$  denotes the term count for  $x^{x^2}$ .

**Later addition (29 September 2026).** Sections 3.14–3.26, at the end of this part, add a later manuscript that closes the missing nonvanishing step at logarithmic deficits  $m = 3, 4$  and improves the lower bound  $\Lambda(n)$  below. They use depth coordinates in which the roles of  $j$  and  $k$  differ from this part’s (Table 3). The text of the original report is unchanged; dated notes like this one point forward.

**Abstract of the original report.** Let  $a(n)$  be the number of nonzero monomials in the fully collected  $n$ th derivative of  $x^{x^2}$ . The conjecture in OEIS A290268 predicts

$$a(n) = U(n) := \frac{n^2}{2} + n + 1 - (n \bmod 2) \left( \frac{1}{2} + \left\lfloor \frac{n+1}{8} \right\rfloor \right).$$

*This article does not establish the equality for all  $n$ . It proves the unconditional bound  $a(n) \leq U(n)$  by identifying every cancellation needed to explain the proposed formula. It also proves positivity throughout the upper half of the coefficient triangle, classifies all zeros on the first two logarithmic-deficit diagonals, and gives explicit quadratic lower bounds. An exact generating-function identity and a finite-difference formula reduce the full conjecture to a precise nonvanishing assertion. Integer arithmetic verifies the support through  $n = 200$ ; two complete modular computations, combined with the proved zero identities, certify  $a(n) = U(n)$  for every  $0 \leq n \leq 3000$ . All source code, finite certificates, and reproduction instructions accompany the article. The remaining nonvanishing step is stated explicitly rather than assumed.*

#### 3.1 The problem and the status of the result

Put

$$f(x) = x^{x^2} = \exp(x^2 \log x), \quad x > 0.$$

The sequence A290268 counts the terms in the fully expanded derivative  $f^{(n)}(x)$ , after identical monomials have been collected. The OEIS entry was contributed by Vladimir Reshetnikov in October 2017; its formula section labels both the generating function and the explicit expression as conjectural. Peter Luschny’s equivalent floor-function form is explicitly conditional on that conjecture [2].

For clarity, throughout this article the letter  $U$  denotes the *proposed* answer, not an already proved formula for  $a$ :

$$U(n) = \frac{n^2}{2} + n + 1 - (n \bmod 2) \left( \frac{1}{2} + \left\lfloor \frac{n+1}{8} \right\rfloor \right). \quad (3.1)$$

Equivalently,

$$U(2r) = 2r^2 + 2r + 1, \quad U(2r+1) = 2(r+1)^2 - \left\lfloor \frac{r+1}{4} \right\rfloor. \quad (3.2)$$

The two central issues are different. One must explain why certain coefficients vanish, and one must also exclude every other cancellation. The first task is completed here. The second is completed in substantial regions and in a large finite range, but not in full generality.

**Theorem 3.1** (Unconditional bounds). *For every integer  $n \geq 0$ ,*

$$\Lambda(n) \leq a(n) \leq U(n), \quad \Lambda(n) = \begin{cases} (3n^2 + 10n + 8)/8, & n \text{ even}, \\ (3n^2 + 12n + 1)/8, & n \text{ odd}. \end{cases} \quad (3.3)$$

*In particular,  $a(n) = \Theta(n^2)$ . The proposed leading constant  $1/2$  is not established by these bounds.*

The upper bound is proved in Section 3.6; the lower bound is proved in Section 3.4. Section 3.9 gives a reproducible finite certification of the stronger equality through 3000. The exact outstanding assertion appears in Section 3.10.

*Update (29 September 2026).* The lower bound  $\Lambda(n)$  is improved by Corollary 3.45 to  $(3n^2 + 26n - 72)/8$  for even  $n \geq 8$  and  $(3n^2 + 28n - 111)/8$  for odd  $n \geq 17$  (Remark 3.46), a linear-term gain of  $2n - 10$ , respectively  $2n - 14$ ; the leading constant  $3/8$  is unchanged. Theorem 3.1 stands as stated.

### 3.2 A canonical polynomial expansion

Set  $z = x^2$  and  $L = \log x$ . Define polynomials  $P_n \in \mathbb{Z}[z, L]$  by

$$f^{(n)}(x) = f(x)x^{-n}P_n(x^2, \log x), \quad P_0 = 1. \quad (3.4)$$

Write

$$P_n(z, L) = \sum_{k,j} c_{n,k,j} z^k L^j. \quad (3.5)$$

Thus  $c_{n,k,j}$  is the coefficient of  $x^{x^2+2k-n}(\log x)^j$  in the derivative. Counting terms means counting nonzero  $c_{n,k,j}$ , rather than counting the summands generated by a particular symbolic differentiation algorithm.

**Lemma 3.2** (Uniqueness of the collected expansion). *A finite identity  $\sum_{k,j} b_{k,j} x^{2k} (\log x)^j = 0$  for all  $x > 0$  forces every  $b_{k,j}$  to vanish. Consequently, the expansion (3.4)–(3.5) is canonical.*

*Proof.* Substitute  $x = e^L$  and group the identity as  $\sum_k e^{2kL} B_k(L) = 0$ , with polynomial  $B_k$ . If  $K$  is the largest index with  $B_K \neq 0$ , division by  $e^{2KL}$  shows that  $B_K(L)$  tends to zero as  $L \rightarrow +\infty$ : all remaining terms are polynomials multiplied by decaying exponentials. A nonzero polynomial cannot tend to zero at  $+\infty$ . This contradiction eliminates  $K$ , and then all indices.  $\square$

The normalized derivative obeys a particularly small recurrence.

**Proposition 3.3** (Polynomial and coefficient recurrences). *For every  $n \geq 0$ ,*

$$P_{n+1} = (z(2L + 1) - n)P_n + 2z \frac{\partial P_n}{\partial z} + \frac{\partial P_n}{\partial L}. \quad (3.6)$$

*With nonexistent coefficients interpreted as zero,*

$$\boxed{c_{n+1,k,j} = (2k - n)c_{n,k,j} + (j + 1)c_{n,k,j+1} + c_{n,k-1,j} + 2c_{n,k-1,j-1}.} \quad (3.7)$$

*For  $n \geq 1$ , the only possible nonzero indices satisfy*

$$1 \leq k \leq n, \quad 0 \leq j \leq k. \quad (3.8)$$

*Proof.* We have  $f'/f = x(2L+1)$ ,  $dz/dx = 2z/x$ , and  $dL/dx = 1/x$ . Differentiate (3.4) and factor out  $f(x)x^{-n-1}$  to obtain (3.6). Taking the coefficient of  $z^k L^j$  gives (3.7). Starting from  $P_0 = 1$ , the first derivative is  $P_1 = z(2L+1)$ . The four operations in (3.6) preserve  $k \geq 1$  and  $j \leq k$ , and increase  $k$  by at most one. Induction proves (3.8) and integrality.  $\square$

For example,

$$\begin{aligned} P_1 &= z(2L+1), \\ P_2 &= z(2L+3) + z^2(2L+1)^2, \\ P_3 &= 2z + z^2(12L^2 + 24L + 9) + z^3(2L+1)^3. \end{aligned}$$

These have respectively 2, 5, and 8 nonzero coefficients, in agreement with the definition. Already  $P_3$  illustrates a degree loss: its  $z$ -coefficient has no  $L$ -term. Later cancellations are less obvious. For instance,

$$[z^2]P_7(z, L) = 28, \quad [z^2]P_9(z, L) = -864L. \quad (3.9)$$

The missing  $L$ -term in the first polynomial and the missing constant in the second will follow from the same reflection principle.

### 3.3 Exact formulas for individual coefficients

#### 3.3.1 An exponential generating function

Define two formal power series

$$A(t) = t(2+t) = (1+t)^2 - 1, \quad B(t) = (1+t)^2 \log(1+t). \quad (3.10)$$

Formal coefficient extraction is denoted by  $[t^n]$ . All logarithms in formal series are expanded at the origin.

**Proposition 3.4** (Generating function). *The complete coefficient array is described by*

$$\sum_{n \geq 0} P_n(z, L) \frac{t^n}{n!} = \exp(z(A(t)L + B(t))). \quad (3.11)$$

In particular, if  $m = k - j \geq 0$ , then

$$c_{n,k,j} = \frac{n!}{j!m!} [t^n] A(t)^j B(t)^m. \quad (3.12)$$

*Proof.* Taylor expansion at the point  $x$  gives

$$\frac{f(x(1+t))}{f(x)} = \sum_{n \geq 0} \frac{x^n f^{(n)}(x)}{f(x)} \frac{t^n}{n!}.$$

The logarithm of the ratio on the left is

$$x^2(((1+t)^2 - 1) \log x + (1+t)^2 \log(1+t)).$$

This proves (3.11) after substitution of  $z, L$ . Alternatively, its coefficients satisfy (3.6), so the identity also holds with  $z, L$  algebraically independent. Expanding separately the two exponentials  $\exp(zA(t)L)$  and  $\exp(zB(t))$  gives (3.12).  $\square$

Because both  $A$  and  $B$  start with a nonzero term of degree one, their product in (3.12) starts at degree  $k$ . This independently explains  $k \leq n$ . The difficulty is that  $B$  does not have only positive coefficients:

$$B(t) = t + \frac{3}{2}t^2 + \frac{1}{3}t^3 - \frac{1}{12}t^4 + \frac{1}{30}t^5 - \dots$$

Consequently, the coefficient formula is not itself a proof of nonvanishing.

### 3.3.2 Falling factorials and finite differences

Let

$$F_n(u) = u^{\underline{n}} = u(u-1)\cdots(u-n+1), \quad F_0(u) = 1,$$

and let  $\Delta_2 G(u) = G(u+2) - G(u)$ . Set  $Q_{n,j}(u) = \Delta_2^j F_n(u)$ .

**Proposition 3.5** (Finite-difference formula). *For  $m = k - j \geq 0$ ,*

$$c_{n,k,j} = \frac{1}{j!m!} \left. \frac{d^m}{du^m} Q_{n,j}(u) \right|_{u=2m} \quad (3.13)$$

$$= \frac{1}{j!} [v^m] \sum_{h=0}^j (-1)^{j-h} \binom{j}{h} F_n(2m + 2h + v). \quad (3.14)$$

*Proof.* The generalized binomial theorem gives  $n![t^n](1+t)^u = F_n(u)$ . Differentiating  $m$  times in  $u$  introduces  $\log^m(1+t)$ . Taking  $j$  forward differences of step two introduces  $((1+t)^2 - 1)^j = A(t)^j$ . Evaluation at  $u = 2m$  introduces  $(1+t)^{2m}$ , which combines with the logarithm to give  $B(t)^m$ . Equation (3.12) now proves (3.13); Taylor's formula proves (3.14).  $\square$

Derivatives of falling factorials also occur in the theory of shifted Stirling numbers and elementary symmetric harmonic sums; Poon gives a broader account of these connections [11]. No general nonvanishing theorem from that literature is being assumed here. Formula (3.14) is also an independent way to compute a coefficient using integer polynomial products.

## 3.4 A transformation proving positivity

A fractional-linear change of variable turns a large region of the coefficient array into a manifestly positive expression. Define

$$H(u) = \frac{\operatorname{atanh} u}{u} = \sum_{r \geq 0} \frac{u^{2r}}{2r+1}, \quad H(0) = 1. \quad (3.15)$$

The value at zero is understood by its formal power series.

**Proposition 3.6** (Transformed coefficient formula). *For  $m = k - j \geq 0$  and  $n \geq k$ ,*

$$c_{n,k,j} = \frac{n!}{j!m!} 2^{2j+m-n} [u^{n-k}] (1-u)^{n-2k-1} (1+u)^{2m} H(u)^m. \quad (3.16)$$

*Negative integer powers of  $1-u$  have their usual formal expansions.*

*Proof.* In the residue expression for  $[t^n] A(t)^j B(t)^m$ , substitute  $t = 2u/(1-u)$ . Then

$$1+t = \frac{1+u}{1-u}, \quad A(t) = \frac{4u}{(1-u)^2}, \quad B(t) = \frac{2(1+u)^2}{(1-u)^2} \operatorname{atanh} u, \quad dt = \frac{2 du}{(1-u)^2}.$$

Using  $[t^n]G(t) = \text{Res}_{t=0} G(t)t^{-n-1} dt$ , the constant factor becomes  $2^{2j+m-n}$ . The remaining power of  $1-u$  is  $n-2k-1$ . The factors  $u^j(\text{atanh } u)^m$  contribute  $u^k H(u)^m$ , so the resulting residue extracts the coefficient of  $u^{n-k}$ . Multiplying by  $n!/(j!m!)$  proves the formula. The formal change-of-variable rule applies because  $2u/(1-u)$  has zero constant term and nonzero linear coefficient.  $\square$

**Theorem 3.7** (A nonvanishing region). *For  $1 \leq k \leq n$  and  $0 \leq j \leq k$ ,*

$$\begin{aligned} n \leq 2k &\implies c_{n,k,j} > 0, \\ n = 2k+1, j < k &\implies c_{n,k,j} > 0. \end{aligned} \tag{3.17}$$

*Proof.* If  $n \leq 2k$ , the factor  $(1-u)^{n-2k-1}$  is  $(1-u)^{-q}$  with  $q \geq 1$ . Every coefficient of this factor is strictly positive. The other two factors in (3.16) have nonnegative coefficients and constant term one. Their product therefore has a strictly positive coefficient in every nonnegative degree.

If  $n = 2k+1$ , that factor disappears. The condition  $j < k$  means  $m \geq 1$ . The series  $H(u)^m$  has a strictly positive coefficient in every even degree. The polynomial  $(1+u)^{2m}$  has positive constant and linear coefficients. An even target degree can be supplied entirely by  $H(u)^m$ ; an odd target degree can use the linear term of  $(1+u)^{2m}$  and an even degree from  $H(u)^m$ . Thus the target coefficient of degree  $n-k$  is strictly positive in both cases. The prefactor in (3.16) is positive.  $\square$

**Corollary 3.8** (Quadratic lower bounds). *The lower bound in Theorem 3.1 holds.*

*Proof.* For  $n = 2r \geq 2$ , Theorem 3.7 supplies all coefficients with  $r \leq k \leq 2r$ , giving

$$a(2r) \geq \sum_{k=r}^{2r} (k+1) = \frac{(r+1)(3r+2)}{2}.$$

For  $n = 2r+1$  with  $r \geq 1$ , it supplies all coefficients with  $r+1 \leq k \leq 2r+1$  and the  $r$  coefficients  $0 \leq j < r$  at  $k = r$ . Hence

$$a(2r+1) \geq r + \sum_{k=r+1}^{2r+1} (k+1) = \frac{3(r+1)(r+2)}{2} - 1.$$

These are precisely (3.3). The cases  $n = 0, 1$  follow directly from  $P_0 = 1$  and  $P_1 = z(2L+1)$ .  $\square$

## 3.5 The two forced families of zeros

### 3.5.1 Degree-forced zeros on the top diagonal

Setting  $j = k$ , and therefore  $m = 0$ , in (3.12) yields an explicit coefficient.

**Proposition 3.9** (Top logarithmic degree). *For  $k \geq 1$  and  $k \leq n \leq 2k$ ,*

$$c_{n,k,k} = \frac{n!}{k!} \binom{k}{n-k} 2^{2k-n} > 0. \tag{3.18}$$

*For  $n > 2k$ , one has  $c_{n,k,k} = 0$ .*

*Proof.* The relevant series is the polynomial  $A(t)^k = t^k(2+t)^k$ , of degree  $2k$ . Its coefficient of  $t^n$  is  $\binom{k}{n-k} 2^{2k-n}$  in the indicated range and zero above it.  $\square$

For a fixed  $n \geq 1$ , this removes exactly  $\lfloor (n-1)/2 \rfloor$  cells from the possible triangle.

### 3.5.2 Reflection-forced cancellations

The less obvious zeros follow from symmetry about the midpoint of the roots of a falling factorial.

**Lemma 3.10** (Reflection of a finite difference). *For all  $n, j \geq 0$ ,*

$$Q_{n,j}(n - 2j - 1 - u) = (-1)^{n-j} Q_{n,j}(u). \quad (3.19)$$

*Proof.* First,  $F_n(n - 1 - u) = (-1)^n F_n(u)$ , by reversing the order of the factors. Expand the left side of (3.19):

$$\sum_{h=0}^j (-1)^{j-h} \binom{j}{h} F_n(n - 2j - 1 - u + 2h).$$

Reflection turns its last factor into  $(-1)^n F_n(u + 2(j - h))$ . Reindexing by  $\ell = j - h$  leaves the factor  $(-1)^{n-j}$  times the usual expansion of  $\Delta_2^j F_n(u)$ .  $\square$

**Theorem 3.11** (Reflection zeros). *Suppose  $1 \leq k \leq n$  and  $0 \leq j < k$ . Then*

$$k \text{ even}, \quad n = 4k - 2j + 1 \quad \implies \quad c_{n,k,j} = 0. \quad (3.20)$$

*Proof.* Put  $m = k - j$ . The reflection center in Lemma 3.10 is  $(n - 2j - 1)/2$ . Under the displayed condition this equals  $2m$ , precisely the evaluation point in (3.13). The parity of  $Q_{n,j}(2m + v)$  as a polynomial in  $v$  is  $n - j$ . Its coefficient of  $v^m$  therefore vanishes whenever  $n - j - m = n - k$  is odd. The relation  $n = 4k - 2j + 1$  makes  $n$  odd, so  $n - k$  is odd when  $k$  is even. Equation (3.14) completes the proof.  $\square$

There is also a direct proof using (3.16). The relation in (3.20) gives  $n - 2k - 1 = 2m$ , so the series whose coefficient is extracted becomes

$$(1 - u^2)^{2m} H(u)^m. \quad (3.21)$$

This series is even. When  $k$  is even,  $n - k$  is odd, and its coefficient of that degree is zero. This second proof makes the cancellation visible without evaluating any large integer coefficient.

*Remark 3.12* (What symmetry does not say). The theorem is an implication, not an equivalence. An even power series has zero odd coefficients, but it can also have zero even coefficients. Similarly, away from the reflection center, a derivative of a polynomial can vanish for reasons unrelated to symmetry. Excluding such additional zeros is a separate requirement of the full conjecture.

## 3.6 Counting the forced zeros

For  $n \geq 1$ , the triangle (3.8) has

$$T(n) = \sum_{k=1}^n (k+1) = \frac{n(n+3)}{2} \quad (3.22)$$

cells. Let  $Z_n^{\text{top}}$  be the top-diagonal cells removed by Proposition 3.9, and let  $Z_n^{\text{ref}}$  be the cells in Theorem 3.11. These sets are disjoint, since the second requires  $j < k$ .

**Lemma 3.13** (Number of reflection zeros). *For even  $n$ ,  $Z_n^{\text{ref}}$  is empty. For odd  $n$ ,*

$$|Z_n^{\text{ref}}| = \left\lfloor \frac{n+1}{8} \right\rfloor. \quad (3.23)$$

*Proof.* Reflection zeros have odd  $n$ . Write  $n = 2r + 1$  and  $k = 2\ell$ . The reflection relation determines  $j = 2k - r = 4\ell - r$ . The inequalities  $0 \leq j < k$  become

$$\frac{r}{4} \leq \ell < \frac{r}{2}.$$

For  $r \geq 1$ , these automatically give  $\ell \geq 1$  whenever the interval contains an integer. Its number of integers is

$$\left\lceil \frac{r}{2} \right\rceil - \left\lceil \frac{r}{4} \right\rceil = \left\lfloor \frac{r+1}{4} \right\rfloor,$$

as is checked in the four residue classes of  $r$  modulo four. For  $r = 0$  both sides are zero. Since  $(r+1)/4 = (n+1)/8$ , the result follows.  $\square$

**Theorem 3.14** (The conjectured expression is an upper bound). *For every  $n \geq 0$ ,*

$$a(n) \leq U(n). \quad (3.24)$$

*Equality holds for a given  $n$  if and only if the coefficient triangle has no zeros other than  $Z_n^{\text{top}} \cup Z_n^{\text{ref}}$ .*

*Proof.* For even  $n = 2r \geq 2$ , subtraction of the top-diagonal zeros from (3.22) gives

$$2r^2 + 3r - (r - 1) = 2r^2 + 2r + 1.$$

For odd  $n = 2r + 1$ , subtraction of both disjoint families gives

$$2r^2 + 5r + 2 - r - \left\lfloor \frac{r+1}{4} \right\rfloor = 2(r+1)^2 - \left\lfloor \frac{r+1}{4} \right\rfloor.$$

These are (3.2). Additional zeros could only decrease the number of nonzero coefficients, proving the inequality and the stated equivalence. Finally  $a(0) = U(0) = 1$ .  $\square$

For example, at  $n = 7$  there are 35 potential cells, three top-diagonal zeros, and the reflection zero  $(k, j) = (2, 1)$ , leaving 31. At  $n = 9$ , there are 54 potential cells, four top-diagonal zeros, and the reflection zero  $(2, 0)$ , leaving 49. The exact expansions in (3.9) display those two different cancellations. The floor term in the proposed formula is thus explained by a proved symmetry, not merely by fitting initial values.

### 3.7 Nonvanishing on the first two deficit diagonals

Call  $m = k - j$  the *logarithmic deficit*. This section gives a complete zero classification for  $m = 1$  and  $m = 2$ . Together with Theorem 3.7, it narrows the unproved portion of the problem.

#### 3.7.1 A beta-integral representation

Suppose  $n > 2k$ ,  $m = k - j \geq 1$ , and put

$$d = n - 2k - 1 \geq 0. \quad (3.25)$$

For sufficiently small complex  $\varepsilon$ , define

$$I_{m,j,d}(\varepsilon) = \int_0^1 t^{2m+\varepsilon} (1-t)^{d-\varepsilon} (2t-1)^j dt. \quad (3.26)$$

The integral and all derivatives required below converge uniformly on compact subsets of  $|\varepsilon| < 1/2$ .

**Lemma 3.15** (Integral formula). *Under these assumptions,*

$$Q_{n,j}(2m + \varepsilon) = (-1)^{n-1} n! \frac{\sin(\pi\varepsilon)}{\pi} I_{m,j,d}(\varepsilon), \quad (3.27)$$

and consequently

$$c_{n,k,j} = \frac{(-1)^{n-1} n!}{j!} [\varepsilon^m] \left( \frac{\sin(\pi\varepsilon)}{\pi} I_{m,j,d}(\varepsilon) \right). \quad (3.28)$$

*Proof.* Euler's reflection formula and beta integral [13, 14] give, initially away from integer  $\alpha$ ,

$$\begin{aligned} F_n(\alpha) &= (-1)^{n-1} \frac{\sin(\pi\alpha)}{\pi} \Gamma(\alpha + 1) \Gamma(n - \alpha) \\ &= (-1)^{n-1} n! \frac{\sin(\pi\alpha)}{\pi} \int_0^1 t^\alpha (1-t)^{n-\alpha-1} dt. \end{aligned}$$

In the  $j$ th difference put  $\alpha = 2m + 2h + \varepsilon$ . The sine factor is independent of the integer  $h$ . The finite sum inside the integral is

$$\sum_{h=0}^j (-1)^{j-h} \binom{j}{h} \left( \frac{t}{1-t} \right)^{2h} = \frac{(2t-1)^j}{(1-t)^{2j}}.$$

The remaining exponent of  $1-t$  is  $d - \varepsilon$ , proving (3.27). All terms are analytic near  $\varepsilon = 0$ , so the identity extends to zero as well. Taking the coefficient of  $\varepsilon^m$  and using (3.14) gives (3.28).  $\square$

### 3.7.2 Deficit one

**Theorem 3.16** (Complete signs for  $m = 1$ ). *Suppose  $m = k - j = 1$  and  $n > 2k$ , and define  $d$  by (3.25). Then*

$$\operatorname{sgn} c_{n,k,j} = (-1)^{n-1} \begin{cases} 1, & j \text{ even,} \\ \operatorname{sgn}(2-d), & j \text{ odd.} \end{cases} \quad (3.29)$$

*In particular the only zeros in this range have  $j$  odd and  $d = 2$ .*

*Proof.* The coefficient of  $\varepsilon$  in (3.28) is

$$c_{n,k,j} = \frac{(-1)^{n-1} n!}{j!} \int_0^1 t^2 (1-t)^d (2t-1)^j dt. \quad (3.30)$$

For even  $j$ , the integrand is nonnegative and positive except at finitely many points, so the integral is strictly positive. For odd  $j$ , pair  $t$  with  $1-t$  and integrate over  $1/2 < t < 1$ . The paired integrand is

$$(2t-1)^j (t^2(1-t)^d - t^d(1-t)^2).$$

The ratio of the two positive terms in parentheses is  $(t/(1-t))^{2-d}$ . Since  $t/(1-t) > 1$ , their difference is identically zero when  $d = 2$  and otherwise has the strict sign of  $2-d$  throughout the open interval. This proves the sign formula.  $\square$

These zeros are exactly the previously identified reflection zeros:  $j$  odd means  $k = j + 1$  is even, and  $d = 2$  means  $n = 2k + 3 = 4k - 2j + 1$ .

### 3.7.3 Deficit two

**Theorem 3.17** (Complete signs for  $m = 2$ ). *Suppose  $m = k - j = 2$  and  $n > 2k$ . Then*

$$\operatorname{sgn} c_{n,k,j} = (-1)^{n-1} \begin{cases} 1, & j \text{ odd}, \\ \operatorname{sgn}(4-d), & j \text{ even}. \end{cases} \quad (3.31)$$

*The only zeros in this range have  $j$  even and  $d = 4$ .*

*Proof.* The coefficient of  $\varepsilon^2$  in (3.28) is

$$c_{n,k,j} = \frac{(-1)^{n-1} n!}{j!} \int_0^1 t^4 (1-t)^d (2t-1)^j \log \frac{t}{1-t} dt. \quad (3.32)$$

For odd  $j$ , the factors  $(2t-1)^j$  and  $\log(t/(1-t))$  have the same sign, so the integral is strictly positive. For even  $j$ , their product is antisymmetric under  $t \mapsto 1-t$ . Pairing the integral as before leaves, on  $1/2 < t < 1$ ,

$$(2t-1)^j \log \frac{t}{1-t} (t^4(1-t)^d - t^d(1-t)^4).$$

The first two factors are positive there. The bracket has the strict sign of  $4-d$ , or is identically zero when  $d = 4$ . This proves the claim.  $\square$

Again the zero condition is exactly reflection:  $k = j + 2$  is even and  $n = 2k + 5 = 4k - 2j + 1$ . For  $n \leq 2k$ , positivity has already been proved, and the borderline  $n = 2k + 1$  is included in both sign arguments. Thus these two theorems, together with Proposition 3.9, classify all zeros for  $k - j \leq 2$  throughout the possible triangle.

### 3.7.4 Why the same sign proof does not settle higher deficits

At  $m = 3$ , formula (3.28) gives the integral of

$$t^6(1-t)^d(2t-1)^j \left( \frac{1}{2} \log^2 \frac{t}{1-t} - \frac{\pi^2}{6} \right) \quad (3.33)$$

multiplied by  $(-1)^{n-1} n! / j!$ . The last factor changes sign inside  $(0, 1)$ . Thus positivity of the weight, or pairing  $t$  with  $1-t$ , no longer excludes cancellation. This is a concrete obstruction to simply extending the proofs above. It is not a counterexample to the conjecture.

*Update (29 September 2026).* At  $m = 3$  and  $m = 4$  the obstruction has since been overcome by a different route: the kernel is written for every deficit as  $R_m/m!$  (Remark 3.31), its sign change is controlled by monotone likelihood-ratio comparisons, and a finite certificate finishes the proof (Theorem 3.23). The obstruction as stated remains for  $m \geq 5$ .

## 3.8 Generating functions and equivalent formulas for the upper bound

All identities in this section are proved identities for  $U(n)$ . They become identities for  $a(n)$  only after the remaining nonvanishing assertion is supplied, or when restricted to a certified finite range.

### 3.8.1 Derivation of the rational generating function

By (3.2), the even and odd subsequences have generating functions

$$E(y) = \sum_{r \geq 0} U(2r)y^r = \frac{(1+y)^2}{(1-y)^3}, \quad (3.34)$$

$$O(y) = \sum_{r \geq 0} U(2r+1)y^r = \frac{2(1+y)}{(1-y)^3} - \frac{y^3}{(1-y)(1-y^4)}. \quad (3.35)$$

For the last term, write  $\lfloor (r+1)/4 \rfloor = \sum_{h \geq 1} \mathbf{1}_{r \geq 4h-1}$  and sum each geometric tail. Therefore

$$\begin{aligned} G(t) &:= \sum_{n \geq 0} U(n)t^n = E(t^2) + tO(t^2) \\ &= \frac{1+t^2}{(1-t)^3(1+t)} - \frac{t^7}{(1-t^2)(1-t^8)} \\ &= \frac{N(t)}{(1-t)(1-t^2)(1-t^8)}, \end{aligned} \quad (3.36)$$

where

$$N(t) = 1 + t + 2t^2 + 2t^3 + 2t^4 + 2t^5 + 2t^6 + t^7 + 2t^8 + t^9. \quad (3.37)$$

Indeed,  $N(t) = (1+t^2)(1+t+\dots+t^7) - t^7(1-t)$ . This is exactly the rational function proposed in the OEIS entry [2]. The derivation shows why period eight appears: it counts the even values of  $k$  on the reflection line, in addition to the parity split of the ordinary triangle.

### 3.8.2 The eight constituent polynomials

Writing  $n = 8q + s$  with  $0 \leq s \leq 7$ , (3.1) becomes the following degree-two quasipolynomial.

| $s$ | $U(8q+s)$          |
|-----|--------------------|
| 0   | $32q^2 + 8q + 1$   |
| 1   | $32q^2 + 15q + 2$  |
| 2   | $32q^2 + 24q + 5$  |
| 3   | $32q^2 + 31q + 8$  |
| 4   | $32q^2 + 40q + 13$ |
| 5   | $32q^2 + 47q + 18$ |
| 6   | $32q^2 + 56q + 25$ |
| 7   | $32q^2 + 63q + 31$ |

The trigonometric expression from the OEIS entry is another way to write the same eight polynomials:

$$U(n) = \frac{8n^2 + 15n + 14 + (n+2)(-1)^n + (2 - 4\sqrt{2}\sin(\pi n/4))\sin(\pi n/2)}{16}. \quad (3.38)$$

To verify the equivalence without an approximation, substitute  $n = 8q + s$ . The sine factors take their exact values at multiples of  $\pi/4$  and  $\pi/2$ . Substitution in each of the eight residue classes yields the polynomials in the table. In particular, no irrational value remains.

### 3.8.3 Linear recurrences

Multiplication of (3.36) by its denominator yields

$$\begin{aligned} U(n) = & U(n-1) + U(n-2) - U(n-3) + U(n-8) \\ & - U(n-9) - U(n-10) + U(n-11), \quad n \geq 11. \end{aligned} \quad (3.39)$$

Multiplying the denominator by  $1+t$  gives the convenient order-twelve form recorded in the OEIS entry:

$$U(n) = 2U(n-2) - U(n-4) + U(n-8) - 2U(n-10) + U(n-12), \quad n \geq 12. \quad (3.40)$$

The twelve initial values are

$$1, 2, 5, 8, 13, 18, 25, 31, 41, 49, 61, 71.$$

Using these recurrences to *generate* the claimed values of  $a(n)$  would be circular in a proof of the conjecture. The computations in the next section instead use the independently derived coefficient recurrence (3.7).

## 3.9 Exact finite verification and certificates

### 3.9.1 Integer arithmetic and independent formula checks

The accompanying Python program applies (3.7) with arbitrary-precision integers. Through  $n = 200$ , it checks not only the term count but the entire zero pattern: a coefficient is zero if and only if it belongs to one of the two proved zero families. This examines 1,373,500 cells with  $1 \leq k \leq n \leq 200$ ,  $0 \leq j \leq k$ .

For  $0 \leq n \leq 16$ , the program independently recomputes 969 triples  $(n, k, j)$ , including the  $k = 0$  boundary, using each of (3.12), (3.14), and (3.16). Fractions are exact rational numbers. All three formulas agree with the integer recurrence. The positivity and deficit-one and -deficit-two sign theorems are also checked throughout the integer range. These checks test the derivations and their implementation; the proofs in the preceding sections do not depend on them.

### 3.9.2 Why modular arithmetic gives a rigorous finite certificate

A nonzero residue modulo any integer  $M \geq 2$  proves that the original integer is nonzero. A zero residue, in contrast, might mean only that  $M$  divides a nonzero integer. We use two moduli,

$$M_1 = 1,000,000,007, \quad M_2 = 1,000,000,009. \quad (3.41)$$

Their primality is not needed for this argument: only exact reduction of an integer recurrence is used, with no modular division.

For each modulus, a C++ program computes every row through  $n = 3000$  and records every zero residue outside the two mathematically proved zero families. There are three such residues modulo  $M_1$  and four modulo  $M_2$ . The two lists are disjoint. Their union and the corresponding residues are as follows.

| $n$  | $k$  | $j$  | $c_{n,k,j} \bmod M_1$ | $c_{n,k,j} \bmod M_2$ |
|------|------|------|-----------------------|-----------------------|
| 1471 | 1046 | 521  | 847578491             | 0                     |
| 1701 | 610  | 505  | 0                     | 439228697             |
| 1854 | 1713 | 549  | 0                     | 366891231             |
| 2336 | 1689 | 145  | 0                     | 782191592             |
| 2543 | 1785 | 663  | 405694396             | 0                     |
| 2807 | 2566 | 502  | 607368440             | 0                     |
| 2842 | 2673 | 2118 | 474540977             | 0                     |

Every entry in this table is therefore a nonzero integer coefficient. All other cells outside the proved zero families have a nonzero residue in both runs. Each modulus run checks 4,509,002,501 potential cells, including the single cell at  $n = 0$ .

**Proposition 3.18** (Finite computational certification). *The accompanying exact computations, together with Proposition 3.9 and Theorem 3.11, certify*

$$a(n) = U(n) \quad \text{for every integer } 0 \leq n \leq 3000. \quad (3.42)$$

*Certificate argument.* The two analytic results prove that every predicted zero is an actual integer zero. For any other coefficient in the finite range, membership in neither, one, or both modular zero lists exhausts the possibilities. The intersection of the lists is empty. Thus at least one of its two residues is nonzero, proving that the integer coefficient is nonzero. Theorem 3.14 now gives the equality for every row in the range.  $\square$

This is a deterministic finite verification, not a probabilistic claim based on the size of the moduli. It is also not a formal proof-assistant certificate: it relies on the supplied ordinary programs and their execution, which can be independently reproduced.

### 3.9.3 Implementation details and limitations

The C++ program stores two coefficient rows in arrays of unsigned 32-bit integers. An update is evaluated in signed 64-bit arithmetic before reduction. Its explicit input bounds guarantee that these intermediates cannot overflow. It never approximates logarithms or transcendental values. The arrays have  $O(N^2)$  entries and the computation uses  $O(N^3)$  fixed-modulus arithmetic operations through order  $N$ .

The Python implementation uses a different update organization: it scatters each existing coefficient into its four successors, whereas the C++ implementation gathers four contributions into each destination. The small-range generating-function and finite-difference calculations provide further independent comparisons. The archive retains both raw modular zero lists, row-by-row support statistics, explicit nonzero witnesses, the combined certificate, and the exact small-range counts.

No finite range, regardless of its size, proves the nonvanishing assertion for all  $n$ . In particular, the certificate supplies neither an induction step beyond 3000 nor a theorem bounding where a first additional zero could occur.

## 3.10 The precise missing step

The coefficient problem can now be stated independently of the original transcendental function.

**Conjecture 3.19** (Complete support classification). Let  $n \geq 1$ ,  $1 \leq k \leq n$ ,  $0 \leq j \leq k$ , and  $m = k - j$ . Then

$$\frac{1}{j!m!} \frac{d^m}{du^m} \Delta_2^j u^n \Big|_{u=2m} = 0 \quad (3.43)$$

if and only if at least one of the following holds:

$$(j = k \text{ and } n > 2k), \quad (k \text{ even and } n = 4k - 2j + 1). \quad (3.44)$$

The implication from (3.44) to (3.43) is proved above. The converse remains unproved in the following precise region:

$$m = k - j \geq 3, \quad n \geq 2k + 2, \quad (3.45)$$

away from the reflection zeros. All other regions have been handled by the preceding analytic theorems.

*Update (29 September 2026).* For  $m = 3$  and  $m = 4$  the converse is now proved, for every  $n$ : Theorem 3.23 shows that on those two deficits the only zeros are the reflection zeros. The region still unproved is therefore

$$m = k - j \geq 5, \quad n \geq 2k + 2,$$

away from the reflection zeros; the full conjecture remains open. At every fixed  $m \geq 3$  the region is also reduced to finitely many cells by Theorem 3.22, but with bounds too large to compute.

**Theorem 3.20** (Exact reduction of the OEIS conjecture). *The equality  $a(n) = U(n)$  for every  $n \geq 0$  is equivalent to Conjecture 3.19. It is also equivalent to proving the nonvanishing assertion only in the remaining region (3.45), excluding the reflection-zero family.*

*Proof.* Formula (3.13) identifies the expression in (3.43) with the actual derivative coefficient. Theorem 3.14 counts exactly the cells left after removal of the two proved, disjoint zero families. Hence any additional zero makes the corresponding inequality strict, and no additional zeros makes it an equality. Theorem 3.7, Proposition 3.9, and Theorems 3.16–3.17 eliminate the complementary regions.  $\square$

A tempting shortcut would assert that a derivative of a falling factorial cannot vanish at an integer other than its reflection center. That assertion is false. Direct differentiation gives

$$\frac{d^5}{du^5} u^8 = 3360(u - 5)(u - 2)(2u - 7). \quad (3.46)$$

It vanishes at  $u = 2$  and  $u = 5$ , as well as at the reflection center  $u = 7/2$ . This is *not* a counterexample to Conjecture 3.19, because the derivative order, evaluation point, and finite-difference order in that conjecture are linked in a special way. It does show why those constraints must be used in a genuine proof of the missing implication.

The transformed formula provides an alternative target. With  $d = n - 2k - 1 \geq 1$ , the outstanding task is to prove that

$$[u^{k+d+1}](1 - u)^d(1 + u)^{2m} \left( \frac{\operatorname{atanh} u}{u} \right)^m \quad (3.47)$$

is nonzero for  $m \geq 3$ , except when  $d = 2m$  and  $k$  is even. The reflection case is transparent because the series then becomes even; the remaining finite-difference cancellations are the essential issue. Equation (3.47) is an equivalent residual lemma, not an additional theorem claimed by this article. (Since the update above it is needed only for  $m \geq 5$ .)

### 3.11 Conclusion

The proposed formula has a rigorous structural explanation. The obvious degree loss removes the top logarithmic coefficients with  $n > 2k$ . A separate reflection symmetry removes exactly  $\lfloor (n+1)/8 \rfloor$  further coefficients for odd  $n$ . Their total produces the OEIS floor expression and its rational generating function as an unconditional upper bound.

A fractional-linear transformation proves that a large region of the coefficient triangle is strictly positive, and beta-integral arguments exclude additional zeros on the first two deficit diagonals. These results prove quadratic growth and reduce the full formula to the explicit residual nonvanishing problem (3.47). Exact finite computations settle every derivative order through 3000. The global equality is not proved here: its remaining obligation is the nonvanishing statement in (3.45), not any of the counting or generating-function manipulations. (Update, 29 September 2026: that statement is now proved at deficits  $m = 3, 4$ , Theorem 3.23; it remains open for  $m \geq 5$ .)

### 3.12 Reproducing the computations

The archive needs Python 3.10 or later and a C++17 compiler for the computations. Python uses only its standard library. The PDF can be rebuilt with a conventional LaTeX installation containing the packages listed in the source preamble.

Run the following commands from the archive directory. The Makefile wraps the same operations.

```
python3 code/verify_exact.py --max-n 200 --formula-n 16
mkdir -p build
g++ -O3 -std=c++17 -Wall -Wextra \
    code/verify_modular.cpp -o build/verify_modular
build/verify_modular 3000 1000000007 data/p1000000007 \
    data/watch_coordinates.csv
build/verify_modular 3000 1000000009 data/p1000000009 \
    data/watch_coordinates.csv
python3 code/combine_certificates.py
pdflatex -interaction=nonstopmode -halt-on-error \
    A290268_partial_results.tex
pdflatex -interaction=nonstopmode -halt-on-error \
    A290268_partial_results.tex
```

The watch file requests explicit residues at the seven exceptional coordinates found in the modular runs. It does not restrict the search: the C++ program checks every coefficient in every row regardless of the watch file. The combiner checks that the complete exceptional zero lists are disjoint and that all listed witnesses are consistent with them.

For a smaller test, the optional watch file can be omitted. For example, `build/verify_modular 100 1000000007 data/test` performs the full modular recurrence through 100. A watch file containing indices beyond the chosen bound is rejected, rather than silently ignored.

### 3.13 Files included in the archive

| File or directory                                                      | Purpose                                                                                             |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <code>article.tex</code>                                               | Complete source of the merged article; this part is its Part 3.                                     |
| <code>article.pdf</code>                                               | The compiled merged article.                                                                        |
| <code>README.md</code> , <code>Makefile</code>                         | Status, dependency notes, and reproduction commands.                                                |
| <code>code/verify_exact.py</code>                                      | Arbitrary-precision recurrence and three independent coefficient-formula checks.                    |
| <code>code/verify_modular.cpp</code>                                   | Complete modular coefficient search with explicit overflow bounds.                                  |
| <code>code/combine_certificates.py</code>                              | Verification and combination of the two finite modular certificates.                                |
| <code>data/exact_counts.csv</code>                                     | Exact term counts through 200.                                                                      |
| <code>data/certified_counts.csv</code>                                 | Certified term counts through 3000.                                                                 |
| <code>data/p1000000007_*</code> and<br><code>data/p1000000009_*</code> | Complete exceptional zero lists, row statistics, witness residues, run logs, and summaries.         |
| <code>data/modular_witnesses.csv</code>                                | The seven coordinates at which one residue vanishes, with the nonzero residue in the other modulus. |
| <code>data/*summary.json</code>                                        | Machine-readable verification scope and results.                                                    |
| <code>data/sample_coefficients.json</code>                             | Fully collected low-order coefficient arrays and cancellation examples.                             |

### 3.14 Depth certificates: deficits three and four

**What this addition is.** Subsections 3.14–3.26 were written into this part on 29 September 2026 from a later manuscript, *From Infinite Nonvanishing to Finite Certificates: Bulk positivity, fixed-depth finiteness, and a four-depth solution for OEIS A290268* (batch 42, manuscript 06; provenance in Section 3.26). It closes the region  $m = k - j \in \{3, 4\}$  of the precise missing step (3.45): at logarithmic deficits three and four, for every  $n$ , the only vanishing coefficients are the reflection zeros of Theorem 3.11. The unproved region is now  $m \geq 5$ ,  $n \geq 2k + 2$ . The full A290268 conjecture is *not* proved.

**Written without this part.** The manuscript was prepared against the repository notes `0eis/A290268/README.md` and the conditional Lean reduction, and never read this report. Several of its results therefore re-derive results printed above. They are printed here once more because their proofs are different, and each is marked as a second route with the earlier result credited:

- its all-depth bulk positivity (Theorem 3.21) is Theorem 3.7 restricted to  $m \geq 1$ ;
- its symmetry holes (Theorem 3.29) are the reflection zeros of Theorem 3.11;
- its depths one and two are Theorems 3.16 and 3.17;
- its upper bound (3.84) is Theorem 3.14;
- its exact depth bridge (Proposition 3.24) is Proposition 3.4 re-indexed, and its branch-cut formula (Proposition 3.30) is Lemma 3.15 with its  $\varepsilon$ -coefficient evaluated in closed form (Remark 3.31).

Its finite rectangles reach only derivative orders  $N \leq 359$ , inside the range  $n \leq 3000$  that Proposition 3.18 already certifies. What is new is the reduction that makes finitely many cells suffice for *every*  $n$ ; see Remark 3.42.

**Notation.** These subsections use the *depth coordinates* of the manuscript, which are also the coordinates of the Lean development. In them  $k$  is the **exponent** of  $\log x$  — the

index called  $j$  everywhere else in this part — and  $d$  is the logarithmic deficit  $m$ , not the  $d$  of (3.25). Table 3 is binding for everything from here to the end of the part.

**Abstract of manuscript 06.** *OEIS A290268 counts the terms in the fully expanded derivatives of  $x^{x^2}$ . Its conjectured quasipolynomial reduces to a nonvanishing problem for  $\gamma(k, d, M) = [w^M](2 + w)^k((1 + w)^2 \log(1 + w))^d$ . The ProveIt investigation identifies symmetry-forced zeros and leaves the higher-depth nonvanishing problem open. We prove positivity throughout the whole bulk  $d \leq M \leq k + 2d + 1$ , uniformly in every depth  $d \geq 1$ . We then prove an effective finiteness theorem: at each fixed depth, every possible additional zero is confined to an explicitly bounded finite rectangle in two integer parameters. The proof combines a centered Meixner–Pollaczek factorization, a branch-cut integral, and monotone likelihood-ratio comparisons on a half interval. At depths three and four the bounds sharpen to rectangles of 494 and 15,540 cells. Two independent integer recurrences, supplemented by modular nonzero witnesses, verify these rectangles exactly. Consequently, for  $1 \leq d \leq 4$  and  $M \geq d$ , the only zeros are  $M = k + 4d + 1$  with  $k + d$  even. This is an infinite-parameter result, not a finite derivative-order experiment. We also bound the number of real zeros of an interpolating Mellin transform by at most  $d$ , derive an unconditional term-count lower bound, and formulate concrete next research targets. The full A290268 formula is not proved here: the remaining nonvanishing problem lies at depths  $d \geq 5$ . The proofs are conventional and computer-assisted where indicated; no Lean kernel certification or independent refereeing is claimed.*

*Editorial note.* “The ProveIt investigation” is the note `Oeis/A290268/README.md`. The bulk positivity claimed in the abstract was already Theorem 3.7 of this part; the manuscript’s proof of it is a second route.

### 3.14.1 Notation: depth coordinates

The manuscript writes the  $N$ th derivative in the lattice of monomials  $x^{x^2+j}(\log x)^k$ . That is the lattice of the Lean development (`coeff n j k`, Section 3.26.3) and of the shipped data files, whose columns are named `d`, `k`, `q`. The subsections below keep those letters rather than translate some four hundred lines of indices, and the price is the table below. Its first four rows are the places where a reader coming from the rest of this part will go wrong.

Two index translations follow from the table and are used repeatedly:

$$c_N(j, k) = c_{N, k+d, k}, \quad N = 2k + 2d + 1 + q \iff n = 2(j_{\text{III}} + m) + 1 + d_{\text{III}},$$

where the subscript III marks the letters of the rest of this part. In particular the tail line  $q = 0$  is the borderline  $n = 2k + 1$  of Theorem 3.7, and the region  $q \geq 1$  is the region  $n \geq 2k + 2$  of (3.45).

*Renamed symbols.* Where the manuscript’s letter names a different object of this part, the object is renamed here, with no change of normalization: the manuscript’s residual polynomial  $P_d(z)$  is  $\Pi_d(\zeta)$  (this part’s  $P_n$  is the canonical polynomial); its centered polynomial  $Q_k^{(r)}$  is  $\mathcal{Q}_k^{(r)}$ ; its  $A_{M,k}(s)$  is  $\mathcal{A}_{M,k}(s)$ ; its base moment  $B_d(r)$  is  $\mathcal{B}_d(r)$  and its bulk count  $B(N)$  is  $\beta(N)$ ; its half-interval logarithm  $L = \log(t/(1-t))$  is  $\tau$  (this part’s  $L$  is  $\log x$ ); its hole indicator  $h_d(N)$  is  $\chi_d(N)$  (the manuscript also used  $h_d$  for a second object, kept below as  $h_d(\tau)$ ); its  $m = \lfloor (d-1)/2 \rfloor$  is  $\nu_d$  and the  $m$  of its count formulas is  $\rho$  (this part’s  $m$  is the deficit); its Mellin transform  $F_{d,k}$  is  $\mathcal{F}_{d,k}$  (this part’s  $F_n$  is the falling factorial); its generating variable  $z$  is  $\xi$ ; the auxiliary combination  $h(t)$  in the proof of Lemma 3.43 is  $\psi(t)$ ; and product indices written  $j$  in the manuscript are  $\ell$ . The manuscript’s falling factorial  $(s)_r$  is printed  $s^x$ , as in the rest of this article. The letters  $s, t, u, w, y$  are local to the paragraph in which they are introduced; as in the manuscript,  $y$  is a centered variable in Subsection 3.15 and  $1-t$  in Subsection 3.17, and  $s$  is both a polynomial variable and  $2t-1$ .

Table 3: Depth coordinates (Subsections 3.14–3.26) against the rest of Part 3. Throughout,  $c_{n,k,j}$  is the coefficient of (3.5).

| Here                           | Meaning                                  | Elsewhere in Part 3           | Tempting false reading                                                         |
|--------------------------------|------------------------------------------|-------------------------------|--------------------------------------------------------------------------------|
| $k$                            | exponent of $\log x$                     | $j$                           | <b>false:</b> the power of $z = x^2$ ,<br>the $k$ of $c_{n,k,j}$               |
| $j$                            | exponent of $x$ in $x^{x^2+j}(\log x)^k$ | $2k - n$                      | <b>false:</b> the exponent of $\log x$                                         |
| $d$                            | depth, $d = (N + j)/2 - k$               | $m = k - j$                   | <b>false:</b> $d = n - 2k - 1$ of (3.25)                                       |
| $q$                            | $M - k - 2d - 1 = N - 2k - 2d - 1$       | $d = n - 2k - 1$ of<br>(3.25) | <b>false:</b> the exponent in<br>$(1 - u)^{-q}$ in the proof of<br>Theorem 3.7 |
| $N$                            | derivative order                         | $n$                           | —                                                                              |
| $M$                            | coefficient index $N - k$                | $n - j$                       | the moduli $M_1, M_2$ of (3.41)                                                |
| $c_N(j, k)$                    | coefficient of $x^{x^2+j}(\log x)^k$     | $c_{N,(N+j)/2,k}$             | —                                                                              |
| $\varphi(w)$                   | $(1 + w)^2 \log(1 + w)$                  | $B(w)$ of (3.10)              | —                                                                              |
| $\gamma(k, d, M)$              | $[w^M](2 + w)^k \varphi(w)^d$            | $[t^{M+k}]A(t)^k B(t)^d$      | —                                                                              |
| $H_{d,k,q}$                    | the integer (3.49)                       | —                             | $H(u) = \operatorname{atanh}(u)/u$ of (3.15),<br>which is unrelated            |
| $r$                            | $M - k$                                  | —                             | $r$ in $n = 2r$ , $n = 2r + 1$                                                 |
| $K_d, Q_d,$<br>$A_d, C_d, E_d$ | constants (3.74)                         | —                             | $A(t)$ of (3.10), $Q_{n,j}$ of<br>Section 3.3                                  |

### 3.14.2 Main results of the addition

Put

$$\varphi(w) = (1 + w)^2 \log(1 + w), \quad \gamma(k, d, M) = [w^M](2 + w)^k \varphi(w)^d. \quad (3.48)$$

The depth is  $d$ , the logarithmic degree is  $k$ , and  $M$  is a coefficient index. All three are nonnegative integers.

**Theorem 3.21** (Bulk positivity; second route to Theorem 3.7). *For every  $d \geq 1$  and  $k \geq 0$ ,*

$$d \leq M \leq k + 2d + 1 \implies \gamma(k, d, M) > 0.$$

In the coordinates of this part the hypothesis reads  $m \geq 1$ ,  $k \leq n \leq 2k + 1$ , so the theorem is exactly the case  $m \geq 1$  of Theorem 3.7; with (3.57) below, which is Proposition 3.9, it is all of it. The manuscript's proof in Subsection 3.16 goes through a centered factorization instead of the transformation (3.16).

For the remaining region it is useful to put

$$q = M - k - 2d - 1 \geq 0, \quad H_{d,k,q} = \frac{(k + 2d + 1 + q)!}{d!} \gamma(k, d, k + 2d + 1 + q). \quad (3.49)$$

The number  $H_{d,k,q}$  is an integer. The  $q = 0$  line overlaps the already positive boundary of the bulk.

**Theorem 3.22** (Effective fixed-depth finiteness). *For each  $d \geq 3$  there are explicit positive integers  $K_d, Q_d$ , given in Subsection 3.19, such that whenever  $k \geq K_d$  or  $q \geq Q_d$ ,*

$$H_{d,k,q} = 0 \iff q = 2d \text{ and } k + d \equiv 0 \pmod{2}.$$

*Thus the complete fixed-depth nonvanishing conjecture is equivalent to a finite, effectively specified collection of integer nonzero tests.*

**Theorem 3.23** (Complete first-four-depth classification). *For  $1 \leq d \leq 4$ ,  $k \geq 0$ , and  $M \geq d$ ,*

$$\gamma(k, d, M) = 0 \iff M = k + 4d + 1 \text{ and } k + d \equiv 0 \pmod{2}. \quad (3.50)$$

*For  $d = 1, 2$  the proof is entirely analytic. For  $d = 3, 4$  it uses finite exact-integer certificates after a proved infinite-to-finite reduction.*

In the coordinates of this part: for  $1 \leq m \leq 4$  and every  $n \geq k$ , the coefficient  $c_{n,k,j}$  vanishes if and only if  $k$  is even and  $n = 4k - 2j + 1$ . For  $m = 1, 2$  this is Theorems 3.7, 3.16 and 3.17; for  $m = 3, 4$  it is new, and it removes  $m = 3, 4$  from the region (3.45).

The practical exceptional rectangles for this theorem are

$$\begin{aligned} d = 3 : \quad & 0 \leq k \leq 12, \quad 0 \leq q \leq 37, \\ d = 4 : \quad & 0 \leq k \leq 83, \quad 0 \leq q \leq 184. \end{aligned} \quad (3.51)$$

They contain 48 symmetry zeros and 15,986 nonzero entries in total. Subsection 3.21 explains every finite obligation and its verification. Subsection 3.22 gives a separate all-depth bound on the number of zeros for each fixed  $k$ .

### 3.14.3 What is and is not new here

The manuscript's own assessment, relative to the repository notes it read: the support reduction, the target quasipolynomial, the reflection idea, and the first two depths are prior repository material [15]. Meixner–Pollaczek polynomials are classical [17]; no new orthogonal-polynomial family is proposed. The contribution is their use to prove all-depth bulk positivity, the effective uniform fixed-depth reduction, and its rigorous completion at depths three and four. The real-zero bound and term-count lower bound are additional consequences.

*Relative to this part* the assessment is narrower. All-depth bulk positivity is Theorem 3.7, and the reflection zeros, the deficit-one and deficit-two classification and the upper bound are all proved above; the manuscript's proofs of them are second routes. New relative to this part are: the centered factorization (Proposition 3.26); the closed form of the branch-cut kernel for every depth and its folding (Subsection 3.17); the monotone comparisons and the effective finite reduction at every fixed depth  $d \geq 3$  (Theorem 3.22); the completion at deficits three and four (Theorem 3.23); the zero-count bound (Theorem 3.44); and the lower bound of Corollary 3.45, which improves  $\Lambda(n)$  of Theorem 3.1.

The manuscript's literature and repository checks support it as a continuation beyond the inspected A290268 notes, not an assertion of absolute historical priority. Related results on generalized Bernoulli polynomials, including integer interlacing in the order-equals-degree family, appear in Štampach's work [18]. Those results concern a different parameter regime and do not by themselves settle the coefficient nonvanishing considered here. The addition is a research draft with explicit proof and computation boundaries, not an announcement that the entire OEIS conjecture has been solved.

## 3.15 Depth coordinates: exact coefficient and polynomial models

### 3.15.1 The lattice recurrence and the depth reduction

For  $x > 0$ , write

$$\frac{d^N}{dx^N} x^{x^2} = x^{x^2} \sum_{j \in \mathbb{Z}} \sum_{k \geq 0} c_N(j, k) x^j (\log x)^k. \quad (3.52)$$

Only finitely many coefficients are nonzero, and all are integers. The sequence in question is  $a(N) = \#\{(j, k) : c_N(j, k) \neq 0\}$ . The OEIS entry, proposed by Reshetnikov in 2017 and subsequently given alternative forms by Luschny, records the conjecture

$$a(2\rho) = 2\rho^2 + 2\rho + 1, \quad a(2\rho - 1) = 2\rho^2 - \lfloor \rho/4 \rfloor \quad (\rho \geq 1), \quad (3.53)$$

which is (3.2) with  $\rho = r + 1$  in the odd case. The equivalent rational generating function remained labeled conjectural in the entry accessed for the manuscript on 29 September 2026 [2]. The precise coefficient model is used rather than the output conventions of any computer algebra system.

The functions  $x^j(\log x)^k$  appearing in a finite expansion are linearly independent; this is Lemma 3.2, and the manuscript's argument is the same one. After setting  $x = e^t$ , a putative relation becomes a finite sum of exponentials  $\exp(jt)$  times polynomials in  $t$ . Divide by the largest exponential and let  $t \rightarrow +\infty$ ; the corresponding polynomial must vanish. Repeat. Thus the coefficient support is an intrinsic object within this normal form.

The repository notes already reduce the support problem to a three-parameter coefficient and prove or discuss several initial strata [15]. The Lean module `Main.lean` explicitly consumes the support characterization as a hypothesis in its support-to-count theorem [16]; that conditional theorem is not treated as an existing proof of the conjecture (Section 3.26.3).

Differentiating a basis term in (3.52) yields

$$\begin{aligned} c_{N+1}(j, k) = & 2c_N(j-1, k-1) + c_N(j-1, k) \\ & + (j+1)c_N(j+1, k) + (k+1)c_N(j+1, k+1), \end{aligned} \quad (3.54)$$

with  $c_0(0, 0) = 1$  and all other initial coefficients zero. Negative logarithmic indices are interpreted as zero. This is the recurrence (3.7) re-indexed, and it is verbatim the definition of `coeff` in the Lean development. In particular,

$$|j| \leq N, \quad j \equiv N \pmod{2}, \quad 0 \leq k \leq (N+j)/2$$

are necessary for a nonzero coefficient.

**Proposition 3.24** (Exact depth bridge). *For an integer pair  $(j, k)$  in this support window, set  $d = (N+j)/2 - k$  and  $M = N - k$ . Then*

$$c_N(j, k) = \frac{N!}{k!d!} \gamma(k, d, M). \quad (3.55)$$

If  $d \geq 1$  and  $j \leq -1$ , so that  $q = -j - 1 \geq 0$ , this becomes the integer identity

$$c_N(j, k) = \binom{N}{k} H_{d,k,q}, \quad N = 2k + 2d + 1 + q. \quad (3.56)$$

*Proof.* Taylor expansion gives

$$\sum_{N \geq 0} \frac{\xi^N}{N!} x^{-x^2} \frac{d^N}{dx^N} x^{x^2} = \exp((2x\xi + \xi^2) \log x) \exp((x + \xi)^2 \log(1 + \xi/x)).$$

Set  $w = \xi/x$ . The contribution of logarithmic degree  $k$  in the first exponential and degree  $d$  in the second is

$$\frac{x^{2k+2d}}{k!d!} w^k (2+w)^k \varphi(w)^d (\log x)^k.$$

Extracting  $\xi^N$  introduces  $x^{-N}$ , so  $j = 2k + 2d - N$  and the remaining coefficient index is  $N - k = M$ . This proves (3.55). Substitution of (3.49) gives (3.56).  $\square$

Since  $A(t)^k B(t)^d = t^k (2+t)^k \varphi(t)^d$ , the bridge (3.55) is (3.12) re-indexed, and the proof above is essentially the Taylor-expansion proof of Proposition 3.4.

For completeness,

$$\gamma(k, 0, M) = \begin{cases} 2^{k-M} \binom{k}{M}, & 0 \leq M \leq k, \\ 0, & M > k, \end{cases} \quad (3.57)$$

which is Proposition 3.9. Also  $\gamma(k, d, M) = 0$  for  $M < d$ , since  $\varphi$  has order one at the origin.

### 3.15.2 A parameter derivative

Define

$$\mathcal{A}_{M,k}(s) = [w^M](2+w)^k(1+w)^s.$$

This is a polynomial in  $s$ . Differentiation with respect to  $s$  gives

$$\gamma(k, d, M) = \left. \frac{\partial^d}{\partial s^d} \mathcal{A}_{M,k}(s) \right|_{s=2d}. \quad (3.58)$$

Thus the question is about Taylor coefficients of explicit polynomials, not the cancellation of numerical transcendental values. (This is the depth-coordinate form of the finite-difference formula (3.13).)

**Definition 3.25** (Centered polynomial). For an integer  $r \geq 0$ , define  $\mathcal{Q}_k^{(r)}(y)$  by

$$\sum_{k \geq 0} \frac{\mathcal{Q}_k^{(r)}(y)}{k!} \xi^k = (1 - \xi^2)^{-(r+1)/2} \exp(2y \operatorname{atanh} \xi). \quad (3.59)$$

**Proposition 3.26** (Centered factorization). If  $M \geq k$  and  $r = M - k$ , then

$$M! \mathcal{A}_{M,k}(s) = s^r \mathcal{Q}_k^{(r)}\left(s - \frac{r-1}{2}\right). \quad (3.60)$$

The polynomial  $\mathcal{Q}_k^{(r)}$  has degree  $k$ , leading coefficient  $2^k$ , nonnegative coefficients, and parity  $k$ . It satisfies

$$\mathcal{Q}_0^{(r)} = 1, \quad \mathcal{Q}_1^{(r)} = 2y, \quad \mathcal{Q}_{k+1}^{(r)} = 2y \mathcal{Q}_k^{(r)} + k(k+r) \mathcal{Q}_{k-1}^{(r)} \quad (k \geq 1). \quad (3.61)$$

*Proof.* Expanding  $(2+w)^k$  and factoring the common falling factorial gives

$$M! \mathcal{A}_{M,k}(s) = s^r \sum_{\ell=0}^k \binom{k}{\ell} 2^\ell \frac{(r+k)!}{(r+\ell)!} (s-r)^\ell.$$

The sum divided by  $k!$  is

$$[t^k](1+t)^{r+k}(1+2t)^{s-r}.$$

Under the formal residue substitution  $t = \xi/(1-\xi)$ , this coefficient is

$$[\xi^k](1+\xi)^{s-r}(1-\xi)^{-s-1}.$$

With  $y = s - (r-1)/2$ , the last generating function is exactly (3.59). This proves the factorization. Both  $(1-\xi^2)^{-(r+1)/2}$  and  $\operatorname{atanh} \xi = \xi + \xi^3/3 + \dots$  have nonnegative coefficients. Expansion of the exponential proves coefficient nonnegativity and parity, and its top  $y$ -degree term gives  $2^k y^k$ . Finally, if  $G$  is the generating function in (3.59), then

$$(1-\xi^2)G'(\xi) = (2y + (r+1)\xi)G(\xi).$$

Coefficient comparison gives (3.61). □

In the standard normalization of the classical Meixner–Pollaczek polynomials  $P_k^{(\lambda)}(x; \phi)$  (classical notation, unrelated to the  $P_n$  of this part), the generating function in [17, Eq. 18.23.7] gives

$$\mathcal{Q}_k^{(r)}(y) = k! i^k P_k^{((r+1)/2)}(-iy; \pi/2). \quad (3.62)$$

All uses of this family in the proof follow directly from Proposition 3.26, so no orthogonality theorem is an unstated dependency.

**Corollary 3.27** (Mixed real and imaginary zeros). *The zeros of  $\mathcal{Q}_k^{(r)}$  are simple and purely imaginary. More precisely, for positive numbers  $\eta_1, \dots, \eta_{\lfloor k/2 \rfloor}$ ,*

$$\mathcal{Q}_k^{(r)}(y) = 2^k y^{k \bmod 2} \prod_{\ell=1}^{\lfloor k/2 \rfloor} (y^2 + \eta_\ell^2).$$

Thus (3.60) separates the consecutive real roots of the falling factorial from a centered imaginary-root factor.

*Proof.* The monic polynomials  $p_k(x) = 2^{-k} i^{-k} \mathcal{Q}_k^{(r)}(ix)$  satisfy

$$p_{k+1}(x) = x p_k(x) - \frac{k(k+r)}{4} p_{k-1}(x).$$

They are the characteristic polynomials of real symmetric tridiagonal matrices with zero diagonal and positive off-diagonal entries  $\sqrt{\ell(\ell+r)}/2$ . The eigenvalues are real. They are simple because an eigenvector is determined by its first coordinate through the tridiagonal recurrence, and a zero first coordinate would force the entire vector to vanish. The zero diagonal and alternating sign conjugation make the spectrum symmetric about zero. Transform back to  $\mathcal{Q}_k^{(r)}$ .  $\square$

### 3.16 Depth coordinates: bulk positivity and symmetry holes

Both results of this subsection were proved earlier in this part, by other means: the bulk is Theorem 3.7 and the holes are Theorem 3.11. The manuscript's proofs are printed as second routes.

*Second proof of Theorem 3.7 for  $m \geq 1$  (manuscript 06): proof of Theorem 3.21.* First suppose  $M \geq k$  and put  $r = M - k$ . The stated range implies  $r \leq 2d + 1$ . In (3.60), set  $s = 2d + u$ . Every factor of

$$(2d + u)^x = \prod_{\ell=0}^{r-1} (u + 2d - \ell)$$

has nonnegative constant term. At most one has constant term zero. If no factor vanishes at  $u = 0$ , all coefficients through degree  $r$  are positive; if one vanishes, all coefficients of degrees 1 through  $r$  are positive. The empty product is one.

The argument of  $\mathcal{Q}_k^{(r)}$  is  $u + y_0$  with  $y_0 = 2d - (r - 1)/2 \geq d > 0$ . Its coefficients through degree  $k$  are all positive: the leading term  $2^k(u + y_0)^k$  already has that property, and all remaining contributions are nonnegative. Since  $1 \leq d \leq r + k = M$ , the coefficient of  $u^d$  in the product is strictly positive. Equation (3.58) proves the claim in this case.

To handle  $M < k$ , induct on  $k$  for the entire claimed range, using

$$\gamma(k, d, M) = 2\gamma(k - 1, d, M) + \gamma(k - 1, d, M - 1). \quad (3.63)$$

The  $k = 0$  case has already been proved. When  $M < k$  and  $M \geq d$ , the first term on the right is positive by induction. The second is positive when  $M - 1 \geq d$  and is zero otherwise. Both indices lie in the preceding bulk range. This completes the induction.  $\square$

**Corollary 3.28** (Positive derivative columns). *Within the structural support window, every coefficient with  $j \geq 0$  is positive. In the  $j = -1$  column every coefficient with  $d \geq 1$  is positive, while the  $d = 0$  coefficient is zero.*

*Proof.* For  $d \geq 1$ , the condition  $M \leq k + 2d + 1$  is exactly  $j \geq -1$  by Proposition 3.24. For  $d = 0$ , use (3.57); its nonzero condition  $M \leq k$  is exactly  $j \geq 0$ .  $\square$

In the coordinates of this part,  $j \geq 0$  is  $n \leq 2k$  and  $j = -1$  is  $n = 2k + 1$ , so the corollary restates Theorem 3.7 together with Proposition 3.9.

**Theorem 3.29** (All-depth symmetry holes; second route to Theorem 3.11). *For  $d \geq 1$  and  $k \geq 0$ ,*

$$k + d \equiv 0 \pmod{2} \implies \gamma(k, d, k + 4d + 1) = 0.$$

*Proof.* Here  $r = 4d + 1$  and the center  $(r - 1)/2$  is  $2d$ . With  $s = 2d + u$ ,

$$s^r = u \prod_{\ell=1}^{2d} (u^2 - \ell^2)$$

is odd, whereas  $\mathcal{Q}_k^{(r)}(u)$  has parity  $k$ . The product has parity  $k + 1$ . Its  $u^d$  coefficient therefore vanishes if  $k + d$  is even. Apply (3.58).  $\square$

This proves the hole direction without relying on the analytic integral below. The polynomial proof also distinguishes the symmetry statement from the separate assertion that there are no other zeros. Translated,  $M = k + 4d + 1$  is  $q = 2d$ , that is  $n = 2k + 2m + 1 = 4k - 2j + 1$  in this part's letters, and  $k + d$  even is “ $k$  even” there; so the hole family is exactly the reflection family of Theorem 3.11, and the proof above is the odd-falling-factorial parity argument that Section 5.4 identifies as common to all three parts.

### 3.17 Depth coordinates: the branch-cut integral and its folding

#### 3.17.1 An exact integral valid in the negative region

Let

$$R_d(v) = \frac{\operatorname{Im}(v + i\pi)^d}{\pi}. \quad (3.64)$$

It is a real polynomial of degree  $d - 1$  with leading coefficient  $d$ .

**Proposition 3.30** (Branch-cut formula). *For  $d \geq 1$ ,  $k \geq 0$ , and  $M = k + 2d + 1 + q$  with integer  $q \geq 0$ ,*

$$\gamma(k, d, M) = (-1)^{M+1} \int_0^1 t^q (1-t)^{2d} (2t-1)^k R_d\left(\log \frac{1-t}{t}\right) dt. \quad (3.65)$$

*The integral is absolutely convergent.*

*Proof.* Apply Cauchy's coefficient formula to  $(2+w)^k(1+w)^{2d}\log^d(1+w)$ , using the principal logarithm and deforming the circle to the cut  $(-\infty, -1]$ . The small circle about  $-1$  gives zero in the limit because its contribution is bounded by a constant times  $\varepsilon^{2d+1}(1+|\log \varepsilon|^d)$ . The large circle vanishes because  $M > k + 2d$ , with bound  $O(R^{k+2d-M}\log^d R)$ .

At  $w = -x$ ,  $x > 1$ , the upper and lower logarithms are  $\log(x-1) \pm i\pi$ . Their jump divided by  $2\pi i$  is  $R_d(\log(x-1))$ . The factor  $w^{-M-1}$  contributes  $(-1)^{M+1}$ . Consequently the coefficient is

$$(-1)^{M+1} \int_1^\infty \frac{(2-x)^k(x-1)^{2d}}{x^{M+1}} R_d(\log(x-1)) dx.$$

The substitution  $t = 1/x$  gives (3.65). At  $t = 0$  and  $t = 1$  the integrand is bounded by a constant times a nonnegative integral power of the distance to the endpoint, multiplied by a fixed power of its logarithm. This is integrable.  $\square$

*Remark 3.31* (Relation to Lemma 3.15; editorial). Formula (3.65) is the  $\varepsilon$ -expansion of Lemma 3.15 carried out for every deficit. Indeed  $\frac{\sin(\pi\varepsilon)}{\pi} \left(\frac{t}{1-t}\right)^\varepsilon = \text{Im} \exp((\sigma + i\pi)\varepsilon)/\pi$  with  $\sigma = \log \frac{t}{1-t}$ , whose coefficient of  $\varepsilon^m$  is  $R_m(\sigma)/m!$ . Hence (3.28) reads

$$c_{n,k,j} = \frac{(-1)^{n-1}n!}{j!m!} \int_0^1 t^{2m}(1-t)^d(2t-1)^j R_m\left(\log \frac{t}{1-t}\right) dt$$

in the letters of this part, and the substitution  $t \mapsto 1-t$  together with the dictionary of Table 3 turns it into (3.56) with (3.65). For  $m = 1, 2, 3$  the kernel  $R_m/m!$  is  $1$ ,  $\sigma$  and  $\frac{1}{2}\sigma^2 - \frac{\pi^2}{6}$ : the integrands (3.30), (3.32) and (3.33). What is new below is the treatment of the sign change that Section 3.7 identified as the obstruction at  $m = 3$ .

### 3.17.2 The residual polynomial

For  $d \geq 1$ , put

$$\nu_d = \left\lfloor \frac{d-1}{2} \right\rfloor, \quad h_d(\tau) = \begin{cases} 1, & d \text{ odd,} \\ \tau, & d \text{ even,} \end{cases}$$

and define the monic polynomial

$$\Pi_d(\zeta) = \prod_{\ell=1}^{\nu_d} \left( \zeta - \pi^2 \cot^2 \frac{\ell\pi}{d} \right). \quad (3.66)$$

The empty product is one. Factoring the imaginary part in (3.64) gives

$$R_d(-\tau) = (-1)^{d-1} d h_d(\tau) \Pi_d(\tau^2) \quad (\tau > 0). \quad (3.67)$$

Indeed, the roots of  $R_d$  are the distinct numbers  $\pi \cot(\ell\pi/d)$  for  $1 \leq \ell \leq d-1$ ; pairing opposite roots proves the identity, including its leading coefficient.

The initial cases are

$$\begin{aligned} \Pi_1(\zeta) &= \Pi_2(\zeta) = 1, \\ \Pi_3(\zeta) &= \zeta - \pi^2/3, \quad \Pi_4(\zeta) = \zeta - \pi^2, \\ \Pi_5(\zeta) &= \zeta^2 - 2\pi^2\zeta + \pi^4/5, \\ \Pi_6(\zeta) &= \zeta^2 - \frac{10}{3}\pi^2\zeta + \pi^4. \end{aligned} \quad (3.68)$$

Thus the first two depths have no residual sign change. At depths three and four the residual polynomial is increasing. At higher depths that last property fails and cannot be silently reused.

### 3.17.3 Positive measures on the half interval

For  $1/2 < t < 1$ , abbreviate

$$y = 1 - t, \quad s = 2t - 1, \quad \tau = \log(t/y), \quad p = 2d, \quad \delta = q - p, \quad b_d(t) = (ty)^p h_d(\tau).$$

Reflect the part  $0 < t < 1/2$  in (3.65). Apart from the sign and positive factor  $(-1)^{d-1}d$ , the result is

$$\int_{1/2}^1 b_d(t) s^k \Pi_d(\tau^2) (t^\delta + (-1)^{k+d-1} y^\delta) dt. \quad (3.69)$$

When  $k + d$  is odd, the bracket is the positive sum  $S_\delta(t) = t^\delta + y^\delta$ . When  $k + d$  is even, it is the difference  $D_\delta(t) = t^\delta - y^\delta$ ; it is identically zero for  $\delta = 0$  and otherwise has constant sign  $\operatorname{sgn} \delta$ .

For a sum case, define a probability measure by normalizing  $b_d s^k S_\delta dt$ . For a nonzero difference case, normalize  $b_d s^k |D_\delta| dt$ . Write  $\mathcal{E}_{d,k,q}$  for expectation with respect to the relevant measure. The following sign rule is exact:

$$\operatorname{sgn} H_{d,k,q} = \begin{cases} (-1)^q \operatorname{sgn} \mathcal{E}_{d,k,q}[\Pi_d(\tau^2)], & k + d \text{ odd}, \\ (-1)^{q+1} \operatorname{sgn}(q - 2d) \operatorname{sgn} \mathcal{E}_{d,k,q}[\Pi_d(\tau^2)], & k + d \text{ even, } q \neq 2d. \end{cases} \quad (3.70)$$

At a symmetry hole  $H_{d,k,2d} = 0$  by Theorem 3.29. All the measures above have finite logarithmic moments of every fixed order.

### 3.18 Depth coordinates: monotone comparisons

**Lemma 3.32** (Covariance comparison). *Let  $\mu$  be a probability measure on an interval, and let  $f, g$  be nondecreasing functions with the needed integrability,  $g > 0$ . Then*

$$\frac{\int f g d\mu}{\int g d\mu} \geq \int f d\mu.$$

*In particular, multiplying a positive density by a nondecreasing function can only increase expectations of nondecreasing functions.*

*Proof.* The covariance equals

$$\frac{1}{2} \iint (f(u) - f(v))(g(u) - g(v)) d\mu(u) d\mu(v) \geq 0.$$

Truncation proves the same conclusion for integrable unbounded functions.  $\square$

**Lemma 3.33** (Radial likelihood ratios). *On  $(1/2, 1)$  the following assertions hold.*

- (i) *For every integer  $\delta$ ,  $S_\delta$  is nondecreasing. The cases  $\delta = 0, 1$  are constant.*
- (ii) *For every nonzero integer  $\delta$ ,  $|D_\delta|/s$  is positive and nondecreasing. The positive cases  $\delta = 1, 2$  are constant.*
- (iii) *If  $\delta \geq 1$  is an integer, both  $S_{\delta+1}/S_\delta$  and  $D_{\delta+1}/D_\delta$  are nondecreasing.*

*Proof.* Write  $v = \frac{1}{2} \log(t/y) > 0$ , so  $t = e^v/(2 \cosh v)$  and  $y = e^{-v}/(2 \cosh v)$ . Direct calculation gives

$$\begin{aligned} \frac{S_{\delta+1}}{S_\delta} &= \frac{1}{2} (1 + \tanh v \tanh(\delta v)), \\ \frac{D_{\delta+1}}{D_\delta} &= \frac{1}{2} (1 + \tanh v \coth(\delta v)). \end{aligned} \quad (3.71)$$

The first expression is nondecreasing for  $\delta \geq 0$ . For  $\delta \geq 1$ ,

$$\frac{d}{dv} \log(\tanh v \coth(\delta v)) = \frac{2}{\sinh(2v)} - \frac{2\delta}{\sinh(2\delta v)} \geq 0,$$

using  $\sinh(\delta x) \geq \delta \sinh x$  for  $\delta \geq 1$ ,  $x > 0$ . This proves (iii). Products of these ratios, starting at  $S_1 = 1$  and  $D_1 = s$ , prove (i) and (ii) for positive  $\delta$ . For a negative integer  $\delta = -a$ , use

$$S_{-a} = (ty)^{-a} S_a, \quad |D_{-a}| = (ty)^{-a} D_a.$$

The extra factor is nondecreasing because  $ty$  decreases on the half interval. The stated constant cases are immediate.  $\square$

For real  $r \geq 0$ , let  $\mu_{d,r}$  denote the probability measure obtained by normalizing  $b_d(t)s^r dt$ , and set

$$\mathcal{B}_d(r) = \mathbb{E}_{\mu_{d,r}}[\tau^2]. \quad (3.72)$$

**Proposition 3.34** (Uniform comparison with a symmetric base). *For every integer  $q \geq 0$  outside a symmetry hole,*

$$\begin{aligned} \mathcal{E}_{d,k,q}[\tau^2] &\geq \mathcal{B}_d(k), & k+d \text{ odd}, \\ \mathcal{E}_{d,k,q}[\tau^2] &\geq \mathcal{B}_d(k+1), & k+d \text{ even}. \end{aligned} \quad (3.73)$$

*The function  $\mathcal{B}_d(r)$  is nondecreasing in  $r$ . Within either parity class of  $k$ , the expectations of every increasing function of  $\tau^2$  increase when  $k$  increases by two. They also increase when  $q$  increases by one in the region  $q \geq 2d+1$ .*

*Proof.* For a sum measure, its ratio to the base density with exponent  $k$  is  $S_\delta$ , which is nondecreasing by Lemma 3.33. For a difference measure, its ratio to the base with exponent  $k+1$  is  $|D_\delta|/s$ , again nondecreasing. Apply Lemma 3.32. Increasing  $r$  multiplies a base density by a positive power of  $s$ . Increasing  $k$  by two multiplies either type of density by  $s^2$ . The assertion about  $q$  follows from the adjacent ratios in (3.71), with  $\delta \geq 1$ .  $\square$

*Remark 3.35* (The role of integer parameters). The global base comparison uses integer  $\delta$ . For noninteger  $0 < \delta < 1$ , the sum  $S_\delta$  has different monotonicity. No conclusion about that interval is being smuggled into the integer coefficient problem. The real-zero theorem in Subsection 3.22 uses a separate argument and does not depend on the integer comparison.

### 3.19 Depth coordinates: an explicit finite reduction at every depth

This subsection proves Theorem 3.22 with deliberately crude but fully explicit bounds. Their purpose is decidability of a fixed-depth stratum, not efficient computation. The next subsection replaces them by small bounds at depths three and four.

#### 3.19.1 A monotone lower bound for every residual polynomial

For  $d \geq 3$ , put

$$\begin{aligned} \nu_d &= \lfloor (d-1)/2 \rfloor, & C_d &= [3(d^2+1)]^{\nu_d} + d^2 + 1, \\ A_d &= 4C_d + 2, & E_d &= (2d+1)A_d + 6d + 2, \\ K_d &= 2^{A_d}(E_d + A_d + 2), \\ Q_d &= 2d + 2^{A_d+1}(E_d + A_d + 6). \end{aligned} \quad (3.74)$$

These are positive integers constructed by elementary integer operations and exponentiation.

**Lemma 3.36** (Affine minorant). *For all  $\zeta \geq 0$ ,*

$$\Pi_d(\zeta) \geq \frac{1}{2}\zeta - C_d. \quad (3.75)$$

*Proof.* Every root  $\beta$  of  $\Pi_d$  satisfies  $0 < \beta < d^2$ , because  $\pi \cot(\pi/d) < d$ . For  $\zeta \geq 2(d^2 + 1)$ , each factor  $\zeta - \beta$  is at least  $\zeta/2$ , and  $\nu_d \geq 1$  implies  $\Pi_d(\zeta) \geq (\zeta/2)^{\nu_d} \geq \zeta/2$ . For  $0 \leq \zeta \leq 2(d^2 + 1)$ , the absolute value of the product is at most  $[3(d^2 + 1)]^{\nu_d}$ . Meanwhile  $\zeta/2 - C_d$  is at most the negative of that quantity. This proves the inequality on both ranges.  $\square$

### 3.19.2 Explicit concentration of the base measure

Write  $A = A_d$ ,  $E = E_d$ , and choose rational points

$$t_0 = 1 - 2^{-A}, \quad t_1 = 1 - 2^{-A-1}, \quad t_2 = 1 - 2^{-A-2}.$$

On  $t \geq t_0$ ,

$$\tau^2 \geq \log^2(2^A - 1) > \frac{(A-1)^2}{4} > 4C_d, \quad (3.76)$$

where  $\log 2 > 1/2$  and  $A = 4C_d + 2$  were used. Below  $t_0$ , the base factor  $b_d$  has integral at most  $A$ : use  $(t(1-t))^{2d} \leq 1$  and, when  $d$  is even,  $\tau < A$ . On  $[t_1, t_2]$ , the interval length is  $2^{-A-2}$ ,  $t(1-t) \geq 2^{-A-3}$ , and  $h_d(\tau) \geq 1$ . Therefore

$$\int_{t_1}^{t_2} b_d(t) dt \geq 2^{-E}. \quad (3.77)$$

**Lemma 3.37** (Large base exponent). *If  $r \geq K_d$ , then  $\mathcal{B}_d(r) > 2C_d$ .*

*Proof.* Let  $s_0 = 2t_0 - 1 = 1 - 2^{1-A}$  and  $s_1 = 2t_1 - 1 = 1 - 2^{-A}$ . The ratio of the unnormalized mass below  $t_0$  to the mass on  $[t_1, t_2]$  is at most

$$A2^E (s_0/s_1)^r \leq A2^E \exp(-2^{-A}r) < 1.$$

For the last inequality, substitute  $r \geq 2^A(E + A + 2)$  and use  $\log A < A$ ,  $\log 2 < 1$ . Thus the normalized mass above  $t_0$  exceeds one half. Equation (3.76) gives  $\mathcal{B}_d(r) > 2C_d$ .  $\square$

### 3.19.3 Explicit concentration in the tail parameter

**Lemma 3.38** (Large coefficient tail). *If  $q \geq Q_d$ , then in both parity types and for every  $k \geq 0$ ,*

$$\mathcal{E}_{d,k,q}[\tau^2] > 2C_d.$$

*Proof.* Within each parity type it suffices, by Proposition 3.34, to use its smallest possible  $k$ , either zero or one. Put  $\delta = q - 2d \geq 1$ . On the part below  $t_0$ , the bracket  $S_\delta$  or  $D_\delta$  is at most  $2t_0^\delta$ , so the unnormalized mass is at most  $2At_0^\delta$ . On  $[t_1, t_2]$ , the factor  $s^k$  is at least  $1/2$ . Both brackets are at least  $t_1^\delta/2$ : for the difference use  $D_\delta = t^\delta(1 - (y/t)^\delta)$  and  $y/t < 1/2$  there. The interval mass is consequently at least  $2^{-E}t_1^\delta/4$ . The ratio of the bad mass to this interval mass is at most

$$8A2^E (t_0/t_1)^\delta \leq 8A2^E \exp(-2^{-A-1}\delta) < 1,$$

by  $\delta \geq 2^{A+1}(E + A + 6)$ . Once again, more than half of the probability lies above  $t_0$ , and (3.76) finishes the proof.  $\square$

*Proof of Theorem 3.22.* If  $k \geq K_d$ , the base comparison and Lemma 3.37 imply  $\mathcal{E}_{d,k,q}[\tau^2] > 2C_d$  for every nonhole  $q \geq 0$ . If  $q \geq Q_d$ , Lemma 3.38 gives the same conclusion for every  $k$ . In either case, Lemma 3.36 gives

$$\mathcal{E}_{d,k,q}[\Pi_d(\tau^2)] \geq \frac{1}{2}\mathcal{E}_{d,k,q}[\tau^2] - C_d > 0.$$

The sign rule (3.70) proves nonvanishing outside the stated symmetry holes. The holes themselves were proved independently. Combined with bulk positivity and the structural zeros, this confines every possible extra zero to  $0 \leq k < K_d$ ,  $0 \leq q < Q_d$ .  $\square$

**Corollary 3.39** (A terminating fixed-depth decision algorithm). *Given  $d \geq 1$ , there is an algorithm that decides whether (3.50) holds for every pair of integer parameters  $(k, M)$  at that depth. For  $d \geq 3$ , it computes the finite rectangle specified by (3.74) using integer polynomial recurrences. If the answer is negative, it returns an exact counterexample.*

*Remark 3.40* (The bound depends on the depth). The bound is uniform in both  $k$  and  $q$ , but depends on  $d$ . It is not a finite verification of the full conjecture over all depths. Nor do the enormous universal constants justify a practical exhaustive computation. Their value is the proved reduction; efficient certificates need sharper analysis, as the next subsection demonstrates.

## 3.20 Depth coordinates: completing depths one through four

### 3.20.1 The first two depths

For  $d = 1, 2$ , the residual polynomial  $\Pi_d$  is identically one. Every positive-weight expectation in (3.70) is strictly positive. Together with bulk positivity and the symmetry theorem, this proves (3.50) at the first two depths. In particular, no higher Peano-kernel argument is needed for depth two. (This is a second route to the zero classification of Theorems 3.16 and 3.17, which prove more: the complete sign pattern. The manuscript's folding (3.69) is the pairing  $t \leftrightarrow 1 - t$  used in their proofs.)

### 3.20.2 A small-certificate criterion for depths three and four

At these depths,  $\Pi_d(\tau^2)$  is an increasing affine function of  $\tau^2$ . The comparisons in Proposition 3.34 apply to its expectation itself, not merely to a lower bound.

**Proposition 3.41** (Three signs and a finite rectangle). *Let  $d \in \{3, 4\}$ . Choose  $R \geq 2$  of parity  $d - 1$  and an integer  $Q \geq 2d + 1$ . Put  $k_s = (d - 1) \bmod 2$  and  $k_a = d \bmod 2$ . Suppose*

$$H_{d,R,2d} > 0, \quad (-1)^Q H_{d,k_s,Q} > 0, \quad (-1)^{Q+1} H_{d,k_a,Q} > 0. \quad (3.78)$$

*Then every nonhole entry with  $k \geq R - 1$  or  $q \geq Q$  is nonzero. Consequently the full depth is settled by testing  $0 \leq k \leq R - 2$ ,  $0 \leq q \leq Q - 1$ .*

*Proof.* Since  $R + d$  is odd and  $2d$  is even, the first sign in (3.78) says exactly that the base expectation  $\mathbb{E}_{\mu_{d,R}} \Pi_d(\tau^2)$  is positive. In a sum case with  $k \geq R - 1$ , parity forces  $k \geq R$ . In a difference case the comparison base has exponent  $k + 1 \geq R$ . Thus all nonhole entries in this  $k$ -range have positive reduced expectation.

The remaining two signs are exactly positive reduced expectations for the smallest sum and difference parity classes at  $q = Q$ ; here  $Q > 2d$ . Increasing  $k$  by two and increasing  $q$  by one preserves that positivity by Proposition 3.34. This covers every  $q \geq Q$ .  $\square$

For  $d = 3$ , take  $(R, Q) = (14, 38)$ . For  $d = 4$ , take  $(R, Q) = (85, 185)$ . The six seed signs and the two rectangles are verified in the next subsection. This proves Theorem 3.23.

### 3.20.3 What the result says about individual coefficients

At depth three the only holes have

$$q = 6, \quad k \text{ odd}; \quad j = -7, \quad N = 2k + 13.$$

At depth four they have

$$q = 8, \quad k \text{ even}; \quad j = -9, \quad N = 2k + 17.$$

(In this part's letters:  $m = 3$ ,  $j$  odd,  $n = 2j + 13$ ; and  $m = 4$ ,  $j$  even,  $n = 2j + 17$  — in both cases  $k = j + m$  even and  $n = 4k - 2j + 1$ .) All other entries at those depths are nonzero, regardless of derivative order. Above the practical  $k$  or  $q$  thresholds their signs are the explicit signs in (3.70) with the reduced expectation replaced by a positive number.

Nonzero does not mean positive. For example, in the  $d = 3, k = 0$  row the reduced expectation is positive at  $q = 0$ , negative for  $1 \leq q \leq 37$ , and positive for all  $q \geq 38$ . These sign changes explain why a naive one-sign induction can fail without producing an integer zero. The exact finite sign data are shipped as `data/02-depth-certificates-sign_thresholds.csv`.

## 3.21 Depth coordinates: the integer certificates and independent checks

### 3.21.1 First recurrence: fixed tail column

Set  $r = 2d + 1 + q$  and work in  $\mathbb{Z}[u]/(u^{d+1})$ . From (3.60),

$$H_{d,k,q} = [u^d](2d + u)^r \mathcal{Q}_k^{(r)}\left(u + \frac{2d - q}{2}\right). \quad (3.79)$$

Although the displayed center may be half-integral, the recurrence only uses the integer linear polynomial  $2u + 2d - q$ . If  $V_k(u)$  denotes the product on the right, truncated after degree  $d$ , then

$$\begin{aligned} V_0(u) &= (2d + u)^r \pmod{u^{d+1}}, \\ V_1(u) &= (2u + 2d - q)V_0(u) \pmod{u^{d+1}}, \\ V_{k+1}(u) &= (2u + 2d - q)V_k(u) + k(k + r)V_{k-1}(u) \pmod{u^{d+1}}. \end{aligned} \quad (3.80)$$

The required integer is the coefficient of  $u^d$  in  $V_k$ . Multiplication by each linear factor of  $V_0$  and every recurrence step are implemented directly on lists of integers. Induction proves that truncation cannot affect any coefficient being tested.

### 3.21.2 Second recurrence: Pascal propagation

Equation (3.63) gives the independent array recurrence

$$H_{d,k+1,q} = 2H_{d,k,q+1} + (k + 2d + 2 + q)H_{d,k,q}, \quad (3.81)$$

with

$$H_{d,0,q} = [u^d](2d + u)^{2d+1+q}.$$

To compute  $0 \leq k \leq K$ ,  $0 \leq q \leq Q$ , initialize the  $k = 0$  row through  $q = Q + K$  and shorten it at each step. This recurrence comes from multiplying by  $(2 + w)$ , rather than from the centered-polynomial three-term recurrence. Every entry of both finite rectangles was recomputed by both methods and compared as an exact integer.

### 3.21.3 Finite obligations and results

| Depth | $k$ maximum | $q$ maximum | Cells  | Symmetry zeros | Nonzero |
|-------|-------------|-------------|--------|----------------|---------|
| 3     | 12          | 37          | 494    | 6              | 488     |
| 4     | 83          | 184         | 15,540 | 42             | 15,498  |
| Total |             |             | 16,034 | 48             | 15,986  |

No additional zero occurs in either rectangle. Every nonzero integer also has a nonzero residue modulo at least one of 1009 and 1013. The first modulus suffices for every nonzero depth-three entry; nineteen depth-four entries require the second modulus. Primality is not needed for this particular witness rule: a nonzero residue modulo any integer greater than one already proves the original integer is nonzero.

The necessary seed signs are

| $d$ | $k$ | $q$ | $\text{sgn } H_{d,k,q}$ | Role            |
|-----|-----|-----|-------------------------|-----------------|
| 3   | 14  | 6   | +                       | central base    |
| 3   | 0   | 38  | +                       | sum tail        |
| 3   | 1   | 38  | −                       | difference tail |
| 4   | 85  | 8   | +                       | central base    |
| 4   | 1   | 185 | −                       | sum tail        |
| 4   | 0   | 185 | +                       | difference tail |

For example,

$$H_{3,14,6} = 9707604961443840000 > 0, \quad H_{3,12,6} = -14670288691200000 < 0.$$

The exact larger seed integers, rather than floating-point approximations, are stored in `data/02-depth-certificates-verification.json`. The preceding central values at  $k = 12$  and  $k = 83$  are negative, so the chosen central exponents are the first positive ones in their relevant parity classes. This observation is not needed for validity of the certificates.

*Remark 3.42* (The rectangles lie inside the range certified in this part; editorial). By  $N = 2k + 2d + 1 + q$ , the depth-three rectangle reaches derivative order  $N \leq 2 \cdot 12 + 7 + 37 = 68$  and the depth-four rectangle  $N \leq 2 \cdot 83 + 9 + 184 = 359$ ; the six seeds sit at  $N = 41, 45, 47$  ( $d = 3$ ) and  $N = 187, 194, 196$  ( $d = 4$ ). All of these lie inside the range  $n \leq 3000$  of Proposition 3.18, whose modular certificate already shows that every rectangle cell outside the reflection family is nonzero. The nonvanishing half of the manuscript’s finite computation is therefore a second, independent confirmation of cells this part had already certified; what Proposition 3.18 does not supply is the six seed *signs* (3.78), which the reduction needs and which the manuscript computes. The contribution of the addition is the proved reduction (Proposition 3.41) that makes a finite set of cells decide every  $n$ , not the finite range itself.

### 3.21.4 Additional cross-check against differentiation

The shipped program independently generates the original lattice using (3.54) through  $N = 80$ . It checks:

| Exact test                                           | Number of cells |
|------------------------------------------------------|-----------------|
| Positive bulk cells                                  | 69,701          |
| Identity $c_N(j, k) = \binom{N}{k} H_{d,k,-j-1}$     | 21,320          |
| First-four-depth zero classification in that lattice | 5,780           |

These are consistency checks on implementation and index translation. They are not the justification for the infinite theorem; that justification is Proposition 3.41 together with the finite obligations just listed.

### 3.21.5 Reproduction and trust boundary

From the report directory, on a copy (see below), run

```
python3 code/02-depth-certificates-minimal_verify.py
python3 code/02-depth-certificates-verify.py
```

The first prints `PASS`; the second prints a JSON report with `"status": "PASS"` and writes nothing. The manuscript's regeneration command, `verify.py -write-data`, writes the unprefixed files `finite_certificate.csv`, `sign_thresholds.csv` and `verification.json` into `data/` beside the shipped prefixed copies, so it should be run only on a copy of the report directory. The code uses only Python's standard library and arbitrary-precision integers. It uses no floating point, symbolic algebra library, random search, external service, or precomputed presumed support formula. Failures raise explicit exceptions, even if Python assertions are disabled. The CSV file supplies the sign and both modular residues for every rectangle cell.

The finite checks were run for the manuscript and returned `PASS`; they were rerun on a copy during the intake and passed again, with byte-identical CSV files. Ordinary program execution is still not a proof-assistant kernel check. A formal version would need the polynomial-model bridge, correctness of one recurrence, and a proved modular checker; the analytic lemmas would also need formalization. No such formalization is claimed. The two exact recurrences reduce the risk of implementation or indexing error, but do not replace independent mathematical review.

## 3.22 Depth coordinates: an all-depth bound on the number of zeros

The branch-cut formula also controls how often cancellation can occur, without any fixed-depth classification assumption. For fixed  $d \geq 1$  and  $k \geq 0$ , define for real  $q > -1$

$$\mathcal{F}_{d,k}(q) = \int_0^1 t^q (1-t)^{2d} (2t-1)^k R_d \left( \log \frac{1-t}{t} \right) dt. \quad (3.82)$$

For integer  $q \geq 0$ , its zeros are precisely those of  $H_{d,k,q}$ .

**Lemma 3.43** (Mellin zero bound from sign changes). *Let  $W$  be a nonzero continuous real weight on  $(0, 1)$  with finitely many zeros and exactly  $J$  sign changes. Assume that  $t^q W(t)$  is integrable for every  $q > -1$ . Then  $F(q) = \int_0^1 t^q W(t) dt$  has at most  $J$  distinct real zeros in  $(-1, \infty)$ . Here a sign change refers to a crossing of sign, not an even-multiplicity zero of the weight.*

*Proof.* Suppose  $q_0 < \dots < q_J$  are zeros. There is a nonzero linear combination  $\psi(t) = \sum_{i=0}^J a_i t^{q_i}$  vanishing at the  $J$  sign-change points of  $W$ : solve the corresponding homogeneous system of  $J$  linear equations. A nonzero linear combination of  $J+1$  distinct real powers has at most  $J$  positive zeros, counting multiplicities. This follows inductively by dividing by the smallest power, differentiating, and applying Rolle's theorem. It also establishes linear independence of the powers.

Hence the prescribed zeros of  $\psi$  are simple and there are no others. After changing the overall sign,  $\psi W$  is nonnegative throughout the interval and positive on a set of positive measure.

Therefore

$$0 < \int_0^1 \psi(t)W(t) dt = \sum_{i=0}^J a_i F(q_i) = 0,$$

a contradiction. Continuity and the finite zero set ensure strict positivity of  $\psi W$  on an interval, as needed above.  $\square$

**Theorem 3.44** (At most  $d$  cancellations in a fixed row). *The number of distinct real zeros of  $\mathcal{F}_{d,k}$  on  $(-1, \infty)$  is at most*

$$J(d, k) = \begin{cases} d-1, & k \text{ even,} \\ d, & k \text{ odd and } d \text{ odd,} \\ d-2, & k \text{ odd and } d \text{ even.} \end{cases} \quad (3.83)$$

*In particular, for fixed  $k, d$  there are at most  $d$  integer zeros in the entire negative-region tail. If a symmetry zero is present, it consumes one of these permitted zeros.*

*Proof.* The polynomial  $R_d$  has  $d-1$  distinct real roots. The logarithmic argument in (3.82) is strictly decreasing from  $+\infty$  to  $-\infty$ , so these initially give  $d-1$  sign changes. If  $k$  is even,  $(2t-1)^k$  adds no sign change. If  $k$  is odd and  $d$  is odd,  $R_d(0) \neq 0$ , and the factor adds one sign change at  $t = 1/2$ . If  $k$  is odd and  $d$  is even,  $R_d$  has a simple zero at zero; its sign change at  $t = 1/2$  is canceled by the odd power of  $2t-1$ , leaving  $d-2$ . The remaining factors are positive. Apply Lemma 3.43.  $\square$

The theorem is a zero-count statement, not an assertion that all zeros are simple or that the upper bound is always attained. It also does not rule out an integer zero by locating nearby real zeros; the finite certificates are what supply that additional information at depths three and four. In this part's letters, it bounds by  $J(m, j)$  the number of  $n$  at which a fixed  $(j, m)$  diagonal can vanish with  $n \geq 2k+1$ .

### 3.23 Depth coordinates: consequences for the term count

#### 3.23.1 An unconditional upper bound (second route to Theorem 3.14)

The triangular support window and (3.57) imply that, in negative columns, the top  $d=0$  term is absent. Removing those structural zeros, the number of remaining cells is  $2\rho^2 + 2\rho + 1$  when  $N = 2\rho$  and  $2\rho^2$  when  $N = 2\rho - 1$ . For even  $N$  no symmetry hole is possible. For odd  $N = 2\rho - 1$ , a hole has

$$k = \rho - 2d - 1 \geq 0, \quad d \equiv \rho - 1 \pmod{2}.$$

Counting these depths gives  $\lfloor \rho/4 \rfloor$  holes (or check the four residue classes of  $\rho$ ). Thus, independently of unknown nonvanishing,

$$a(2\rho) \leq 2\rho^2 + 2\rho + 1, \quad a(2\rho - 1) \leq 2\rho^2 - \lfloor \rho/4 \rfloor. \quad (3.84)$$

The conjecture is exactly equality in these bounds. This is Theorem 3.14 and its equality clause, with the hole count  $\lfloor \rho/4 \rfloor = \lfloor (n+1)/8 \rfloor$  of Lemma 3.13 counted by depth instead of by  $k$ .

### 3.23.2 A certified lower bound from the proved regions

Let  $\beta(N)$  count the proved positive bulk cells, including  $d = 0$  where it is nonzero. Direct summation gives

$$\beta(2\rho) = \frac{(\rho+1)(3\rho+2)}{2}, \quad \beta(2\rho-1) = \frac{3\rho^2+3\rho-2}{2} \quad (\rho \geq 1). \quad (3.85)$$

These are exactly the two cases of  $\Lambda(N)$  in (3.3): the bulk is the region of Theorem 3.7, and Corollary 3.8 counts the same cells. For a fixed depth  $1 \leq d \leq 4$ , the cells outside this bulk have  $q \geq 1$ . The identity  $N = 2k + 2d + 1 + q$  shows that their number before holes is  $\max(0, \lfloor N/2 \rfloor - d)$ . Define

$$\chi_d(N) = \mathbf{1}\{N \geq 4d + 1 \text{ and } N \equiv 2d + 1 \pmod{4}\}. \quad (3.86)$$

This is exactly the number, zero or one, of symmetry holes at that depth and derivative order.

**Corollary 3.45** (Explicit unconditional lower bound). *For all  $N \geq 0$ ,*

$$a(N) \geq \beta(N) + \sum_{d=1}^4 \max(0, \lfloor N/2 \rfloor - d) - \sum_{d=1}^4 \chi_d(N). \quad (3.87)$$

*In particular, the lower bound has leading term  $3N^2/8$ , whereas the conjectured upper bound has leading term  $N^2/2$ .*

*Proof.* Count the positive bulk and the disjoint first-four-depth cells with  $q \geq 1$ . By Theorem 3.23, only the indicated holes must be removed. The counts give (3.87).  $\square$

For example, for even  $N = 2\rho$  with  $\rho \geq 4$ , the lower bound simplifies to  $(3\rho^2 + 13\rho - 18)/2$ . It agrees with the conjectured value at  $N = 8, 10$  and is 84 at  $N = 12$ , compared with the conjectured upper value 85. The unresolved cell there has depth five. These examples illustrate the precise boundary of the argument; they are not evidence that depth five fails. No asymptotic equivalence  $a(N) \sim N^2/2$  is deduced here.

*Remark 3.46* (How much this improves  $\Lambda$ ; editorial). Since  $\beta = \Lambda$ , the gain of (3.87) over Theorem 3.1 is the sum of the last two terms. For even  $N \geq 8$  no hole occurs and every maximum is attained by  $N/2 - d \geq 0$ ; for odd  $N \geq 17$  each maximum is  $(N-1)/2 - d > 0$  and exactly two indicators equal one (at  $d = 1, 3$  if  $N \equiv 3$ , at  $d = 2, 4$  if  $N \equiv 1 \pmod{4}$ ). Hence

$$a(N) \geq \begin{cases} \Lambda(N) + 2N - 10 = \frac{3N^2 + 26N - 72}{8}, & N \text{ even, } N \geq 8, \\ \Lambda(N) + 2N - 14 = \frac{3N^2 + 28N - 111}{8}, & N \text{ odd, } N \geq 17. \end{cases}$$

The leading constant  $3/8$  is unchanged; the linear term improves. Below these thresholds the gain is read off (3.87) directly; the bound equals  $U(N)$  for every  $N \leq 11$ . Of the even gain  $2N - 10$ , the part  $N - 3$  from depths one and two already followed from Theorems 3.16–3.17, although Part 3 did not add it to  $\Lambda$ ; the part  $N - 7$  from depths three and four is new. At odd  $N \geq 17$  the split is  $(N - 5) + (N - 9)$ . The two closed forms were checked against a direct count of the proved cells for every  $N < 400$ .  $\Lambda(n)$  remains the lower bound stated in Theorem 3.1; the certified range  $n \leq 3000$  of Proposition 3.18 makes both bounds irrelevant there, since  $a(n) = U(n)$  is known for those  $n$ .

### 3.24 Depth coordinates: further research questions

#### 3.24.1 Depths five and six: a stronger moment criterion

The explicit quadratics in (3.68) suggest a practical next step. Since  $\mathbb{E}[Z^2] \geq (\mathbb{E}Z)^2$ , where  $Z = \tau^2$ , sufficient conditions for positive reduced expectation are

$$\begin{aligned} d = 5 : \quad \mathbb{E}Z &> \pi^2(1 + 2/\sqrt{5}), \\ d = 6 : \quad \mathbb{E}Z &> 3\pi^2. \end{aligned} \tag{3.88}$$

These follow by applying the variance inequality to the two quadratic polynomials and placing the mean above their largest root. Unlike the crude affine minorant, these criteria retain the exact quadratic structure. Their usefulness is that  $\mathbb{E}Z$  is monotone under all the proved likelihood-ratio comparisons even though  $\Pi_5$  and  $\Pi_6$  are not monotone on the entire positive axis.

**Question 1.** Determine certified practical  $k$  and  $q$  cutoffs from (3.88), and settle the resulting finite rectangles at depths five and six. A successful computation must be accompanied by the mean-moment bounds; an unbounded inference from a numerical scan would not suffice. (Editorial: the range  $n \leq 3000$  of Proposition 3.18 already certifies nonvanishing of every depth-five and depth-six cell with  $n \leq 3000$ ; what is missing is the reduction, and the seed signs it will require.)

#### 3.24.2 Sharpening the fixed-depth reduction

**Question 2.** How small can uniform fixed-depth cutoffs be made as functions of  $d$ ? The constants in (3.74) are intentionally wasteful. The largest residual root is  $\pi^2 \cot^2(\pi/d)$ , of order  $d^2$ , so endpoint concentration on a logarithmic scale is a natural quantity to analyze. This observation does not yet prove an optimal cutoff growth rate.

**Question 3.** Can lower bounds involving the first few moments of  $\tau^2$ , or positivity in an orthogonal-polynomial basis adapted to  $\mu_{d,r}$ , replace the affine minorant for all depths? The target is a small finite certificate rather than an enormous merely terminating algorithm.

#### 3.24.3 The remaining full conjecture

**Question 4.** Prove that every fixed-depth rectangle has only the symmetry zeros, uniformly in depth, or exhibit the first exact counterexample. The present results show where additional zeros must be searched for at each depth, but they do not supply an induction in  $d$ . The polynomial  $\Pi_d$  acquires additional oscillations, which must be addressed rather than ignored. (This is Conjecture 3.19 restricted to  $m \geq 5$ .)

**Question 5.** Can the real-zero bound (3.83) be combined with arithmetic separation to exclude integer zeros uniformly? The Mellin argument limits the number of candidates in each row, but integer nonvanishing needs a second ingredient. An effective distance-from-integers theorem for its zeros would be particularly valuable.

**Question 6.** Establish the conjectured leading term  $a(N) \sim N^2/2$  without first proving every individual nonvanishing claim. This would require bounding the total number of extra zeros across growing depths. The fixed-row bound alone does not give such a bound after summing over the whole triangular array.

### 3.24.4 Formal proof and symbolic computation

**Question 7.** Formalize the polynomial factorization, bulk theorem, and integer certificate in the existing A290268 Lean development. The algebraic route has a small interface: coefficient extraction, falling factorials, the recurrence (3.61), and a finite modular nonzero checker. The branch-cut and likelihood-ratio proofs are a separate analytic layer. Keeping these layers explicit would avoid confusing a verified finite table with a verified infinite theorem. (Section 3.26.3 lists what that development has and which of its hypotheses these results would discharge.)

**Question 8.** Extend the method to derivative expansions of  $x^{x^r}$  for integer  $r \geq 3$ , or to other exponential-logarithmic families. The correct depth generating function and the symmetry center must first be rederived. The success of the square-exponent case does not justify assuming an analogous support quasipolynomial elsewhere.

### 3.24.5 Suggested order of work

The most direct continuation is to formalize the bulk factorization while independently implementing certified moment bounds for depths five and six. After those bounds determine practical finite rectangles, the existing integer recurrence can generate exact certificates immediately. Only then should one search for a depth-uniform induction or arithmetic zero-separation theorem. This separates achievable finite obligations from the genuinely new ingredient needed for the complete conjecture.

## 3.25 Depth coordinates: conclusion, verifier and source audit

### 3.25.1 Conclusion of the addition

The obstruction in A290268 is not merely the existence of a few symbolic cancellations. Its coefficients combine a consecutive-root factor with an imaginary-root factor, and their tail values are signed Mellin moments of an explicitly oscillating logarithmic polynomial. Making both structures visible yields three concrete advances in the manuscript’s account: positivity in the whole bulk at every depth, an explicit finite decision problem at every fixed depth, and a complete nonvanishing classification at depths three and four, supplementing the first two depths. Of these, the first was already Theorem 3.7 of this part, and the second and third are new.

The 16,034-cell verification is logically effective because the manuscript first proves that it is exhaustive for those two infinite-parameter strata. It is not extrapolation from a finite sample. The full counting formula remains a separate target at depths five and higher, with the remaining questions and the precise proof boundary now substantially narrowed.

### 3.25.2 A compact standalone finite verifier

The following standalone version verifies the two exact rectangles and six seed signs; it is shipped, byte for byte, as `code/02-depth-certificates-minimal_verify.py` (whose docstring calls this listing “Appendix A of `article.tex`”, the manuscript’s numbering). The fuller program `code/02-depth-certificates-verify.py` additionally checks the independent Pascal recurrence, produces the CSV witnesses, and cross-checks the original differentiation lattice.

```
def must(test):
```

```

    if not test:
        raise ArithmeticError("certificate failure")

def column(d, q, K):
    r = 2*d + 1 + q
    a = [1] + [0]*d
    for j in range(r):
        a = [(2*d-j)*a[i] + (a[i-1] if i else 0)
              for i in range(d+1)]
    out = [a[d]]
    if K == 0:
        return out
    b = [(2*d-q)*a[i] + (2*a[i-1] if i else 0)
          for i in range(d+1)]
    out.append(b[d])
    for k in range(1, K):
        c = [(2*d-q)*b[i] + (2*b[i-1] if i else 0)
              + k*(k+r)*a[i] for i in range(d+1)]
        out.append(c[d])
        a, b = b, c
    return out

for d, R, Q in [(3, 14, 38), (4, 85, 185)]:
    for q in range(Q):
        for k, value in enumerate(column(d, q, R-2)):
            hole = (q == 2*d and (k+d) % 2 == 0)
            must((value == 0) == hole)
            if not hole:
                must(value % 1009 != 0 or value % 1013 != 0)
        must(column(d, 2*d, R)[R] > 0)
        ks, ka = (d-1) % 2, d % 2
        must((-1)**Q * column(d, Q, ks)[ks] > 0)
        must((-1)**(Q+1) * column(d, Q, ka)[ka] > 0)
print("PASS")

```

The loop bounds are conclusions of the analytic reduction, not values chosen merely because the program happened to finish. Each arithmetic step is the recurrence (3.80) in the ring  $\mathbb{Z}[u]/(u^{d+1})$ . (The Python variable  $j$  in `column` is a loop index over the factors of the falling factorial, not either meaning of  $j$  in Table 3.) Thus the program is a finite, reproducible arithmetic certificate with a fully specified mathematical meaning.

### 3.25.3 Source audit

The manuscript's inspected repository snapshot is `afb2d1227d8bc5df3beffc1463e958db3544960b`. At that snapshot the A290268 research note explicitly identified higher-depth negative-region nonvanishing and the general- $k$  bulk as open parts of its analysis; the general- $k$  bulk was in fact already Theorem 3.7 of this report, which the manuscript did not inspect. The `Main.lean` file makes the support characterization a hypothesis. The proof in the manuscript does not depend on uninspected formal claims or on the consistency of unrelated parts of the repository.

The manuscript's scoped directory listing contains the core, closed-form, counting, structural, table, and main modules. The research note also discusses other module names and development intentions. Those descriptions are not treated as evidence that every discussed file is present or that every corresponding theorem is kernel-certified. (At the intake, the three module names

that note lists beyond these six, A290268.Hole, A290268.DepthOne and A290268.Series, were confirmed absent from the repository’s entire history.)

External checks included the live OEIS entry, the NIST generating function, and the primary paper by Štampach, whose bibliographic data the manuscript cross-checked against a paper of R. B. Paris (bibliographic verification only, recorded in 02-depth-certificates-SOURCE\_AUDIT.md). Search results did not supply a prior published proof of the specific fixed-depth statements proved here. That limited search is not a proof of universal novelty.

## 3.26 Depth certificates: provenance, files and formal status

### 3.26.1 Provenance

Subsections 3.14–3.26 were built from **one** manuscript: batch 42, manuscript 06, archive ProveIt\_A290268\_Finite\_Certificates.zip (22 pages, dated 29 September 2026), pinned to ProveIt commit afb2d1227d8bc5df3beffc1463e958db3544960b, and placed in this report by commit 3609d0473. Its Section 1 is split between Subsection 3.14 (main results, novelty) and the opening of Subsection 3.15 (the conjecture, linear independence); its Sections 2–11 map one for one onto Subsections 3.15–3.24; its Section 12 and its two appendices form Subsection 3.25. Every theorem, proposition, lemma, corollary, definition, remark and question of the manuscript is printed; none was weakened or strengthened. The editorial choices were these.

- It is placed as subsections at the end of Part 3 rather than as a new part, so that no existing section, theorem or equation number moves.
- Its bulk positivity, symmetry holes, depths one and two, upper bound, depth bridge, linear independence and branch-cut formula are marked as second routes to, or re-indexings of, results of this part, which the manuscript did not read. The editorial Remarks 3.31, 3.42 and 3.46 and the parenthetical translations are additions of the intake, not of the manuscript.
- The manuscript’s letters are kept, with the renamings of Subsection 3.14.1; its reproduction commands are rewritten to the shipped file names; its archive table is replaced by the file list below; its OEIS citation is merged with this part’s.
- Its statement that the finite checks “have actually been run” is kept, with the intake’s rerun recorded beside it.

Not shipped: the manuscript’s `article.tex`, `article.pdf`, `README.md` and `SHA256SUMS.txt` (verified at placement, 10 of 10 entries); the archive itself survives in the history of commit 8315d24e3.

### 3.26.2 Files

The manuscript’s code and data are shipped with the prefix 02-depth-certificates-, byte-identical to the delivery:

| Shipped path (delivery name)                                                         | Purpose                                                                              |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <code>code/02-depth-certificates-verify.py</code><br>( <code>code/verify.py</code> ) | Full exact verifier: both recurrences, the lattice cross-check, and data generation. |

| Shipped path (delivery name)                                                                                   | Purpose                                                                  |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <code>code/02-depth-certificates-minimal_verify.py</code><br>( <code>code/minimal_verify.py</code> )           | The standalone verifier of Subsection 3.25.2.                            |
| <code>code/02-depth-certificates-Makefile</code><br>( <code>Makefile</code> )                                  | The delivery's make targets; see the README before using it.             |
| <code>data/02-depth-certificates-finite_certificate.csv</code><br>( <code>data/finite_certificate.csv</code> ) | All 16,034 rectangle cells with signs and residues modulo 1009 and 1013. |
| <code>data/02-depth-certificates-sign_thresholds.csv</code><br>( <code>data/sign_thresholds.csv</code> )       | Observed sign transitions of the reduced expectation in each tested row. |
| <code>data/02-depth-certificates-verification.json</code><br>( <code>data/verification.json</code> )           | Executed-check summary and the exact seed integers.                      |
| <code>02-depth-certificates-SOURCE_AUDIT.md</code><br>( <code>SOURCE_AUDIT.md</code> )                         | Source identity, inspected scope, and limitations.                       |

The two CSV files keep the delivered CRLF line endings.

### 3.26.3 Formal status

Nothing in this addition is formalized, and its placement beside a Lean development confers no formal status. The Lean development `Combinatorics/DerivativeExpansions/A290268/Lean` (library A290268, namespace `LeanProofs.A290268`, imported by the root `ProveIt.lean`) relates to it as follows.

- `coeff n j k` (`Core.lean`) is the recurrence (3.54) in the same coordinates; `S n` (`Count.lean`) is the conjectured support, and `card_S` proves  $|S_n|$  equal to the closed form unconditionally.
- The manuscript relies on the *statements* of the three conditional theorems of `Main.lean`: `a_eq_closedForm_of_support` (hypothesis: the support of `coeff n` is `S n`), `a_eq_closedForm_of_two_sided` (the structural and nonvanishing inclusions as separate hypotheses) and `a_eq_closedForm_of_support_all` (the support hypothesis at every order). It reads them, correctly, as conditional.
- `Structural.mem_S_of_coeff_ne_zero` proves the structural inclusion from a hypothesis `hhole`: vanishing on the line  $N$  odd,  $-j \equiv N \pmod{4}$ ,  $-j \geq 3$ ,  $2(-j) - 1 \leq N$ ,  $2k = N + 2j + 1$ . With  $-j = 2d + 1$  that line is exactly the hole family of Theorem 3.29, and so of Theorem 3.11; either proof would discharge `hhole` on paper. Neither is formalized, and the module `A290268.Hole` that the declaration's docstring names as the discharge does not exist in the repository.
- Theorem 3.23 would discharge the nonvanishing inclusion only on cells of depth at most four; depths  $d \geq 5$  remain, so the `Main.lean` theorems stay conditional.
- The only unconditional statement about the sequence itself is `Table.a_values_le_53`,  $a(n)$  equal to the closed form for  $n \leq 53$ , a `native_decide` certificate.

## 4 $x^{x^x}$ : OEIS A281434

**Framework.** This part instantiates Theorem 1.2(a) with the three-variable frame  $u = x^x$ ,  $v = x^{-1}$ ,  $z = \log x$ ; Proposition 4.2 below is that instance, derived here independently, and it is the only one of the three whose frame is not two-dimensional. Because  $u = x^x$  is not of the form  $x^\lambda$  with  $\lambda$  constant, the independence statement this part needs is *not* a special case of Theorem 1.4; the asymptotic argument that replaces it is given in full below.

**Notation.** In this part  $z$  **denotes**  $\log x$ , not  $x^2$ ; compare Part 3. See Section 1.5. Throughout this part  $a(n)$  denotes the term count for  $x^{x^x}$ .

**Abstract of the original report.** Let  $a(n)$  be the number of nonzero monomials in the fully expanded  $n$ th derivative of  $x^{x^x}$ , after collecting identical terms. We replace repeated symbolic differentiation by a seven-transition integer recurrence on three exponents. It computes the prefix through  $n$  in  $O(n^4)$  integer arithmetic operations and  $O(n^3)$  live integers. A faster implementation performs the recurrence modulo a machine-size integer and verifies every possible zero by an independent exact coefficient formula; this method is deterministic, not a probabilistic zero test. We prove the explicit bounds

$$\frac{7n^3 + 39n^2 + 26n + 3\varepsilon_n(n-1)}{24} \leq a(n) \leq \frac{2n^3 + 3n^2 + 4n}{3}, \quad n \geq 1,$$

where  $\varepsilon_n$  is 1 for odd  $n$  and 0 for even  $n$ . In particular,  $a(n) = \Theta(n^3)$ . The upper bound counts an explicit lattice envelope; the lower bound uses nonvanishing leading coefficients and high-multiplicity roots of coefficient polynomials. We also identify an unavoidable family of logarithm-free cancellations and an infinite family with positive logarithm exponent. The stronger statement  $a(n) \sim \frac{2}{3}n^3$ , and the still stronger linear-deficit conjecture, are not proved here. Exact code, tests, terms through  $n = 100$ , and selected computations through  $n = 200$  accompany this article.

### 4.1 The problem and the scope of the results

The sequence A281434 is defined in the OEIS as the number of terms in the fully expanded  $n$ th derivative of  $x^{x^x}$ , with offset 0 [3]. Exponentiation is right-associated. Thus the function in question is

$$f(x) = x^{(x^x)} = \exp(x^x \log x), \quad x > 0,$$

not  $(x^x)^x = x^{x^2}$ . Its initial term counts are

$$1, 3, 12, 30, 64, 113, 188, 285, 415, 577, 780, 1017, 1312, \dots$$

The positivity restriction supplies an unambiguous real logarithm; all subsequent coefficient manipulations are exact polynomial identities.

The phrase “fully expanded” must include collecting like terms and deleting zero coefficients. Counting product-rule branches is a different problem. Even recording every monomial that could be reached by differentiation gives an overcount: distinct contributions can cancel exactly. For  $n \geq 1$ , define

$$B(n) = \frac{2n^3 + 3n^2 + 4n}{3}, \quad \Delta(n) = B(n) - a(n). \quad (4.1)$$

The polynomial  $B(n)$  is A037237( $n - 1$ ) [6]. The related OEIS entry A352697 records the differences  $\Delta(n)$  and conjectures both  $\Delta(n) = O(n)$  and infinitely many zero differences [5]. These observations are useful motivation, but neither is needed by our algorithms or proofs.

**Theorem 4.1** (Main results). *There is a deterministic exact algorithm for A281434 requiring  $O(n^4)$  integer arithmetic operations and  $O(n^3)$  stored integers to compute  $a(0), \dots, a(n)$ . There is also a machine-modular implementation for a single  $a(n)$ , with exact verification of every candidate cancellation. For  $n \geq 1$ ,*

$$L(n) := \frac{7n^3 + 39n^2 + 26n + 3\varepsilon_n(n-1)}{24} \leq a(n) \leq B(n), \quad \varepsilon_n = \frac{1 - (-1)^n}{2}. \quad (4.2)$$

Consequently

$$a(n) = \Theta(n^3), \quad \lim_{n \rightarrow \infty} \frac{\log a(n)}{\log n} = 3. \quad (4.3)$$

The distinction between a growth *order* and an asymptotic *equivalent* is important. Equation (4.3) is unconditional. An equivalent  $a(n) \sim \frac{2}{3}n^3$  would additionally require  $\Delta(n) = o(n^3)$ ; the present bounds do not establish that. The computations are consistent with a much smaller deficit, but this numerical observation will not be substituted for a nonvanishing theorem.

## 4.2 A canonical differential-polynomial representation

### 4.2.1 Three auxiliary variables

Put

$$u = x^x, \quad v = x^{-1}, \quad z = \log x, \quad A = z(z+1).$$

The three elementary derivatives and the logarithmic derivative of  $f$  are

$$u' = u(z+1), \quad v' = -v^2, \quad z' = v, \quad \frac{f'}{f} = u(A+v). \quad (4.4)$$

There is a unique polynomial  $P_n \in \mathbb{Z}[u, v, z]$  such that

$$f^{(n)}(x) = f(x)P_n(x^x, x^{-1}, \log x). \quad (4.5)$$

Write

$$P_n(u, v, z) = \sum_{k,j,\ell \geq 0} c_n(k, j, \ell) u^k v^j z^\ell. \quad (4.6)$$

The corresponding expanded term of the original derivative is

$$c_n(k, j, \ell) x^{x^x + kx - j} (\log x)^\ell.$$

Therefore

$$a(n) = \#\{(k, j, \ell) : c_n(k, j, \ell) \neq 0\}. \quad (4.7)$$

In particular, a negative coefficient contributes one term, not minus one.

### 4.2.2 Why the representation is unambiguous

Suppose a finite linear combination of  $u^k v^j z^\ell$ , after substitution, vanishes for all sufficiently large positive  $x$ . Group it by  $k$  and choose the largest  $k$  having a nonzero Laurent-polynomial coefficient in  $x$  and  $\log x$ . Division by  $x^{kx}$  makes every smaller- $k$  group exponentially small relative to that coefficient. Within the largest group, choose the smallest  $j$ , and then the largest  $\ell$  with nonzero coefficient. Its asymptotic magnitude cannot be canceled by the remaining Laurent-logarithmic terms. This is a contradiction. Hence the displayed monomials are linearly independent in the required finite sense.

This argument is also a safeguard against a representation-dependent term count: we are counting a specified canonical basis, not the summands in an arbitrarily factored expression.

### 4.2.3 The recurrence and its proof

**Proposition 4.2** (Differential recurrence). *Starting from  $P_0 = 1$ , the polynomials satisfy*

$$P_{n+1} = u(z^2 + z + v)P_n + u(z + 1)\partial_u P_n - v^2\partial_v P_n + v\partial_z P_n. \quad (4.8)$$

*Proof.* Differentiate (4.5), apply the product rule, and substitute (4.4). The result is exactly (4.8). The base case is  $f^{(0)} = f$ . Induction proves the identity at every order, and also proves integrality of every coefficient.  $\square$

For one coefficient  $c = c_n(k, j, \ell)$  the seven contributions are as follows. All contributions with the same destination must be added before zero coefficients are removed.

| Destination at order $n + 1$ | Contribution | Source in the derivative    |
|------------------------------|--------------|-----------------------------|
| $(k + 1, j, \ell + 2)$       | $c$          | $f'/f$ : $uz^2$             |
| $(k + 1, j, \ell + 1)$       | $c$          | $f'/f$ : $uz$               |
| $(k + 1, j + 1, \ell)$       | $c$          | $f'/f$ : $uv$               |
| $(k, j, \ell + 1)$           | $kc$         | differentiation of $u^k$    |
| $(k, j, \ell)$               | $kc$         | differentiation of $u^k$    |
| $(k, j + 1, \ell)$           | $-jc$        | differentiation of $v^j$    |
| $(k, j + 1, \ell - 1)$       | $\ell c$     | differentiation of $z^\ell$ |

Transitions with a zero multiplier are omitted. In particular, no negative logarithm exponent is ever introduced.

### 4.2.4 The first cancellation

The first two polynomials have the compact forms

$$P_1 = u(A + v), \quad (4.9)$$

$$P_2 = u^2(A + v)^2 + u((z + 1)A + (3z + 2)v - v^2). \quad (4.10)$$

Expansion gives  $a(1) = 3$  and  $a(2) = 12$ . At the next order, the coefficient polynomial of  $uv^2$  is

$$-(z + 1) + (\partial_z - 1)(3z + 2) = -4z.$$

Its constant term vanishes. The candidate envelope has 31 cells at order 3, but the derivative has 30 terms. A Boolean reachability calculation would fail at this very early example.

## 4.3 The support envelope and the cubic upper bound

**Definition 4.3.** For  $n \geq 1$ ,  $1 \leq k \leq n$ , and  $0 \leq j \leq n$ , put

$$m_{n,k,j} = \max(0, k - j), \quad (4.11)$$

$$U_{n,k,j} = \begin{cases} n + k, & j = 0, \\ \min(n + k - j - 1, 2n - 2j), & j \geq 1. \end{cases} \quad (4.12)$$

Let  $\mathcal{E}_n$  consist of the integer triples with these ranges and  $m_{n,k,j} \leq \ell \leq U_{n,k,j}$ .

**Proposition 4.4** (Support containment). *For  $n \geq 1$ ,  $\text{supp } P_n \subseteq \mathcal{E}_n$ .*

*Proof.* Consider a nonzero differentiation path, and let  $a, b, c, d, e, r, s$  count its uses of the seven transitions in the displayed table, in that order. Then

$$k = a + b + c, \quad j = c + r + s, \quad \ell = 2a + b + d - s, \quad n = a + b + c + d + e + r + s.$$

In particular,

$$\begin{aligned} n - j &= a + b + d + e, \\ \ell - (k - j) &= a + d + r \geq 0, \\ 2(n - j) - \ell &= b + d + 2e + s \geq 0, \\ n + k - j - \ell &= b + c + e + s \geq 0. \end{aligned}$$

If  $j > 0$ , at least one transition of type  $c$  or  $s$  must occur: the transition of type  $r$  has multiplier  $-j$  and cannot create the first positive  $j$ . Thus  $n + k - j - \ell \geq 1$  when  $j > 0$ . The remaining ranges follow immediately from the transition table and the first differentiation. Every coefficient is a sum over such paths, so cancellation can remove cells but cannot add a cell outside these constraints.  $\square$

**Proposition 4.5** (Size of the envelope). *For  $n \geq 1$ ,  $\#\mathcal{E}_n = B(n)$ .*

*Proof.* For  $j = 0$  there are  $n + 1$  logarithm exponents for each of  $n$  values of  $k$ , giving  $n(n + 1)$  cells. For  $j \geq 1$ , set  $d = n - j$ , so  $0 \leq d \leq n - 1$ . The number of cells in this  $j$ -slice is

$$\sum_{k=1}^n (\min(d + k, 2d + 1) - \max(0, k - n + d)) = n(2d + 1) - d(d + 1).$$

Indeed, replacing each minimum by  $2d + 1$  overcounts by  $d(d + 1)/2$ , and the sum of the subtracted maxima is another  $d(d + 1)/2$ . Consequently

$$\begin{aligned} \#\mathcal{E}_n &= n(n + 1) + \sum_{d=0}^{n-1} (n(2d + 1) - d(d + 1)) \\ &= n(n + 1) + n^3 - \frac{(n - 1)n(n + 1)}{3} \\ &= \frac{2n^3 + 3n^2 + 4n}{3}. \end{aligned}$$

$\square$

This proves the upper bound in (4.2). More precisely,  $\Delta(n)$  is exactly the number of zero coefficients *inside*  $\mathcal{E}_n$ . The polynomial  $B(n)$  is not being guessed by interpolation: it is the exact count of a proved finite set of candidate monomials.

#### 4.4 An independent exact formula for every coefficient

The recurrence is already an exact algorithm. An independent formula is valuable for two other reasons: it certifies possible cancellations after a modular computation, and it proves that every  $(k, j)$  row is nonzero.

#### 4.4.1 Notation for Stirling numbers and auxiliary polynomials

We use signed Stirling numbers  $s(n, k)$  of the first kind and Stirling numbers  $S(n, k)$  of the second kind, with the conventions

$$y^n = \prod_{i=0}^{n-1} (y - i) = \sum_k s(n, k) y^k, \quad y^n = \sum_k S(n, k) y^{\underline{k}}.$$

These conventions agree with DLMF §26.8 [12]. In particular,  $s(n, k)$  has sign  $(-1)^{n-k}$  and is nonzero for  $1 \leq k \leq n$ ;  $S(n, k) > 0$  on the same range.

For nonnegative  $d, p, q$ , define the noncentral Stirling numbers

$$T_d(p; q) = \sum_{r=0}^d \binom{d}{r} q^{d-r} S(r, p). \quad (4.13)$$

They can be computed without rational arithmetic:

$$T_0(p; q) = \mathbf{1}_{p=0}, \quad T_{d+1}(p; q) = (p + q)T_d(p; q) + T_d(p - 1; q). \quad (4.14)$$

Indices  $p < 0$  or  $p > d$  have value zero. The useful boundary values are

$$T_d(0; q) = q^d, \quad T_d(d; q) = 1, \quad T_d(p; 0) = S(d, p), \quad T_d(p; 1) = S(d + 1, p + 1),$$

where  $0^0 = 1$ . Equivalently,

$$e^{-w} \sum_{m \geq 0} (m + q)^d \frac{w^m}{m!} = \sum_{p=0}^d T_d(p; q) w^p. \quad (4.15)$$

To verify this identity, expand  $(m + q)^d$ , express powers of  $m$  in falling factorials, and sum  $\sum_m m^p w^m / m! = w^p e^w$ .

Define

$$Q_{j,r}(\beta) = \beta^{\underline{j}} (\beta + 1)^{\overline{r}} = \prod_{i=0}^{j-1} (\beta - i) \prod_{i=1}^r (\beta + i). \quad (4.16)$$

These integer polynomials are easy to build by multiplying by linear factors. Notice that  $\deg Q_{j,r} = j + r$ .

#### 4.4.2 A parameterized derivative identity

For parameters  $\alpha, \beta$ , Taylor expansion with the increment  $xt$  gives

$$\frac{(x + xt)^{\alpha(x+xt)+\beta}}{x^{\alpha x + \beta}} = (1 + t)^\beta \exp(\alpha x [tz + (1 + t) \log(1 + t)]). \quad (4.17)$$

Comparing coefficients of  $t^n$  and then of  $(\alpha x)^d$  yields

$$\frac{d^n}{dx^n} x^{\alpha x + \beta} = x^{\alpha x + \beta - n} \sum_{d=0}^n (\alpha x)^d C_{n,d}(z, \beta), \quad (4.18)$$

where, with  $j = n - d$ ,

$$C_{n,d}(z, \beta) = \sum_{s=0}^d \binom{n}{s} \frac{z^s}{(d - s)!} \partial_\beta^{d-s} Q_{j,d-s}(\beta). \quad (4.19)$$

For completeness, choose  $s$  factors  $tz$  from the  $d$ th power of the bracket in (4.17). The remaining factors contribute  $(1+t)^{d-s} \log^{d-s}(1+t)$ . Write the logarithm power as  $\partial_\beta^{d-s}$  acting on  $(1+t)^{\beta+d-s}$  and use

$$[t^{n-s}](1+t)^{\beta+d-s} = \frac{(\beta+d-s)^{n-s}}{(n-s)!} = \frac{Q_{j,d-s}(\beta)}{(n-s)!}.$$

The factorial prefactors simplify to  $\binom{n}{s}/(d-s)!$ , proving the formula.

#### 4.4.3 The coefficient oracle

**Theorem 4.6** (Exact coefficient formula). *Suppose  $(k, j, \ell) \in \mathcal{E}_n$ , and put*

$$d = n - j, \quad h = d + k - \ell,$$

$$q_- = \max(0, k - \ell, k - d), \quad q_+ = \min(k, h, j).$$

Then

$$c_n(k, j, \ell) = \sum_{q=q_-}^{q_+} \binom{n}{\ell - k + q} \binom{h}{q} T_d(k - q; q) [\beta^h] Q_{j, h-q}(\beta). \quad (4.20)$$

An empty sum is zero. Outside the envelope, the coefficient is zero.

*Proof.* Expand the outer exponential:

$$f(x) = \sum_{m \geq 0} \frac{x^{mx} z^m}{m!}, \quad x^{mx} z^m = \partial_\beta^m x^{mx+\beta} \Big|_{\beta=0}.$$

Apply (4.18) with  $\alpha = m$ . For any polynomial  $C(\beta)$ ,

$$\partial_\beta^m (e^{\beta z} C(\beta)) \Big|_{\beta=0} = \sum_{q=0}^m \frac{m!}{(m-q)!} z^{m-q} [\beta^q] C(\beta).$$

After division by  $f = e^{uz}$ , the sum over  $m$  occurring for fixed  $d, q$  is

$$\begin{aligned} e^{-uz} \sum_{m \geq q} \frac{m^d u^m z^{m-q}}{(m-q)!} &= u^q e^{-uz} \sum_{r \geq 0} (r+q)^d \frac{(uz)^r}{r!} \\ &= u^q \sum_{p=0}^d T_d(p; q) (uz)^p. \end{aligned}$$

To obtain  $u^k z^\ell$ , take  $p = k - q$  and take  $s = \ell - k + q$  from (4.19). Set  $r = d - s = h - q$ . Finally,

$$[\beta^q] \frac{\partial_\beta^r Q_{j,r}(\beta)}{r!} = \binom{q+r}{q} [\beta^{q+r}] Q_{j,r}(\beta),$$

and  $q + r = h$ . This proves (4.20); the stated bounds on  $q$  enforce  $p \in [0, d]$ ,  $s \in [0, d]$ , and  $q \leq j$ .

The infinite series can be read as locally normally convergent analytic series for positive  $x$  near any fixed point, or as formal exponential identities followed by extraction of the finitely many relevant coefficients.  $\square$

#### 4.4.4 Three boundary checks

Useful independently checkable consequences are

$$[v^0]P_n = (z+1)^n \sum_{k=1}^n S(n, k)(uz)^k, \quad n \geq 1, \quad (4.21)$$

$$[u^n]P_n = (z(z+1) + v)^n, \quad (4.22)$$

$$[v^n]P_n = u^n. \quad (4.23)$$

The first follows either from (4.20) or from setting  $v = 0$  in the recurrence. The second follows because reaching  $u$ -degree  $n$  requires every step to differentiate the outer factor  $f$ . For the third, the coefficient of the next highest  $v$ -power is obtained by multiplication by  $u - n$ , starting from 1.

### 4.5 A rigorous cubic lower bound

An upper bound alone does not establish cubic growth. We now show that a positive cubic number of the candidate cells must survive, without assuming that cancellations are rare.

#### 4.5.1 Every row has its advertised highest coefficient

Let

$$R_{n,k,j}(z) = [u^k v^j]P_n(u, v, z).$$

**Lemma 4.7** (Nonzero leading coefficient). *For every  $1 \leq k \leq n$  and  $0 \leq j \leq n$ , the polynomial  $R_{n,k,j}$  is nonzero and has degree exactly  $U_{n,k,j}$ .*

*Proof.* For  $j = 0$ , equation (4.21) gives

$$R_{n,k,0}(z) = S(n, k)z^k(z+1)^n.$$

Now let  $j \geq 1$  and put  $d = n - j$ . If  $k \leq d$ , the proposed highest exponent is  $\ell = d + k - 1$ , so  $h = 1$  in (4.20). Only  $q = 0, 1$  contribute. The coefficient is

$$(-1)^{j-1}(j-1)! \left( \binom{n}{d-1} S(d, k) + \binom{n}{d} S(d+1, k) \right), \quad (4.24)$$

which is nonzero. Here we used  $[\beta]Q_{j,0} = [\beta]Q_{j,1} = (-1)^{j-1}(j-1)!$ .

If  $k \geq d+1$ , the highest exponent is  $\ell = 2d$ , and  $h = k - d$ . The only possible summand is  $q = h$ , giving

$$[z^{2d}]R_{n,k,j}(z) = \binom{n}{d} s(j, k-d). \quad (4.25)$$

It is nonzero because  $1 \leq k - d \leq j$ . The support envelope already rules out higher exponents, completing the proof.  $\square$

#### 4.5.2 A high-multiplicity root

**Lemma 4.8** (Divisibility). *The row polynomial is divisible by*

$$z^{\max(k-j,0)}(z+1)^{M_{n,k,j}}, \quad M_{n,k,j} = \max(n-2j, k-j, 0). \quad (4.26)$$

*Proof.* Extracting  $u^k v^j$  from (4.8) gives

$$R_{n+1,k,j} = z(z+1)R_{n,k-1,j} + R_{n,k-1,j-1} + k(z+1)R_{n,k,j} \\ + (\partial_z - (j-1))R_{n,k,j-1}. \quad (4.27)$$

Rows with out-of-range indices are zero. Begin with  $R_{0,0,0} = 1$ . For the factor  $z+1$ , the first and third terms increase the inherited multiplicity by one. The second term has inherited multiplicity at least  $\max(n-2j+2, k-j, 0)$ . The derivative in the last term can reduce its inherited multiplicity by at most one; before differentiation it is at least  $\max(n-2j+2, k-j+1, 0)$ . Each is sufficient for the target exponent  $\max(n+1-2j, k-j, 0)$ . This proves the  $(z+1)$  assertion by induction. The same argument with the factor  $z$  proves the exponent  $\max(k-j, 0)$ . Since  $z$  and  $z+1$  are relatively prime, both divisibilities hold together.  $\square$

**Lemma 4.9** (Multiplicity forces many terms). *A nonzero polynomial with a nonzero root of multiplicity at least  $M$  has at least  $M+1$  nonzero coefficients.*

*Proof.* Suppose  $p(z) = \sum_{i=1}^t b_i z^{e_i}$ , with distinct nonnegative integers  $e_i$  and  $b_i \neq 0$ . If a root  $\alpha \neq 0$  has multiplicity at least  $t$ , then  $(z\partial_z)^r p(\alpha) = 0$  for  $0 \leq r < t$ . Thus

$$\sum_{i=1}^t e_i^r b_i \alpha^{e_i} = 0, \quad 0 \leq r < t.$$

The matrix  $(e_i^r)$  is an invertible Vandermonde matrix, contradicting  $b_i \alpha^{e_i} \neq 0$ . Therefore the multiplicity is at most  $t-1$ .  $\square$

### 4.5.3 Summing the mandatory terms

By Lemmas 4.7–4.9, each  $(k, j)$  row contributes at least  $1 + M_{n,k,j}$  nonzero monomials. Rows are disjoint in the canonical basis, so

$$a(n) \geq \sum_{k=1}^n \sum_{j=0}^n (1 + \max(n-2j, k-j, 0)). \quad (4.28)$$

For fixed  $j$ , the sum of the maxima over  $k$  is

$$\sum_{k=1}^n M_{n,k,j} = \begin{cases} n(n-2j) + \frac{j(j+1)}{2}, & 0 \leq j \leq \lfloor n/2 \rfloor, \\ \frac{(n-j)(n-j+1)}{2}, & j > \lfloor n/2 \rfloor. \end{cases} \quad (4.29)$$

In the first case, start with  $n-2j$  in each row and add the excess  $1 + \dots + j$ . In the second case, only  $k > j$  contributes. Summing these elementary progressions, and adding the  $n(n+1)$  ones in (4.28), gives exactly  $L(n)$  in (4.2). This completes the proof of the growth bounds in Theorem 4.1.

In particular,

$$\frac{7}{24} \leq \liminf_{n \rightarrow \infty} \frac{a(n)}{n^3} \leq \limsup_{n \rightarrow \infty} \frac{a(n)}{n^3} \leq \frac{2}{3}. \quad (4.30)$$

These constants are proved bounds, not estimates of an experimentally fitted leading coefficient.

## 4.6 Exact cancellation families

### 4.6.1 Logarithm-free coefficients and derivatives of a falling factorial

Let  $F_n(y) = y^n$ . In (4.20), setting  $\ell = 0$  forces  $q = k$ , so for  $d = n - j$  and  $1 \leq k \leq j$ ,

$$c_n(k, j, 0) = \frac{k^d}{d! k!} F_n^{(d+k)}(d). \quad (4.31)$$

Although written using factorial quotients, this expression is an integer; the implementation instead uses integer polynomial coefficients and binomial coefficients. Formula (4.31) reduces every logarithm-free cancellation to an exact derivative evaluation of a single elementary polynomial.

**Proposition 4.10** (Central cancellations at every odd order). *For odd  $n \geq 1$ ,*

$$\Delta(n) \geq \left\lfloor \frac{n+1}{4} \right\rfloor. \quad (4.32)$$

*Proof.* Write  $n = 2d + 1$  and set  $j = d + 1$ . Then

$$F_n(d + \beta) = \beta \prod_{r=1}^d (\beta^2 - r^2)$$

is odd. For each  $1 \leq k \leq d + 1$  with  $k \equiv d \pmod{2}$ , its coefficient of  $\beta^{d+k}$  vanishes. Formula (4.31) gives a missing monomial  $(k, d + 1, 0)$ . There are  $\lfloor (d + 1)/2 \rfloor = \lfloor (n + 1)/4 \rfloor$  such values of  $k$ .  $\square$

Thus  $a(n) = B(n)$  is impossible at every odd  $n \geq 3$ . This does not resolve whether equality occurs infinitely often at even orders. Nor does it classify all logarithm-free cancellations: additional integer derivative zeros occur away from the central symmetry.

### 4.6.2 An infinite family with positive logarithm exponent

It is tempting to guess that all cancellations are logarithm-free. The following family disproves that guess.

Since  $f = e^g$  with  $g = uz$ , the  $u^{n-1}$  part of  $f^{(n)}/f$  consists of the partition type having one block of size two and all other blocks of size one. Equivalently, induction in (4.8) gives

$$[u^{n-1}]P_n = \binom{n}{2} (A + v)^{n-2} ((z + 1)A + (3z + 2)v - v^2), \quad n \geq 2. \quad (4.33)$$

For  $2 \leq j \leq n - 1$ , put  $m = n - j - 1$  and

$$a = \binom{n-2}{j}, \quad b = \binom{n-2}{j-1}, \quad c = \binom{n-2}{j-2}.$$

The coefficient of  $v^j$ , after removing the common factor  $\binom{n}{2}$ , is

$$z^m (z + 1)^m [-cz^2 + (a + 3b - c)z + (a + 2b)]. \quad (4.34)$$

Its coefficient of  $z^{2m+1}$  is

$$a + 3b - (m + 1)c. \quad (4.35)$$

Using

$$\frac{a}{c} = \frac{m(m+1)}{j(j-1)}, \quad \frac{b}{c} = \frac{m+1}{j-1},$$

this vanishes exactly when  $m = j^2 - 4j$  in the stated range.

**Proposition 4.11** (Non-logarithm-free cancellation family). *For every integer  $t \geq 2$ , let*

$$n = t^2 + t - 1, \quad k = n - 1, \quad j = t + 2, \quad \ell = 2t^2 - 7. \quad (4.36)$$

*Then  $(k, j, \ell) \in \mathcal{E}_n$ ,  $\ell > 0$ , and  $c_n(k, j, \ell) = 0$ .*

*Proof.* Substitute  $j = t + 2$  into  $m = j^2 - 4j$ : it gives  $m = t^2 - 4$  and  $n = m + j + 1 = t^2 + t - 1$ . Then  $\ell = 2m + 1$  is positive, is inside the envelope, and its coefficient vanishes by (4.35). The next higher coefficient is  $-\binom{n}{2}c \neq 0$ , so this is a genuine internal missing term.  $\square$

The first corresponding orders are 5, 11, 19, 29, 41, 55, ..., with missing triples

$$(4, 4, 1), (10, 5, 11), (18, 6, 25), (28, 7, 43), \dots$$

These are additional to the logarithm-free central cancellations. Since all orders in (4.36) are odd, this proves  $\Delta(n) \geq \lfloor (n+1)/4 \rfloor + 1$  along that infinite subsequence.

## 4.7 Algorithms and complexity

### 4.7.1 The dependency-free exact recurrence

The most direct implementation stores a dictionary from triples to integer coefficients. At each differentiation it applies the seven transitions, combines equal keys, and removes zeros. A short reference version is given in Section 4.10. It uses no symbolic-algebra system and no numerical approximation.

The supplied production version packs a triple into one integer key. For a requested maximum order  $N$ , take

$$b = N + 2, \quad c = 2N + 3, \quad \text{key}(k, j, \ell) = (kb + j)c + \ell.$$

Incrementing  $k, j, \ell$  corresponds to adding  $bc, c, 1$ , respectively. The chosen bases exceed every possible exponent, so no carries can identify different monomials. The packed representation reduces tuple allocation and dictionary overhead; it does not change the mathematics.

**Proposition 4.12** (Arithmetic complexity). *The exact recurrence computes the entire prefix through  $N$  in  $O(N^4)$  integer arithmetic operations and  $O(N^3)$  live integer coefficients. Each coefficient has  $O(N \log N)$  bits.*

*Proof.* At order  $n$  there are at most  $B(n) = O(n^3)$  terms, and each contributes to at most seven destinations. Summing through  $N$  gives  $O(N^4)$  operations. Only the current and next dictionaries are needed, giving  $O(N^3)$  storage.

Let  $H_n$  be the sum of the absolute values of the coefficients of  $P_n$ . The absolute sum of all multipliers leaving a term is  $3 + 2k + j + \ell \leq 3 + 5n$ . Therefore

$$H_{n+1} \leq (5n + 3)H_n, \quad H_0 = 1, \quad H_n \leq \prod_{r=0}^{n-1} (5r + 3) \leq 5^n n!.$$

Every individual coefficient satisfies this bound, so its bit length is  $O(n \log n)$ .  $\square$

This is a unit-cost *integer-arithmetic* bound, not a claim that arbitrarily large integers are machine words. With elementary small-integer multiplication, a conservative bit bound is  $O(N^5 \log^2 N)$  time and  $O(N^4 \log N)$  coefficient-storage bits. Hashing and object overhead affect practical memory. We do not claim the algorithm is asymptotically optimal.

#### 4.7.2 A deterministic modular sieve with exact refinement

Most coefficients need never be constructed as large integers. Fix any integer modulus  $M \geq 2$ , and run the same recurrence in  $\mathbb{Z}/M\mathbb{Z}$ . If a residue is nonzero, the original integer coefficient is certainly nonzero. Only zero residues need further attention.

The exact single-term algorithm is:

1. Compute  $P_n \bmod M$  by the seven-transition recurrence.
2. Enumerate the triples in  $\mathcal{E}_n$  whose residues are zero.
3. Evaluate each such coefficient with (4.20), using integer arithmetic, and count only the coefficients that are actually zero.
4. Return  $B(n)$  minus that exact zero count.

A zero residue is *never* accepted as proof of a zero coefficient. Consequently this algorithm is correct for every admissible modulus, including small or composite moduli. No randomness, primality assumption, or conjecture about support is required. Tests deliberately use moduli 2, 7, and 12 to force many false modular zeros.

The supplied NumPy implementation uses signed 64-bit arrays and the modulus 1 000 000 007 by default. At order  $m$ , the array has shape  $(m+1, m+1, 2m+1)$ . The next array has shape  $(m+2, m+2, 2m+3)$ , and each transition becomes a shifted slice addition. All entries are reduced modulo  $M$  after each step. The implementation checks

$$(10n+10)M < 2^{63}$$

before starting; this conservative condition bounds the intermediate products and sums before modular reduction. Overflow would invalidate the certificate, so the check is part of correctness rather than merely an optimization detail.

The dense modular pass takes  $O(n^4)$  fixed-width operations and  $O(n^3)$  fixed-width words. Let  $Z_M(n)$  denote the number of zero residues inside the envelope. The oracle tables  $Q_{j,r}$  and  $T_d(p;q)$ , and Pascal's triangle for binomial coefficients, are built lazily. For a single target  $n$ , filling all potentially needed tables takes  $O(n^3)$  integer arithmetic operations and storage; each candidate coefficient then requires at most  $O(n)$  integer products and additions. Thus a conservative oracle bound is

$$O(n^3 + nZ_M(n)) \subseteq O(n^4).$$

In practice the lazy tables are much smaller when the modular pass leaves few candidates. Unlike the reference recurrence, the oracle may multiply two large integers, so its unit-arithmetic bound should again not be read as a bit-operation bound.

#### 4.7.3 Which implementation to use

For a whole prefix, the dependency-free recurrence has the simplest predictable cost and automatically supplies every intermediate count. For an isolated larger index, the modular

method usually avoids nearly all large-integer coefficient work. Running a new modular pass separately for every prefix index would discard this advantage and incur repeated work.

Neither method differentiates a transcendental expression in a computer algebra system. Both return the exact OEIS term count, including signed cancellations, rather than an expression-size estimate.

## 4.8 Computational validation and results

### 4.8.1 Checks actually performed

The accompanying files record ten passing test groups. The tests compare the reference recurrence with all 43 published A281434 values through  $n = 42$ , and compare the generated prefix with all 80 published A352697 deficits through  $n = 80$  [3, 5]. Every candidate coefficient through  $n = 12$  is checked against the independent formula (4.20). The two methods are also compared through  $n = 12$  for each of four moduli, including the deliberately small and composite choices described above.

Additional tests verify the envelope, its exact cardinality, the lower-bound sum, the  $(z + 1)$  multiplicities, the two leading-coefficient formulas, the odd central cancellations, and initial instances of the infinite non-logarithm-free family. Such finite tests check the implementations and help detect algebraic mistakes; the universal statements are justified by the proofs, not by the tests.

The integer recurrence was run through  $n = 100$ . The modular implementation was separately checked against it at  $n = 20, 40, 60, 80, 100$ . Selected larger computations used the modular pass and exact verification of all candidate zeros. The archive contains the complete generated prefix, the tests, and the selected missing triples.

### 4.8.2 Selected exact values

| $n$ | $a(n)$    | $B(n)$    | $\Delta(n)$ |
|-----|-----------|-----------|-------------|
| 0   | 1         | –         | –           |
| 1   | 3         | 3         | 0           |
| 2   | 12        | 12        | 0           |
| 3   | 30        | 31        | 1           |
| 5   | 113       | 115       | 2           |
| 11  | 1 017     | 1 023     | 6           |
| 20  | 5 760     | 5 760     | 0           |
| 40  | 44 320    | 44 320    | 0           |
| 60  | 147 680   | 147 680   | 0           |
| 80  | 347 838   | 347 840   | 2           |
| 100 | 676 800   | 676 800   | 0           |
| 101 | 697 178   | 697 203   | 25          |
| 150 | 2 272 700 | 2 272 700 | 0           |
| 151 | 2 318 265 | 2 318 303 | 38          |
| 200 | 5 373 600 | 5 373 600 | 0           |

For example, the two missing triples at  $n = 80$  are  $(33, 36, 0)$  and  $(42, 45, 0)$ . At  $n = 101$ , all 25 missing terms are supplied by the central logarithm-free symmetry in Proposition 4.10. These are exact statements about the recorded finite computations.

### 4.8.3 Measured performance

On the execution environment used for this report, the packed integer recurrence generated the full prefix  $0 \leq n \leq 100$  in approximately 26.16 seconds. Selected single-index modular runs were:

| $n$ | Modular zero candidates | Actual zeros | Elapsed seconds |
|-----|-------------------------|--------------|-----------------|
| 40  | 0                       | 0            | 0.021           |
| 60  | 0                       | 0            | 0.096           |
| 80  | 2                       | 2            | 0.342           |
| 100 | 0                       | 0            | 0.820           |
| 101 | 25                      | 25           | 0.698           |
| 150 | 0                       | 0            | 4.970           |
| 151 | 38                      | 38           | 5.460           |
| 200 | 0                       | 0            | 17.769          |

These are individual measurements, not a controlled benchmark suite or a promise of timings on other hardware. In particular, computing one target by the modular method and computing an entire prefix by the integer method are different workloads. The exact timings, interpreter information, and modulus are saved in `data/benchmarks.csv` and `data/environment.json`. The fact that every zero residue in this small selection was a genuine zero is not assumed by the algorithm. The small-modulus tests exercise the opposite situation and confirm that false modular zeros are rejected by the exact oracle.

## 4.9 What is and is not established asymptotically

The unconditional conclusion is cubic-order growth, with the fully explicit bounds (4.2). Consequently,  $a(n) = n^{3+o(1)}$  in the logarithmic-exponent sense. The lower bound remains valid even if many more cancellations occur at orders far beyond the computed range.

There are three progressively stronger claims:

$$\begin{array}{ll}
 \text{Proved here:} & a(n) = \Theta(n^3). \\
 \text{Not proved here:} & a(n) \sim \frac{2}{3}n^3. \\
 \text{Stronger OEIS conjecture:} & a(n) = \frac{2}{3}n^3 + n^2 + O(n).
 \end{array}$$

The last statement is equivalent to  $\Delta(n) = O(n)$ , given the exact formula for  $B(n)$ , and implies the middle statement. Neither follows from agreement with  $B(n)$  at many computed indices.

A proof of the middle statement needs only a subcubic bound on the number of holes; it need not classify every cancellation. In particular, the entire plane  $\ell = 0$  has only  $O(n^2)$  candidate cells. Thus even a complete failure to classify the logarithm-free derivative zeros would not by itself prevent proving the leading equivalent. The remaining issue is a uniform bound on cancellations with  $\ell > 0$  across the three-dimensional envelope. Formula (4.20) supplies an explicit integer expression for those coefficients, and the family in Proposition 4.11 shows why a blanket nonvanishing assertion would be false.

For the stronger linear-deficit conjecture, even the logarithm-free plane requires more delicate control. Its central symmetry already forces  $\Delta(n)$  to be at least a positive linear function along the odd orders, so an  $o(n)$  bound is impossible. The appropriate target is an upper bound of matching order, not an assertion that the derivative eventually fills its whole envelope.

The computation problem is therefore resolved by unconditional exact algorithms, and the polynomial growth exponent is established. Determining the exact leading asymptotic constant remains a separate nonvanishing problem in this analysis.

#### 4.10 A compact exact implementation

The following version emphasizes transparency. The archive's `code/a281434.py` additionally provides packed keys, the coefficient oracle, modular arrays, input validation, and a command-line interface.

```
from collections import defaultdict

def a281434_prefix(N):
    if not isinstance(N, int) or isinstance(N, bool) or N < 0:
        raise ValueError("N must be a nonnegative integer")
    p = {(0, 0, 0): 1}
    values = [1]
    for _ in range(N):
        q = defaultdict(int)
        for (k, j, ell), c in p.items():
            q[k+1, j, ell+2] += c
            q[k+1, j, ell+1] += c
            q[k+1, j+1, ell] += c
            if k:
                q[k, j, ell+1] += k*c
                q[k, j, ell] += k*c
            if j:
                q[k, j+1, ell] -= j*c
            if ell:
                q[k, j+1, ell-1] += ell*c
        p = {key: c for key, c in q.items() if c}
        values.append(len(p))
    return values
```

The only potentially negative transition is the one with multiplier  $-j$ . Deleting it, taking absolute values before accumulation, or replacing the dictionary by a set changes the sequence.

#### 4.11 Reproducing the supplied computations

Python 3.9 or later is sufficient for the dependency-free method. The modular method additionally requires NumPy. From the archive's `code` directory, the commands below produce a single term, inspect missing monomials, create a prefix, and run the tests.

```
python a281434.py 100 --method modular
python a281434.py 80 --method modular --holes
python a281434.py 100 --prefix
python -m unittest -v test_a281434.py
python run_experiments.py --reference
python run_experiments.py 20 40 60 80 100 101 150 151 200
```

The first command returns 676800. The second returns 347838 and the two missing triples listed in the article. The full prefix and related deficits are in `data/b281434.txt` and `data/b352697.txt`. These are generated data files accompanying this report, not claims

that the OEIS has incorporated the additional terms.

The LaTeX source builds with

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

Repeated runs resolve the table of contents and cross-references. The bundled `build.sh` and `build.ps1` perform these commands.

## 5 The three answers side by side

### 5.1 What was proved

|                   | Part 2: $x^x$<br>A293239                                                                                           | Part 3: $x^{x^2}$<br>A290268                                                                                                                  | Part 4: $x^{x^x}$<br>A281434                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Frame             | $t = x^{-1}$ , $L = \log x$<br>(2 variables)                                                                       | $z = x^2$ , $L = \log x$<br>(2 variables)                                                                                                     | $u = x^x$ , $v = x^{-1}$ , $z = \log x$<br>(3 variables)                                              |
| Clause of Thm 1.2 | (a)                                                                                                                | (b)                                                                                                                                           | (a)                                                                                                   |
| Growth order      | $\Theta(n^2)$                                                                                                      | $\Theta(n^2)$                                                                                                                                 | $\Theta(n^3)$                                                                                         |
| Upper bound       | $a(n) \leq q(n)$ , the proposed value                                                                              | $a(n) \leq U(n)$ , the conjectured value                                                                                                      | $a(n) \leq \frac{2n^3 + 3n^2 + 4n}{3}$<br>( $n \geq 1$ ; at $n = 0$ it reads $0 \geq 1$ )             |
| Lower bound       | $n + 1 + \lfloor n^2/4 \rfloor$                                                                                    | $(3n^2 + 10n + 8)/8$ even,<br>$(3n^2 + 12n + 1)/8$ odd                                                                                        | $\frac{7n^3 + 39n^2 + 26n + 3\varepsilon_n(n-1)}{24}$                                                 |
| Exact structure   | $a(n) = 1 + \binom{n+1}{2} - Z(n)$ ,<br>$Z$ counting zeros of the<br>Lehmer–Comtet triangle<br>A008296             | $a(n) = U(n) -$<br>(extra zeros), all pre-<br>dicted zeros explained                                                                          | $a(n) =  \mathcal{E}_n  - \Delta(n)$ , $\Delta$<br>the in-envelope cancella-<br>tion count            |
| Settled           | the reduction; the $\Theta$ ; a<br><i>disproof</i> of the recurrence<br>printed with range $n > 6$ ,<br>at $n = 8$ | the upper bound; the can-<br>cellation mechanism; the<br>$\Theta$ ; a complete zero clas-<br>sification on the first two<br>deficit diagonals | the $\Theta$ ; two exact algo-<br>rithms; central cancella-<br>tions at every odd order<br>$n \geq 3$ |
| Not settled       | the principal closed<br>formula: needs Conjec-<br>ture 2.9                                                         | the conjecture itself, and<br>the constant $\frac{1}{2}$ : needs<br>Conjecture 3.19                                                           | the constant: $a(n) \sim \frac{2}{3}n^3$<br>would need $\Delta(n) = o(n^3)$                           |

*Update (29 September 2026).* For  $x^{x^2}$  the table records Part 3 as first written. Its later Sections 3.14–3.26 add to the settled column the complete zero classification on the third and fourth deficit diagonals, and improve the lower bound to  $(3n^2 + 26n - 72)/8$  for even  $n \geq 8$  and  $(3n^2 + 28n - 111)/8$  for odd  $n \geq 17$  (Remark 3.46); the leading constant  $\frac{1}{2}$  is still not settled, and Conjecture 3.19 is still needed, now only on deficits  $m \geq 5$ .

### 5.2 Why the orders differ

The growth orders line up with the number of variables in the frame, and Part 1 explains why the upper half of that observation must hold. In a frame of  $m$  variables, each exponent in  $\text{supp } P_n$  is bounded by a constant multiple of  $n$ , because each application of the recurrence raises any one exponent by a bounded amount. The envelope therefore lies in a box of side  $O(n)$  in  $m$  dimensions, so  $|\mathcal{E}_n| = O(n^m)$ ; with Corollary 1.5 for the two frames it covers, and Part 4’s own independence argument for the third, Proposition 1.6 then gives  $a(n) = O(n^m)$ . With  $m = 2$  for  $x^x$  and  $x^{x^2}$  and  $m = 3$  for  $x^{x^x}$ , the observed orders  $n^2$ ,  $n^2$ ,  $n^3$  are exactly the envelope dimensions.

Two cautions are worth stating, because the pattern invites more than it supports.

First, the matching *lower* bounds are not a consequence of the frame at all. A frame bounds the support from outside; it says nothing about how much of the envelope survives cancellation, and in principle almost all of it could vanish. Each part proves its own lower bound by

exhibiting coefficients that are nonzero, and those arguments have nothing in common. Part 2 counts nonzero entries along each diagonal of its triangle, bounding the zeros there by the degree of a polynomial that cannot vanish identically. Part 3 proves strict *positivity* on a region, through a substitution that turns its coefficients into an integral with a sign-definite kernel. Part 4 forces a root of high multiplicity and then uses the fact that a polynomial with a nonzero root of multiplicity  $M$  has at least  $M + 1$  nonzero coefficients. Three mechanisms, no common method; the framework supplies none of them.

Second, the frame is a choice, not an invariant. Nothing stops one from embedding any of these problems in a larger frame and obtaining a worse upper bound; what the three parts exhibit is the smallest frame each author found in which the derivative closes. The claim “the order is the dimension” is therefore a statement about these three constructions, not a theorem about power towers, and it should not be extrapolated to  $x^{x^x}$  without doing the work.

### 5.3 What the shared framework buys, and what it does not

It buys four things, with two qualifications that should be stated rather than glossed. The count is an invariant of the function rather than of a differentiation algorithm — by Corollary 1.5 for Parts 2 and 3, and by Part 4’s own argument for the third frame, which that corollary does not cover. The coefficients are integers, so the computations are exact and a nonzero residue is a proof of nonvanishing (Proposition 1.8). The *envelope* bound is free once the envelope is described (Proposition 1.6) — though only Part 4’s advertised upper bound is the bare envelope count; the bounds listed for the other two in Section 5.1 are the envelope minus proved cancellation families, and those families are real theorems, not framework. And the problem is reduced from “count the terms” to “find the zeros”, which is the form all three parts actually work in.

It buys nothing at all about the zeros. That is where each of the three problems becomes its own problem, and where all three still have something open. In Part 2 the zeros are the vanishing entries of a classical triangle and the missing step is their complete classification. In Part 3 the zeros are forced by a reflection symmetry and the missing step is that there are no others. In Part 4 the zeros are not even conjecturally classified, and the open question is the weaker one of their density. These three statements are not special cases of one another, and no argument in any part transfers to any other.

### 5.4 What the three do share, beyond the framework

There is more common structure than the framework alone, and it is worth setting down, because it is the one thing a reader could take from these three parts that none of them states on its own.

Every cancellation family that any of the three parts *proves* comes from the same source: the falling factorial  $y^{\underline{n}} = y(y-1)\cdots(y-n+1)$  is symmetric about its centre, so that  $(n-1-y)^{\underline{n}} = (-1)^n y^{\underline{n}}$ , and a polynomial that is odd about a point has no even coefficient there. Each part reaches that fact in its own coordinates:

- Part 2, Proposition 2.8, centres at  $r = 2h$  to get  $F_{4h+1}(2h+u) = u \prod_{s=1}^{2h} (u^2 - s^2)$ , an odd polynomial, killing the coefficient of  $u^{2h}$ ;
- Part 3, Lemma 3.10, uses  $F_n(n-1-u) = (-1)^n F_n(u)$  directly, transported through a 2-step difference;
- Part 4, Proposition 4.10, gets  $F_n(d+\beta) = \beta \prod_{r=1}^d (\beta^2 - r^2)$ , again odd.

So all three counting problems are, at bottom, questions about the integer zeros of derivatives and finite differences of one classical polynomial, and every zero any of them can currently explain is a parity zero of that polynomial about its centre. What is *not* shared is everything past that point: the triangles the parity acts on are different, the difference operators are different, and — this is the crux — in no case is the parity family known to be the whole zero set, which is exactly what each part’s open question asks.

Given that, a caution about the two  $\Theta(n^2)$  results. Parts 2 and 3 both bound a two-dimensional envelope and subtract cancellations, and both leave the leading constant open, and it would be easy to read them as one result twice. They are not: the arrays are different, one is normalized by  $x^{-n}$  and the other is not, and the open statements are statements about different objects. What can be said honestly is the metamathematical thing: no argument given in either part carries over to the other. Whether either open conjecture *implies* the other is not something this article can settle, and it does not claim to.

### 5.5 What would settle each one

- $x^x$  (**Part 2**). Conjecture 2.9: that for  $1 \leq r \leq m$  the only vanishing first-kind Lehmer–Comtet entries are the sporadic  $b(8, 5)$  and the symmetry family  $b(4h + 1, 2h)$ ,  $h \geq 1$ . The sporadic entry and the whole family are both *proved* to vanish in Part 2 — and the sporadic one is genuinely sporadic, not the first term of a second family, which is part of what makes the converse hard; the open direction is that there are no others. A proof would give the closed formula for  $a(n)$  and, with it, the corrected recurrence for the actual derivative count on its true range  $n \geq 14$ .
- $x^{x^2}$  (**Part 3**). The nonvanishing statement isolated in Conjecture 3.19: that no coefficient vanishes outside the *two* explained families — the degree-forced top diagonal ( $j = k$  with  $n > 2k$ ) and the reflection family ( $k$  even,  $n = 4k - 2j + 1$ ). Coefficients do vanish in bulk outside the reflection family alone; it is the union that is conjecturally complete. Theorem 3.20 shows the statement is exactly equivalent to the OEIS conjecture, so nothing weaker will do. Since the later addition to that part it is proved on deficits  $m \leq 4$  (Theorem 3.23); what remains is  $m \geq 5$ .
- $x^{x^x}$  (**Part 4**). A uniform bound  $\Delta(n) = o(n^3)$  would give the equivalent  $a(n) \sim \frac{2}{3}n^3$ ; the stronger  $\Delta(n) = O(n)$ , conjectured not in A281434 itself but in the companion entry A352697 which records the deficits, needs control even on the logarithm-free plane, where only  $O(n^2)$  cells are candidates. Whether equality  $\Delta(n) = 0$  recurs infinitely often at even orders is open.

The three obstructions do not even have the same shape. In Parts 2 and 3 the open statement is a nonvanishing statement: no coefficient outside the explained families is zero. In Part 4 it cannot be, because additional zeros provably exist — Proposition 4.11 exhibits an infinite family of them with positive logarithm exponent, and Proposition 4.10 forces more at every odd order  $n \geq 3$ . There the open question is only *how many*: a density statement,  $\Delta(n) = o(n^3)$ , not a nonvanishing one. That difference is the sharpest thing the comparison has to offer, and it is the reason the three investigations are printed here as three.

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